Modelling and Simulation of Mechatronic Systems

Complete Study Notes

Based on A. Macchelli's lectures — UniBO DISI

Study Notes

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Part I

Foundations of Bond Graph Modelling

Chapter 1

Physical Modelling and the Bond Graph Philosophy

1.1 The systems viewpoint

A **system** is any entity that can be conceptually separated from its environment by a physical or conceptual boundary. Inside the boundary are interacting subsystems and components.

Key Insight: The modelling trade-off

No system can be modelled exactly. The art is to choose the **right level of complexity**. A model that is too simple will miss important dynamics; one that is too detailed will obscure the physics and be computationally intractable. This judgement is acquired through practice and physical intuition.

Dynamic models serve four distinct purposes:

- **Analysis**: Predict the future given known inputs and initial conditions. This can be data-based (black box) or physics-law based (white box) depending on how the component is characterized.
- **Identification**: Find the model from measured input/output histories.
- **Synthesis**: Find the system that turns a given input into a desired output.
- **Control**: Design a feedback law that drives the system to a desired behaviour.

1.2 Power as the universal currency

The central idea of bond graph theory is that **all physical interactions involve the exchange of power**. When a motor drives a pump, when a spring pushes a mass, when a voltage source drives a current through a resistor, in every case, energy flows across a boundary, and the instantaneous rate of that flow is power. By making power the primary language, bond graphs work equally well in every energy domain. At every interconnection point between subsystems, called a **port**, two variables are defined whose product is instantaneous power:

$$P(t) = e(t)\, f(t) \tag{1.1}$$

These are called the **effort** e and the **flow** f. Their specific physical meaning depends on the domain:

Domain	**Effort** e	**Flow** f	**Power** ef
Mechanical (translation)	Force F [N]	Velocity V [m/s]	[W]
Mechanical (rotation)	Torque τ [N·m]	Ang. velocity ω [rad/s]	[W]
Electrical	Voltage e [V]	Current i [A]	[W]
Hydraulic	Pressure P [Pa]	Volume flow Q [m^3/s]	[W]

Two additional variables complete the picture. The **momentum** p (generalised) is the time integral of effort; the **displacement** q is the time integral of flow:

$$p(t) = p_0 + \int_{t_0}^{t} e(\tau)\, d\tau, \qquad q(t) = q_0 + \int_{t_0}^{t} f(\tau)\, d\tau, \qquad \dot{p} = e, \quad \dot{q} = f. \tag{1.2}$$

These are the **energy variables** (or state variables). While power is the instantaneous flow, energy flow is the time integral of that power:

$$E(t) = \int^t P(\tau)d\tau = \int^t e(\tau)f(\tau)d\tau$$

In cases where an effort is a function of displacement, the energy is

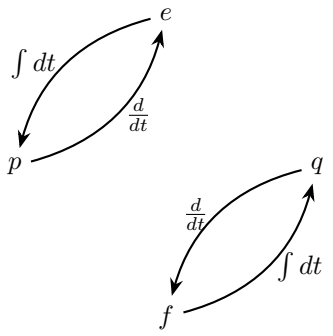
$$E(q) = \int e(q)dq$$

Conversely, when flow is a function of momentum,

$$E(p) = \int f(p)dp$$

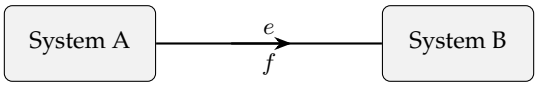
.

The relationship between these four variables is elegantly summarized by the **Tetrahedron of State**:



1.3 Bonds, ports and multiports

A **subsystem with one or more ports** is called a **multiport**. Each port is a place where power can flow in or out. Familiar examples: an electric motor is a 3-port (field winding, armature winding, mechanical shaft); a hydraulic pump is a 2-port (shaft, fluid line); a mechanical spring is a 1-port. When two multiports are connected, meaning one port of each is joined, power flows between them and the effort and flow at the shared port become identical for both. This shared port is a **bond**, drawn as a line with a half-arrow indicating the assumed positive direction of power flow:



When major sub-systems are represented by words instead of internal details, the resulting graph is known as a **word bond graph**. The half-arrow is *not* the direction of causality and *not necessarily* the direction energy actually flows at any given instant, it simply sets the sign convention: power is positive when it flows in the direction of the arrow.

Definition: Bond

A bond is a line connecting two ports. It carries *both* an effort e and a flow f simultaneously. The product ef is the instantaneous power flowing in the direction of the half-arrow. A bond is a 2-wire connection in the metaphorical sense: it always transmits a pair of conjugate variables.

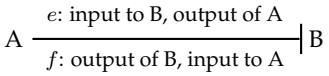
1.4 Causality: inputs, outputs and causal strokes

At any port, one of the two power variables is the **input** (cause) and the other is the **output** (effect). One can either impose a force and observe the resulting velocity, or impose a velocity and observe the resulting force, but **never both simultaneously**. This choice is called the **causality** of the port.

Causality is indicated on a bond graph by a **causal stroke**: a short perpendicular bar at one end of the bond. It is critical to remember that the half-arrow power direction indications and the causal stroke are **totally uncorrelated**. The rule is:

- The causal stroke marks the end toward which the **effort** is being directed (effort is an input to that end).

- The **flow** signal is then directed *away* from the stroked end (flow is an output of that end).



Causal assignment is not free, each type of element has constraints on which causalities are admissible. The choice of causality throughout the bond graph propagates and constrains subsequent choices, and ultimately determines whether the system equations are explicit ODEs, implicit DAEs, or contain algebraic loops. This is detailed in Chapter 4.

1.5 Pure signal flow and active bonds

Not all interconnections transmit significant power. A sensor measuring a velocity transmits negligible power back to the mechanism it measures; an electronic amplifier takes a tiny input signal and produces a much larger output, with power supplied by an external source. In these cases, one of the two power variables is negligible, and the bond **degenerates to a single signal**, called an **active bond**, drawn with a full arrow rather than a half-arrow. Because their description is functional rather than physical, ideal instruments and amplifiers represented by these bonds violate many physical laws. Active bonds are used to represent signal processing, controllers, and ideal sensors within an otherwise power-based bond graph. The combination of bond graphs (for the physical plant) and block diagrams (for the controller) is the natural language for mechatronic system design.

Chapter 2

Basic Bond Graph Elements

A remarkably small set of ideal elements suffices to model the dynamics of systems across all energy domains. There are **nine** fundamental bond graph elements: three 1-port passive elements (R, C, I), two source elements (Se, Sf), two 2-port elements (TF, GY), and two junction elements (0-junction, 1-junction). Every physical system model is assembled from these building blocks.

Real devices can be considered as subsystems from the point of view of power exchanges and external port variables. These basic multiport elements are idealized mathematical models,or conceptual "entities",used to model specific physical effects within a device. The bond graph notation often allows one to visualize aspects of the system structure more easily than with state equations or graphical notations designed strictly for a single energy domain.

2.1 The 1-port resistor *R*: energy dissipation

A resistor is a 1-port in which effort and flow are related by a **static constitutive law** $e = \mathcal{R}(f)$ (or equivalently $f = \mathcal{R}^{-1}(e)$).

No energy is stored: whatever power enters the port is dissipated instantly (converted to heat in the thermal domain, which is usually neglected in the model). In the e–f plane, a passive resistor's constitutive curve lies in the first and third quadrants, ensuring $ef \geq 0$ (power always flows in). Power flows into the port when the product of e and f is positive according to the assigned sign convention. Because energy seems to disappear from the electrical, mechanical, or hydraulic domains, it is in reality transferred into the thermal domain, whose presence is typically neglected in these basic dynamic models.

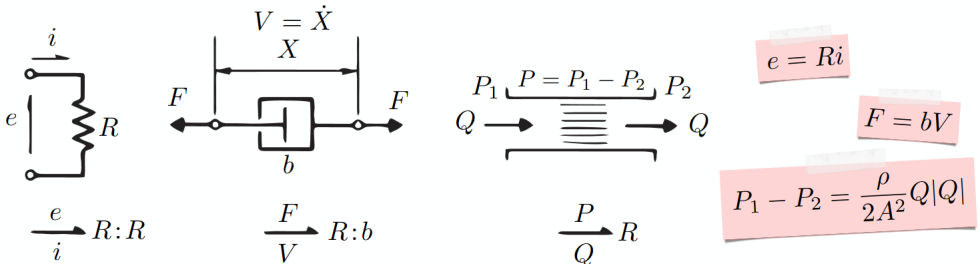


Figure 2.1: Different types of Resistive 1-port elements.

Physical realisations: electrical resistance $e = Ri$; viscous damper $F = bV$; fluid resistance $\Delta P = R_f Q$ (linear); orifice flow $\Delta P = kQ|Q|$ (nonlinear).

In the bond graph:

The half-arrow points toward R (positive power convention). **For a linear element: appending `R:R` (or `R:b` for a damper) names the parameter**.

Key Insight: Causality of R

The causality of a resistor is not fixed: it can be either effort-in or flow-in. This is because the constitutive relation is static, so if you impose one variable, the other is immediately determined without any integration or differentiation.

2.2 The 1-port capacitor *C*: potential energy storage

A capacitor (or compliance) stores **potential energy** via a static relationship between effort and displacement:

$$e = \mathcal{C}^{-1}(q), \qquad q = \int f\, dt. \tag{2.1}$$

In the linear case, $e = q/C$. The linear compliance C is defined as the inverse of the slope of the e versus q constitutive curve.

Stored energy:

$$E(q) = \int_{q_0}^{q} e(\bar{q})\, d\bar{q} \quad \longrightarrow \quad E = \frac{q^2}{2C} \text{ (linear)}. \tag{2.2}$$

Usually E_0 is set to zero by offsetting the energy reference point.

Since the capacitor stores energy without loss, the power flow into the port can reverse, meaning power flows back out into the system. During this process, energy is strictly conserved.

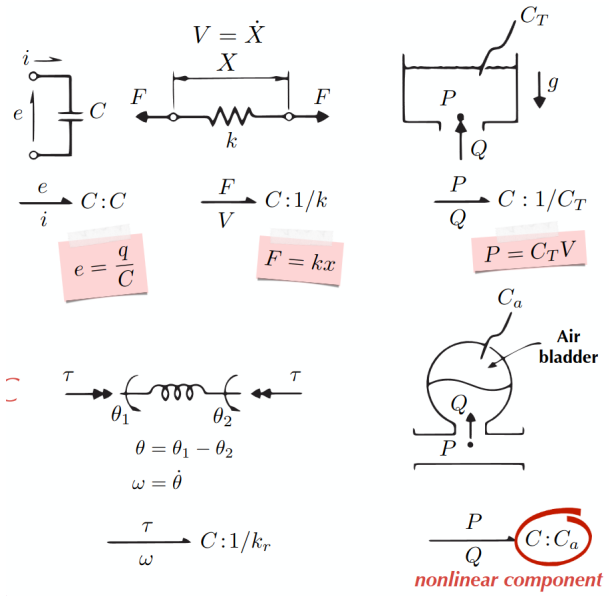


Figure 2.2: Different types of Capacitive 1-port elements.

Physical realisations: electrical capacitor $e = q/C$; spring $F = kx$ (so $C = 1/k$); hydraulic accumulator $P = V/C_a$; torsional spring $\tau = k_r\theta$.

Key Insight: Preferred causality of C

A capacitor can either be in **integral causality** and in **derivative causality**.

$$e = \mathcal{C}^{-1}\left(\int f\, dt\right), \qquad f = \frac{d}{dt}\mathcal{C}(e).$$

The natural causality for a capacitor is **flow-in, effort-out**: flow is integrated to give displacement, from which effort is computed. This is called *integral causality*. If the bond graph forces effort-in on C (derivative causality), the element does not contribute an independent state variable, which has implications for the system's order.

Intuitively, from a simulation perspective, you impose the flow (e.g., current or velocity) which the element integrates into displacement (charge or position), and this displacement yields the effort (voltage or force) as an output.

2.3 The 1-port inertia *I*: kinetic energy storage

An inertia (inductor) stores **kinetic energy** via a static relationship between flow and momentum:

$$f = \mathcal{I}^{-1}(p), \qquad p = \int e\, dt. \tag{2.3}$$

In the linear case, $f = p/I$. Here, the linear inertia I is defined as the inverse of the slope of the f versus p curve.

Stored energy:

$$E(p) = \int_{p_0}^{p} f(\bar{p})\, d\bar{p} \quad \longrightarrow \quad E = \frac{p^2}{2I} \text{ (linear).} \tag{2.4}$$

Similarly to the capacitor, the E_0 reference point is typically set to zero.

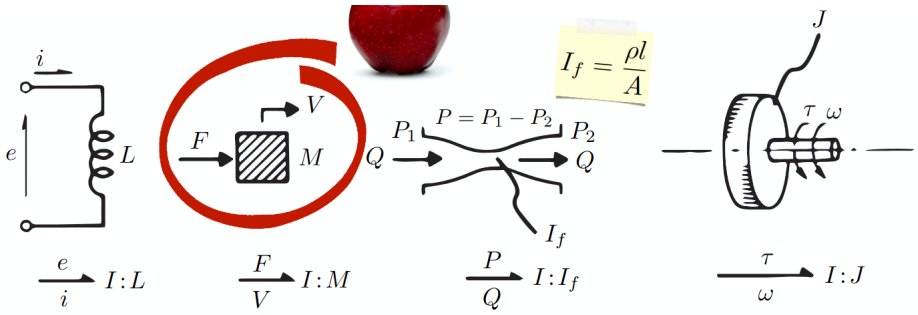


Figure 2.3: Different types of Inertial 1-port elements.

Physical realisations: point mass $F = ma \Rightarrow V = P/m$ (so $I = m$); rotating body $\tau = J\dot{\omega} \Rightarrow \omega = p_\tau/J$; fluid inertance $\Delta P = I_f \dot{Q}$.

Key Insight: Preferred causality of I

An inertia integrates force to give momentum, which means that it accumulates the effect of effort over time. The formula is $v = p/I$, so the flow (velocity) is derived from the momentum, which is $p = \int e\, dt$. If it was otherwise, and you imposed the flow (velocity), it would mean that imposing an abrupt change in velocity would require an infinite force, which is not physically realizable.

$$f = \mathcal{I}^{-1}\left(\int e\, dt\right), \qquad e = \frac{d}{dt}\mathcal{I}(f). \tag{2.5}$$

The natural causality for an inertia is **effort-in, flow-out**: effort is integrated to give momentum, from which flow is computed. Again, this is integral causality. Newton's second law is a direct statement of this: force (effort) integrates to momentum, from which velocity (flow) is recovered.
As with the capacitor, imposing flow on an inertia leads to derivative causality because the imposed flow must be differentiated to find the required effort, thus reducing the number of independent state variables (you lose position, hence memory).

The tetrahedron of state variables summarises the four types and their relationships: $\dot{p} = e$, $\dot{q} = f$, with C linking e–q and I linking f–p.

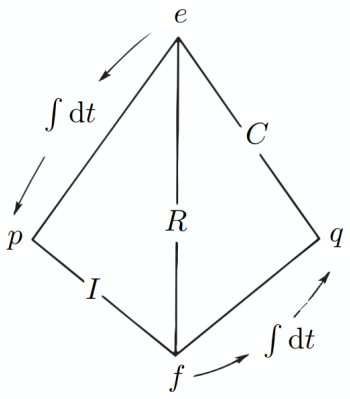


Figure 2.4: Tetrahedron of state variables.

2.4 Source elements *Se* and *Sf*

An **effort source** Se maintains a specified effort $e(t)$ at its port, regardless of the flow (infinite power if needed, the idealisation of a stiff voltage/force/pressure supply). A **flow source** Sf dually maintains a specified flow.

Their causality is fixed: Se always imposes effort (causal stroke away from Se); Sf always imposes flow.

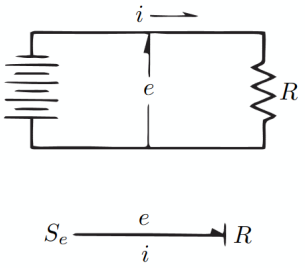


Figure 2.5: Example of source elemets. Voltage generator and resistor in series, with Se imposing voltage and R determining current.

In reality, components always find a compromise. For instance, when you set the voltage of a real battery, the act of drawing current inherently lowers the voltage itself. Because ideal sources do not model this compromise in one go, a real-world component like a battery must be modeled using an ideal voltage source (Se) in series with a resistance (R) to accurately simulate the voltage drop under load.

When the source value is itself a function of a signal from another part of the graph, it becomes a **modulated source** MSe or MSf. This is how controlled actuators, variable-flow pumps, and vibration shakers are represented.

Figure 2.6: Modulated source of effort (MSe) and modulated source of flow (MSf).

2.5 The transformer *TF*: effort–effort, flow–flow exchange

A transformer is a **power-conserving 2-port**: $e_1 f_1 = e_2 f_2$ at all times. Its constitutive relations scale effort and flow by a constant (or time-varying) modulus m:

$$\begin{cases} e_1 = m\, e_2 \\ m\, f_1 = f_2 \end{cases} \tag{2.6}$$

Effort is scaled up by m, flow is scaled down by m (or vice versa).

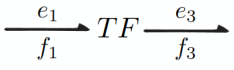


Figure 2.7: Bond graph symbol for the transformer element.

Physical examples: electrical transformer ($m = N_1/N_2$, turns ratio); lever ($m = b/a$, arm ratio); hydraulic piston ($F = AP$, $AV = Q$, so $m = A$); gear pair ($m = r_1/r_2$); rack and pinion.

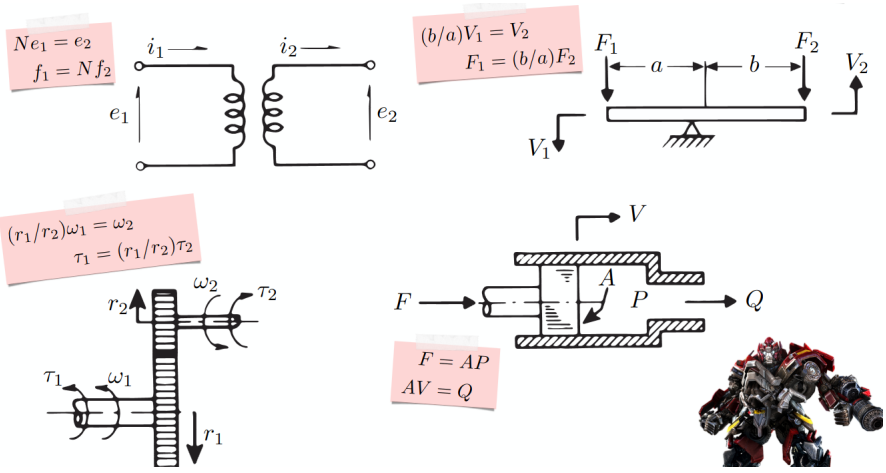


Figure 2.8: Physical examples of transformer elements.

Key Insight: TF causality

Two causal forms are valid for TF: either effort-in on port 1 (and effort-out on port 2), or effort-in on port 2. The causal stroke must appear on exactly one of the two ports.

If the geometry changes (e.g. a linkage), m becomes time-dependent, the element becomes a **modulated transformer** MTF.

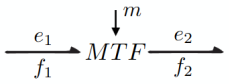


Figure 2.9: Modulated transformer (MTF).

In robotics, the Jacobian matrix plays the role of the MTF modulus. The Jacobian expresses the relationship between end-effector and joint velocities ($v = J(\theta)\dot{\theta}$). Assuming power conservation, it naturally also relates the end-effector applied force to the joint torques ($\tau = J^T(\theta)F$).

2.6 The gyrator *GY*: effort–flow exchange

A gyrator is also power-conserving, but it exchanges the roles of effort and flow:

$$e_1 = r\, f_2, \qquad r\, f_1 = e_2. \tag{2.7}$$

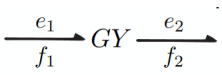


Figure 2.10: Bond graph symbol for the gyrator element.

This is fundamentally different from TF: while a transformer maps effort to effort and flow to flow through a constant, the gyrator maps a flow to an effort and vice versa through a constant.

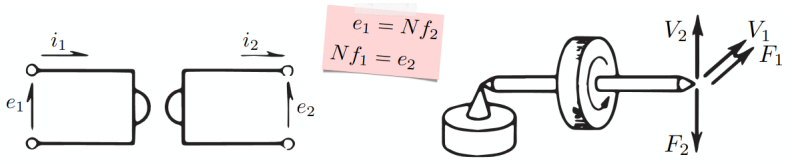


Figure 2.11: Example of gyrator element.

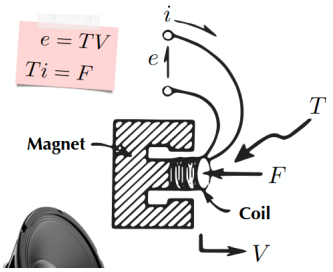


Figure 2.12: More example of gyrator element.

Physical examples: DC motor (back-EMF $e_a = T\omega$, torque $Ti_a = \tau$, so $r = T$); gyroscope (a force in one direction induces an angular velocity in the perpendicular direction).

Key Insight: Gyrator is more fundamental than transformer

The gyrator can sometimes seem like a mysterious element, but it is actually more fundamental than a transformer. Two cascaded gyrators produce a transformer.

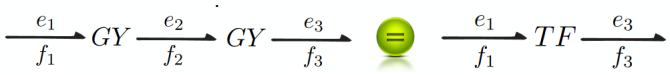


Figure 2.13: Gyrator as a fundamental element.

More profoundly, a gyrator essentially interchanges the roles of effort and flow, so it maps C-elements into I-elements: a capacitor seen through a gyrator behaves as an inertia. This means one does not fundamentally need two types of energy-storage elements, the gyrator generates one from the other. Historically, ignoring gyrators and forcing gyroscopic systems into transformer notation led to incorrect models.

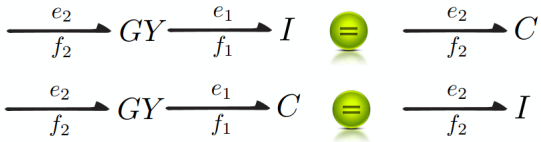


Figure 2.14: Gyrator mapping C to I and vice versa.

When the gyration ratio r depends on a signal (e.g. motor torque constant varying with field current), the element becomes a modulated gyrator **MGY**.

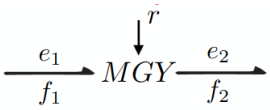


Figure 2.15: Modulated gyrator (MGY).

2.7 Junction elements: 0-junction and 1-junction

Junctions are **power-conserving** multi-port elements that distribute power among connected elements. They enforce algebraic constraints on efforts and flows. Fundamentally, zero and one junctions are simply mathematical **conditions** applied to the system (such as force balance, flow conservation, or parallel branches sharing the same voltage).

2.7.1 The 0-junction: common effort

At a 0-junction all connected bonds share the **same effort**, and the flows sum algebraically to zero (with signs set by the half-arrow directions):

$$e_1 = e_2 = e_3 = \cdots, \qquad \sum_k \pm f_k = 0. \tag{2.8}$$

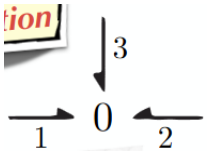


Figure 2.16: Bond graph symbol for the 0-junction element.

Electrical analogue: a **node** (Kirchhoff's current law, same voltage). Mechanical analogue: a point where a single force acts on several elements with velocities that sum to zero (geometric compatibility, spring and damper in parallel). Here, we are imposing a relation between relative velocities. Hydraulic analogue: a junction of pipes (conservation of volume flow).

2.7.2 The 1-junction: common flow

At a 1-junction all connected bonds share the **same flow**, and the efforts sum algebraically to zero:

$$f_1 = f_2 = f_3 = \cdots, \qquad \sum_k \pm e_k = 0. \tag{2.9}$$

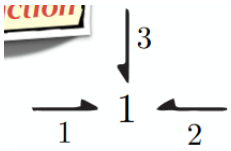


Figure 2.17: Bond graph symbol for the 1-junction element.

Electrical analogue: a **loop** (Kirchhoff's voltage law, same current). Mechanical analogue: Newton's law for a mass, all forces acting on an inertia sharing a single velocity must balance. It encodes the action-reaction principle and effectively acts as a rigid connection among the mechanical elements. Hydraulic analogue: pressure drops around a single-flow circuit sum to zero.

Watch out: Sign conventions at junctions

The algebraic sum at a junction depends on the *direction of the half-arrows*. One convention: at a 0-junction, the half-arrow of the "driving" bond points *toward* the junction, and all others point away, then the sum of all f_k equals the driving flow

2.7.3 Bond graph identities and simplification

A 0-junction with only two bonds connecting to it reduces to direct connection (same effort, same flow). A 1-junction with only two bonds similarly reduces. These **bond graph identities** allow systematic simplification of an initial, verbose bond graph into a compact one. They are used heavily when eliminating zero-power bonds (e.g. after connecting to ground in a mechanical model) and when simplifying "reducible loops" that appear when spring and damper share the same relative velocity.

For **generalizing n-port components** there exist specific rules.

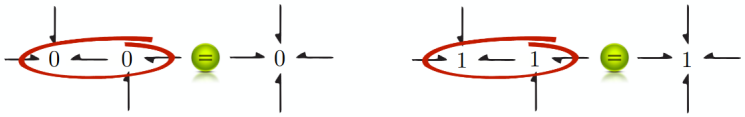


Figure 2.18: Rules for simplifying n-port elements

2.8 Causality summary for basic elements

For effort and flow sources, causality is trivial and fixed. For basic 2-ports (TF, GY), there are two possible causal assignments; as soon as one variable is assigned as an input, the other is constrained by the constitutive relation. Similar logic naturally extends to 3-ports (junctions).

Element	Preferred causality	Alternative	Notes
Se	effort out (fixed)	-	always imposes e
Sf	flow out (fixed)	-	always imposes f
R	either	either	indifferent (usually)
C	flow in (integral)	effort in (derivative)	derivative $\Rightarrow$ dependent state
I	effort in (integral)	flow in (derivative)	derivative $\Rightarrow$ dependent state
TF	one of two forms	-	must be one in, one out
GY	both in *or* both out	-	two valid forms
0-jct	one effort in, rest out	-	uniquely determined
1-jct	one flow in, rest out	-	uniquely determined

Causality and bond graphs are very important. Indeed it's possible to predict many important features of the system even before the constituitve laws for all the elements have been specified.

It's possible to predict the number of state variables and it's useful for writing state equations.

Once the causal stroke is added, it's possible to transform the bond graph into a block diagram. They are more complex since a single bond represent two signal flows.

Chapter 3

System Models in Different Domains

3.1 Electrical circuits and networks

Any electrical circuit can be modelled by a bond graph containing elements of the set $\{0, 1, R, C, I, Se, Sf\}$. For electrical circuits the mapping to bond graph variables is direct: voltage is effort, current is flow. The 0-junction encodes Kirchhoff's current law at a node (same voltage, currents sum to zero); the 1-junction encodes Kirchhoff's voltage law around a loop (same current, voltage drops sum to zero).

TF and GY are not included since they are used to represent electrical networks.

- 0 Junction: Same voltage
- 1 Junction: Same current

Construction procedure. Constructing the overall bond graph model is much easier if a procedure can be followed that ensures success regardless of the complexity.

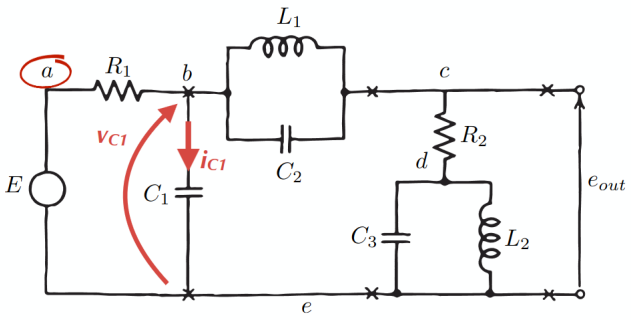


Figure 3.1: Example of electric circuit

1. **Assign a power convention**: for each circuit element choose positive current direction and positive voltage drop. Talking about power in BG, we typically ensure that, when positive, power is flowing *towards* the passive 1-port element. Note that active elements are elements providing power (sources, but also energy-storing elements freeing their energy). In some situations, sources may even absorb power.
2. **One 0-junction per node voltage**: place a 0-junction for every distinct node voltage (including what will be the ground). Every bond touching a 0-junction carries the same effort (node voltage).
3. **One 1-junction per element**: for each circuit element (R, L, C, source), insert a 1-junction to represent the voltage difference across it. By properly directing the half-arrows on 1-junctions, the voltage drop can be established across each element. Connect the 1-junction between the two 0-junctions at its terminals. Attach the bond graph element (R, I, C, Se, Sf) to the 1-junction.
4. **Ground = zero effort source**: choose a ground node and connect the ground 0-junction to a Sf : 0 (zero-flow source) or Se : 0 (zero-voltage), which in practice means deleting all bonds with zero power at that junction. Note that different bond graphs will result for a given circuit if we choose different ground nodes.7

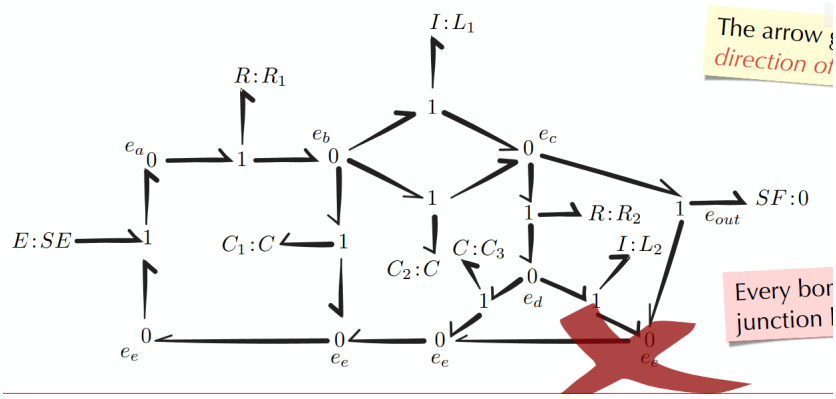


Figure 3.2: Example of bond graph construction for the circuit in Fig. 3.1.

5. **Simplify**: apply bond graph identities to remove redundant junctions and bonds.

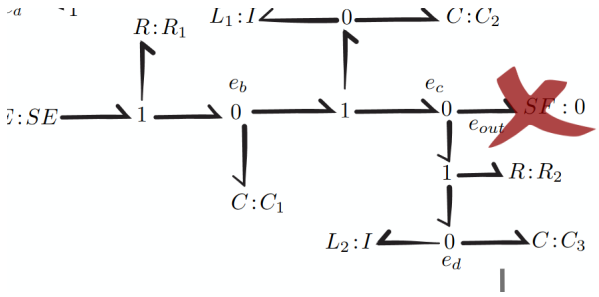


Figure 3.3: Example of bond graph simplification.

An electrical network is an extension of electrical circuits to include idealised transformers TF and gyrators GYG.

Electrical Transformers

It's used used to to step voltages up or down.

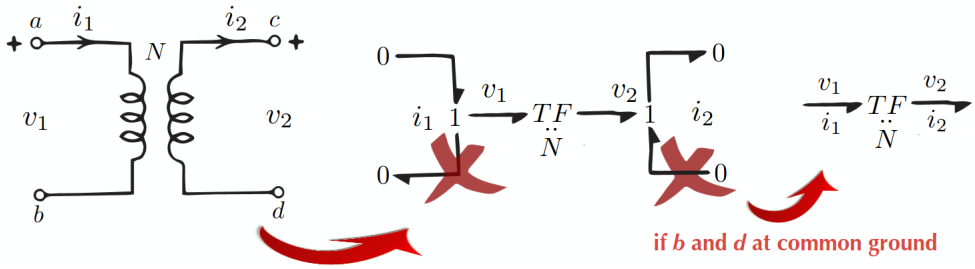


Figure 3.4: Ideal transformer with turns ratio N.

Electrical Gyrators

Electrical gyrators are exhibited in Hall effect transducers, where voltage across a semiconductor material is related to a current through the material perpendicular to the voltage drop direction. The same construction method extends to **electrical networks** containing transformers (TF) and gyrators (GY). The transformer element inserts between two 1-junctions; the turns ratio N becomes the TF modulus. The basic rules for bond graph construction remain unchanged.

Key Insight: Ladder and T/pi structures

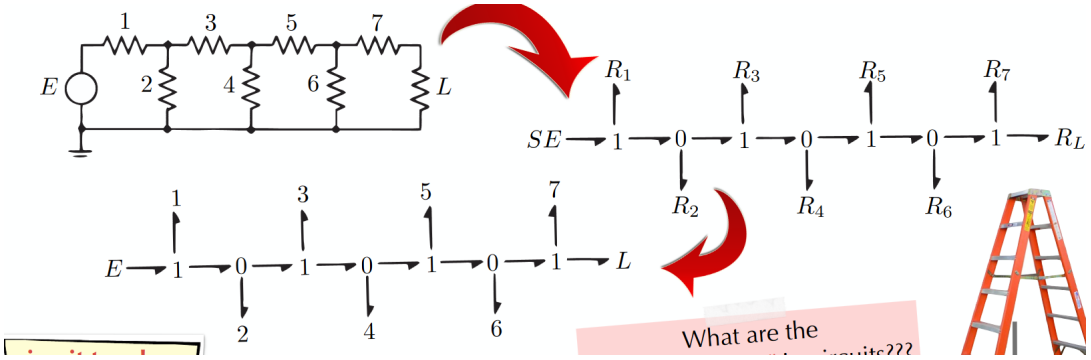


Figure 3.5: Resistive ladder with resistors

Transmission lines and long hydraulic lines have lumped approximations as ladder networks cascaded T or pi sections. In bond graph terms, the T-section corresponds to alternating 1-junctions (series inductors) and 0-junctions (shunt capacitors). The topological dual (exchanging 0 and 1 junctions) gives the pi-section. Notice that the difference between these two schemes is the role of current and voltage is interchanged. This technique holds in general when we want to find the topological dual of even complex circuits: exchange 0 and 1 junctions in the BG! The pi and tee structures become truncations of the ladder structure that allow for reasonable inclusion of some of the **transmission line dynamics without having the overall system become too large**. Understanding this duality is a powerful shortcut when building bond graphs by inspection.

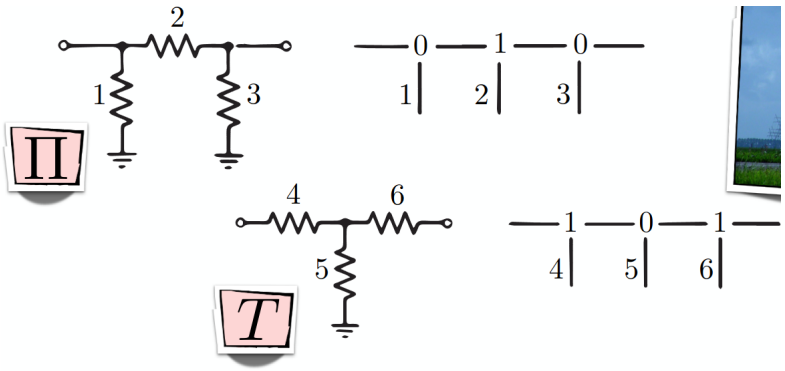


Figure 3.6: pi and tee sections as truncations of the ladder structure.

3.2 Mechanical translation

Mechanical systems are those composed of components such as masses, springs, dampers, levers, flywheels, gears, shafts, and so forth. When dynamic systems are put together from these components, we must interconnect rotating and translating inertial elements with axial and rotational springs and dampers, and we must appropriately account for the kinematics of the systems.

For mechanical translation, velocity is flow and force is effort. The analogy with electrical circuits is *dual*: in electrical circuits, nodes share voltage (0-junction); in mechanics, rigid connections share velocity (1-junction).

- 0 Junction: Relative velocity (force-generating elements)
- 1 Junction: Same velocity

Construction procedure for Mechanical Translation.

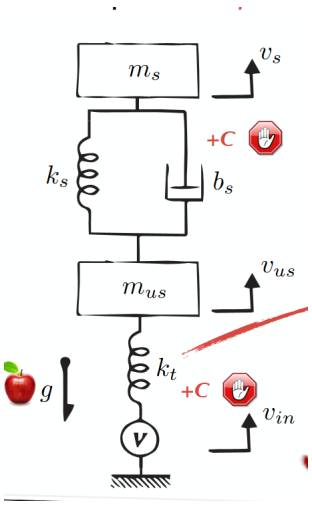


Figure 3.7: Example of mechanical system with translation.

1. **Indicate velocity directions**: on a free-body-type sketch, use arrows and symbols to indicate the positive direction of absolute velocity components. Fixing conventional directions in this scheme is the first crucial step! State whether force-generating elements are positive in tension or compression using symbols such as +T or +C.

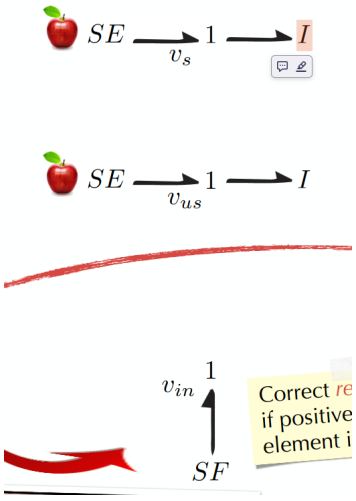


Figure 3.8: Example of free-body sketch with velocity directions and force conventions.

2. **One 1-junction per distinct velocity**: each inertia is attached to its absolute velocity 1-junction.

3. **0-junctions for relative velocities**: spring and damper forces react to the rate of extension, meaning they depend on the *relative* velocity across them. A 0-junction connecting two 1-junctions produces the relative velocity difference needed by the compliant/resistive element. The sign at the 0-junction must be chosen so that the relative velocity is positive when the element is in the assumed tension/compression state. When forces are positive in tension, power will flow toward the elements.

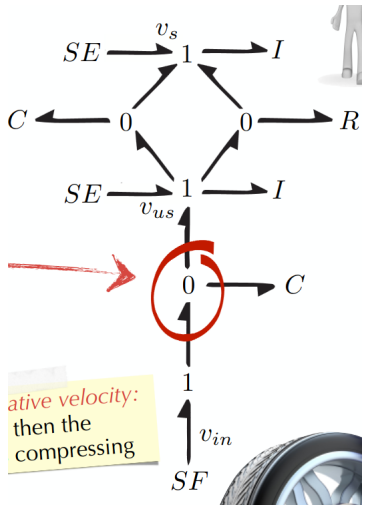


Figure 3.9: Example of bond graph construction for the mechanical system in Fig. 3.7.

4. **Reference (ground)**: the inertial frame is fixed in zero velocity, represented by a Sf : 0, which is deleted after establishing relative velocities.

5. **Attach elements**: inertias on 1-junctions; springs (C : $1/k$) and dampers (R : b) on their relative-velocity 0-junctions; gravity and other absolute forces as Se sources on velocity 1-junctions.

The *beauty* of power conservation in junctions is that we need only to constrain the velocities, and the forces will be automatically balanced without further effort. Newton's second law and the action-reaction principle are built in without explicitly writing them. We can perform a double check to see that the force balance is realized by the 1-junctions.

Reducible loop. When a spring and damper are in parallel (same relative velocity), the 0-junction connecting both their velocity 1-junctions can be simplified using bond graph identities into a single 0-junction carrying both the C and R elements. This is the **reducible loop** , a very common pattern that emphasizes that several elements share exactly the same relative velocity.

Levers, pulleys and kinematic linkages. With minor efforts, it is possible to deal with translational systems containing levers, pulleys, and other simple motion-force transforming devices. These introduce position-dependent force/velocity transformations, represented by modulated transformers (MTF) or fixed TF when the geometry is constant. The MTF modulus is the relevant geometric ratio (e.g. the Jacobian in robotics).

3.3 Fixed-axis rotation

Fixed-axis rotation occurs in the mechanics of rotating machines, such as electric motors and internal combustion engines, gear boxes, transmissions, transfer cases, differentials, and drive shafts.

Only minor modifications to the procedure for mechanical translation allow handling systems with fixed-axis rotation. It is entirely analogous to translation with the variable substitution $V \to \omega$, $F \to \tau$, $m \to J$, $k \to k_r$, $b \to b_r$.

The construction procedure is identical: 1-junctions for distinct absolute angular velocities, 0-junctions for relative angular velocity across compliant or resistive rotational elements, TF for gear pairs (modulus = gear ratio r_1/r_2), and eliminating the bonds with zero angular velocity.

In the fixed-axis rotation framework, it's not really relevant to talk about +T or +C; what is relevant is to maintain coherence with the relative velocities across all the schemes (e.g., choosing one velocity as the upstream master velocity and the other as the downstream slave velocity).

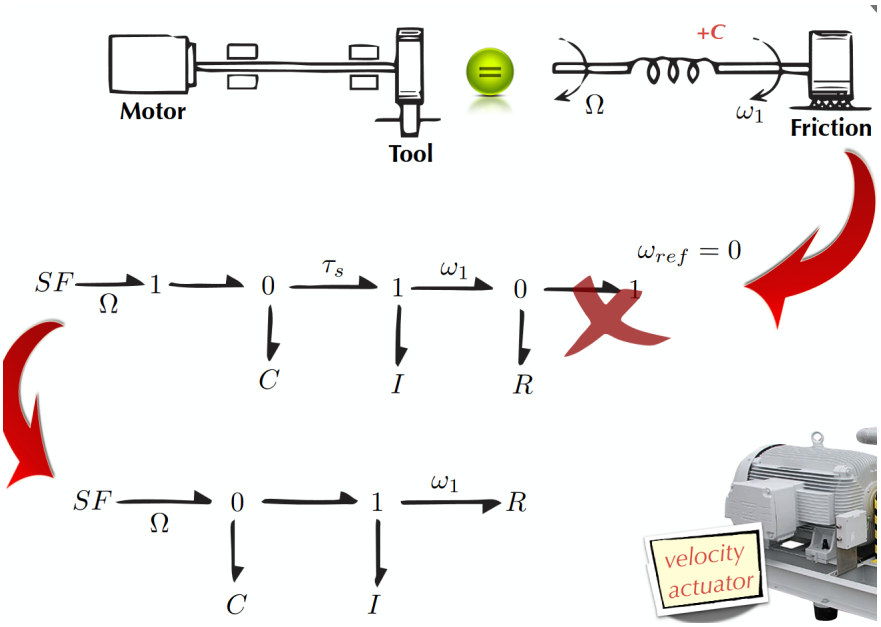


Figure 3.10: Example of bond graph construction for a fixed-axis rotational system.

3.4 Plane motion of rigid bodies

Mechanical bodies with finite mass both translate and rotate in real applications; their dynamics are governed by a simultaneous combination of translation and rotation. 3-D motion is not trivial, so plane motion is much simpler. Plane motion results when the inertial bodies of a physical system are constrained to translate in two dimensions and to rotate only about an axis perpendicular to the plane of motion. The motion is always w.r.t. an inertial frame (with eventually coordinate transformations in the middle).

When a body both translates and rotates in a plane, three absolute velocity components are needed: v_x, v_y (translational velocities of the centre of mass) and ω (angular velocity). Newton's law in the plane gives three 1-junctions, one per component, each with its own inertia element:

$$m\dot{v}_x = F_x, \quad m\dot{v}_y = F_y, \quad J\dot{\omega} = \tau, \tag{3.1}$$

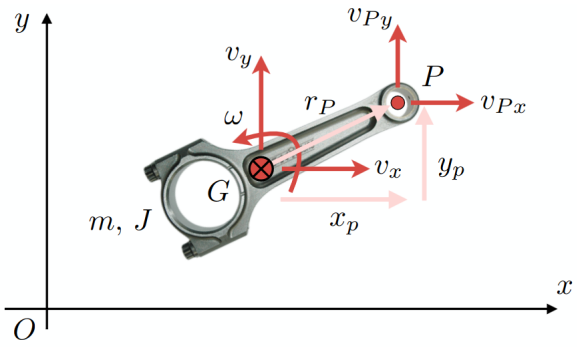


Figure 3.11: Mechanical body

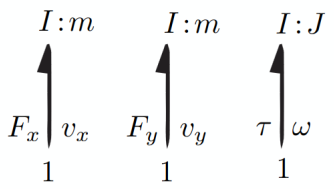


Figure 3.12: Bond graph representation of the plane motion of a rigid body.

so the kinetic energy is $T = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}J\omega^2$ with three I-elements. This dynamic representation does not involve directly any position information, which is coherent with the fact that this object stores kinetic energy, not potential energy.

Devices are attached at points P offset from the centre of mass by $(r_P \cos\theta, r_P \sin\theta)$ in the inertial frame. The velocity of point P is:

$$v_{Px} = v_x - (r_P \sin\theta)\omega, \qquad v_{Py} = v_y + (r_P \cos\theta)\omega. \tag{3.2}$$

These are linear combinations of the three body-velocity components, implemented with MTF elements (moduli $r_P \sin\theta$ and $r_P \cos\theta$, time-varying through θ). By power conservation ($\Delta F_x v_x + F_y v_y + \tau\omega = F_{Px}v_{Px} + F_{Py}v_{Py} + \tau_P\omega_P$), the corresponding forces and torques at the body's ports are automatically balanced without writing free-body equations explicitly. The result is an "object" with a force/velocity interface toward the environment (what is outside the system). All the port variables are expressed with respect to the inertial frame.

Key Insight: Parenthesis: the Body Frame!

To characterize the energy of a rigid body, it is necessary to know the velocity of the center of mass and the angular velocity. The velocities discussed so far are expressed in the inertial reference, but for multi-body dynamics, it is sometimes convenient to introduce body-fixed coordinates. This coordinate frame remains attached to the center of mass and moves with the body. The advantage is that the inertial tensor is constant, unlike in the inertial frame where it varies with the configuration of the mechanism. However, expressing absolute accelerations in the body frame requires additional terms (e.g., $a_x = \dot{v}_X - \omega v_Y$). While body frames are generally preferred to have a constant inertial tensor, in the planar case, inertial frames can be used without any problems.
We can see the body dynamics as a "black box" with three ports (two for translation and one for rotation) that can be connected to the environment. The internal structure of the body is not relevant for the external behavior, which is fully characterized by the kinetic energy and the force/velocity relations at the ports. This is a powerful abstraction that allows us to model complex systems without needing to know all the internal details of each component.

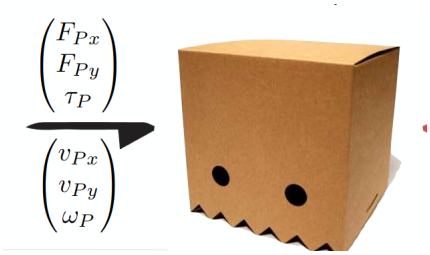


Figure 3.13: Black-box representation of a rigid body with three ports.

Connecting rigid bodies: joints. Going back to the formulation in the inertial frame, to interconnect rigid bodies and obtain complex mechanisms, it is necessary to define joints that constrain motion in specific directions. A **rotational joint** constrains the two connected bodies to have the same position (zero relative linear velocity) while leaving angular velocity free. Any difference in angular velocity ($\omega_r = \omega_B - \omega_A$) means a torque exists.

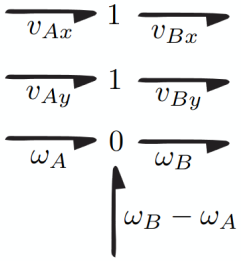


Figure 3.14: Rotational joint connecting two bodies.

A **translational joint** constrains relative motion to one direction. Since velocities are in the inertial frame, this direction is not fixed and is related to an angle ϕ. The relative velocities must be transformed into the direction of the joint. Both are represented by setting appropriate velocity-difference bonds to zero and using MTF elements for the direction transformation in the inertial frame. This is the basis for the bond graph modelling of robot manipulators and mechanisms.

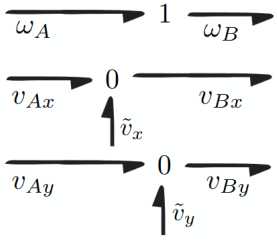


Figure 3.15: Translational joint connecting two bodies.

Figure 3.16: Rotation of the relative velocity. Instead of dividing by cos and sin, we use a trick with a MTF

3.5 Hydraulic and acoustic circuits

- One category is **hydrostatic systems**. Typical examples are pumps, motors, pipes, pistons, valves, filters, and accumulators that use nearly incompressible fluids. These systems have high pressures and low fluid velocities so that the **static pressure dominates the dynamic pressure**. Pressure P is effort, volume flow rate Q is flow.
- A second category involves compressible gases where the pressure deviations are small enough that the **acoustic approximation** applies. In this regime, the gas behaves linearly, sound waves can be described using simplified wave equations, and we can avoid considering thermodynamic effects. Pneumatic systems with large pressure variations need a more complicated modeling treatment.

Acoustic approximation refers to a simplified model used for compressible gases, like air, when pressure variations are very small compared to the ambient pressure. In this regime, the gas behaves linearly, and sound waves can be described using simplified wave equations.

$$\frac{\partial^2 p}{\partial t^2} = c^2 \nabla^2 p$$

Where p is the acoustic pressure, and c is the speed of sound in the medium.

3.5.1 Fluid Resistance

A 1-port fluid resistor relates a **pressure drop** $P_3 = P_1 - P_2$ to a volume flow rate. **The value of the resistance can be measured experimentally.**

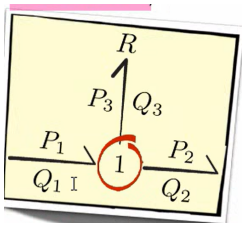


Figure 3.17: Fluid resistor connecting two fluid nodes.

$$Q_1 = Q_2 = Q_3$$
$$P_3 = P_1 - P_2 = \Phi_R(Q_3) = R_3 Q_3 \quad \text{(linear)}$$

The Reynold number determines if the flow is laminar or turbulent and thus the relationship between pressure drop and flow rate

$$Re = \frac{4\rho Q}{\pi d \mu}$$

- If the Reynolds number is low ($Re < 2000$), $P = RQ$, where R is the linear resistance coefficient. This is the regime of laminar flow, where the fluid flows in parallel layers without turbulence.

$$R = 128\mu l/(\pi d^4)$$

- If the Reynolds number is high ($Re > 5000$), the flow is turbulent, and the pressure drop is given by $P = a_t Q|Q|^{3/4}$, where a_t is a turbulent resistance coefficient.
- For $2000 < Re < 5000$, the flow is in a transitional regime, and the relationship between pressure drop and flow rate is more complex, often requiring empirical data or more sophisticated models to accurately describe the behavior.

Orifices

Short obstructions like orifices and valves are important cases where a pressure drop occurs over short lengths. For an orifice with a fixed area, the pressure drop is proportional to the square of the volume flow rate

$$\Delta P = \frac{\rho Q|Q|}{2C_d^2 A_0^2}$$

where the discharge coefficient C_d depends on the geometry and nonidealities.

Valves

For valves, the area $A(x)$ depends on a position coordinate. The position of a mechanical part influences the hydraulic resistance, acting as a transducer linking the mechanical and hydrostatic domains. They can be modeled by variable-resistance elements modulated by a displacement $\dot{x}$.

$$\Delta P = \frac{\rho Q|Q|}{2C_d^2(x) A_0^2(x)}$$

It's important to note that elecetrovalevs modeling is a challenging task.

3.5.2 Fluid Capacitance

A fluid capacitor imposes a relationship between a **pressure and the integral of the volume flow rate**. Essentially is the volume of fluid.

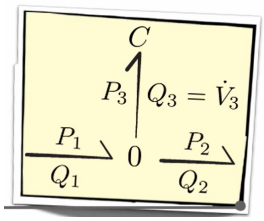


Figure 3.18: Fluid capacitor.

$$P_3 = \Phi_C^{-1}(V_3) = \frac{V_3}{C}$$
$$V_3 = \int Q_3 dt$$

For a straight-walled water tank in a gravity field,

$$P_3 = \rho g h = \rho g \frac{V_3}{A_3}$$

$$C_3 = \frac{A_3}{\rho g}$$

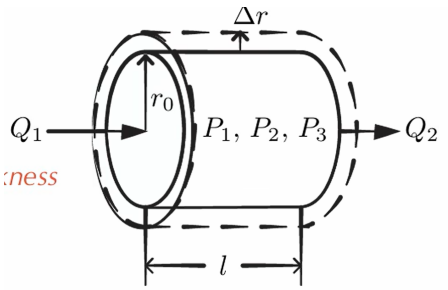


Figure 3.19: Fluid capacitance due to vessel elasticity.

Key Insight: Fluid capacitance as "pressure inertia"

The fluid capacitance can be thought as a "Pressure Inertia". Just as a mass resists changes in velocity, a fluid capacitor resists changes in pressure. The larger the capacitance, the more fluid volume is needed to produce a given pressure change, which is analogous to how a larger mass requires more force to achieve the same acceleration.

Fluid Compressibility

If and are not equal, represents the rate of compression of the fluid in the pipe. and V_3 represents the decrease in volume of the fluid in the segment.

$$dP = -B\frac{dV}{V_0}$$
$$\Delta P = \frac{B}{V_0}\Delta V_3 \Longrightarrow C = \frac{V_0}{B}$$

Elasticity of the vessel

In the other case, the fluid is incompressible, but the vessel containing it is elastic. The pressure change is related to the change in volume by the bulk modulus of the vessel material. The bulk modulus B is defined as the ratio of the infinitesimal pressure increase to the resulting relative decrease in volume:

The bulk modulus is expressed as:

$$B = -V_0\frac{dP}{dV}$$

$$\sigma = \frac{r_0 P}{t_w} \longrightarrow \text{hoop stress}$$
$$\epsilon = \frac{\sigma}{E} = \frac{r_0 P}{t_w E} = \frac{\Delta r}{r_0} \longrightarrow \text{circumferential strain}$$

So the change in volume is:

$$\Delta V = \Delta(\pi r^2 l) = \cdots = \frac{2r_0 V_0}{t_w E}P$$
$$P_3 = \frac{t_w E}{2r_0 V_0}V_3 \Longrightarrow C = \frac{2r_0 V_0}{t_w E}$$

Key Insight: Capacitance in parallel

For cases in which both effect (fluid compressibility and vessel elasticity) contribute, the total capacitance is the sum of the individual capacitances, as they are in parallel. This is because both effects contribute to the overall change in volume for a given pressure change, and their contributions add together to determine the total capacitance of the system.

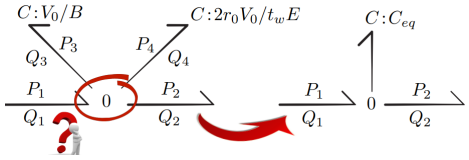


Figure 3.20: Parallel combination of fluid compressibility and vessel elasticity.

If both fluid compressibility and pipe elasticity contribute, an equivalent compliance can be constructed: $C_{eq} = V_0(\frac{1}{B} + \frac{2r_0}{t_w E})$.

When fluid compliance cannot be modeled with linear behavior, nonlinear models are essential. An example is a gas accumulator with a rubber bladder containing compressed gas. Using the isentropic pressure-volume law

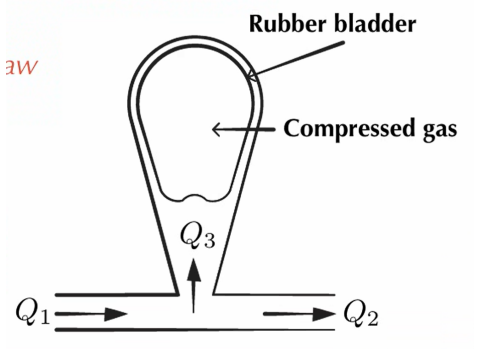


Figure 3.21: Nonlinear fluid capacitor with isentropic behavior.

$$PV^{\gamma} = const$$

The pressure is a nonlinear function of the volume: $P_3 = \frac{P_0 V_0^{\gamma}}{(V_0 - V)^{\gamma}}$.

3.5.3 Fluid Inertia

Fluid systems have inertia effects analogous to mechanical systems: fluid elements store kinetic energy. The inertia coefficient is related to the parameters of a pipe segment.

$$p_{P_3} = \int P_3 d\tau = \int (P_1 - P_2) d\tau \leftrightarrow I_3 Q_3 = p_{P_3}$$

From Newton's law, a linear inertia coefficient for a pipe of constant cross-section A and length l is $I = \rho \frac{l}{A}$. For a pipe with a non-constant section, $I = \int_0^l \frac{\rho}{A(s)} ds$.

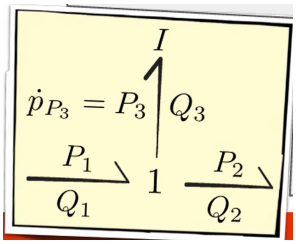


Figure 3.22: Fluid inertia in a pipe segment.

Key Insight: Explaining the fluid inertia

The fluid inertia is used to explain the phenomenon that happens when, for instance, a valve is suddenly closed. The fluid in the pipe has kinetic energy (a buildup of momentum), and when the valve is closed, the momentum cannot dissipate immediately, it has to keep flowing until it is dissipated.

Construction of Hydraulic Bond Graphs

For both, the construction procedure follows the same 0/1-junction rules as electrical circuits:

1. Indicate positive directions for fluid flows and pressures.
2. For each distinct absolute pressure introduce a 0-junction. Attach elements if they react to one of the absolute pressures. If necessary, include a 0-junction for atmospheric pressure (which can be eliminated if "gauge" pressures are used. Namely, it can be elimnated if all the compoents has a base, the atmospheric pressure).

3. Insert R-, C-, and I-elements between appropriate 0-junctions using 1-junctions that establish pressure differences acting on the passive elements.

4. Simplify the graph.

3.6 Transducers & multi-energy domain models

Transducers are devices that couple subsystems of distinct energy domains. A major reason for studying bond graphs is that they provide a uniform and precise way to represent systems when several forms of power and energy are involved.

Transformer transducers. An ideal transformer transducer, like a hydraulic ram (piston), converts mechanical power to hydraulic power.

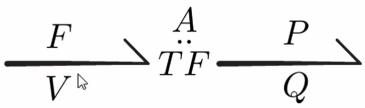


Figure 3.23: Ideal transformer transducer (piston) connecting mechanical and hydraulic domains.

- Transformer is for the fact that the relation between the two domains is a linear scaling of effort and flow variables.
- Transducer is for the fact that it converts power from one domain to another.

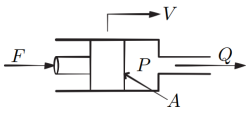


Figure 3.24: Ideal transformer transducer (piston) with idealized power flow.

$$F = AP$$
$$V = \frac{Q}{A}$$

This applies only to the ideal effects at the face of the piston. Real devices exhibit power losses and energy-storing mechanisms. The process consists of enriching the idealized model. On the mechanical side, we add a 1-junction with a resistor to include friction (which may be dry friction, $F_f = F_0 sgn(V)$, or linear, $F_f = bV$) and an I-element for the piston's mass. On the hydraulic side, a 0-junction and resistance represent leakage, and a C-element models fluid compliance (which may be nonlinear and depend on the nominal extension x_0). This complexity can lead to oscillatory phenomena that are damped by the resistance.

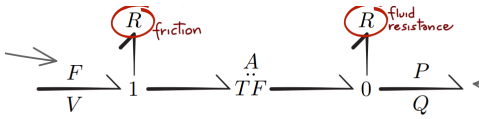


Figure 3.25: Enriched transformer transducer model with losses and energy storage.

Positive displacement hydraulic pumps and motors consist of pistons moved by a shaft. With many pistons, the angular speed is related to the flow rate by a nearly constant coefficient:

$$P_B - P_A = T\tau$$
$$TQ = \omega$$

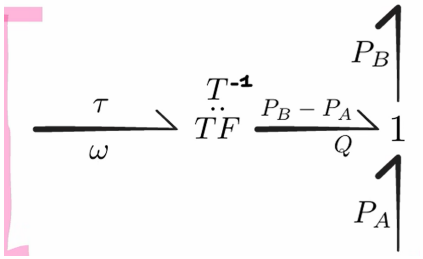


Figure 3.26: Positive displacement hydraulic pump/motor model.

Key Insight

Enriching the ideal transformer model The ideal transformer model captures the fundamental power conversion mechanism, but it does not account for real-world complexities such as friction, inertia, leakage, and compliance. By enriching the model with additional elements (resistors for losses, I-elements for inertia, C-elements for compliance), we can capture a more accurate representation of the system's behavior. The augmented model can exhibit also non linear phenomena.

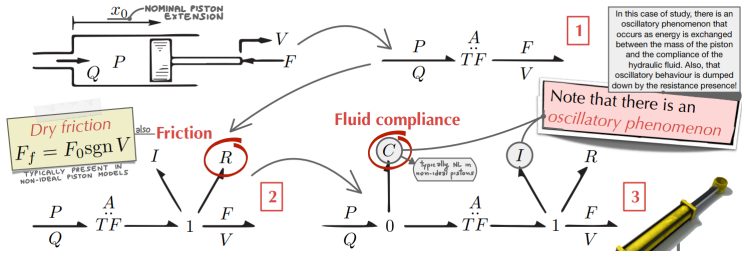


Figure 3.27: Enriched transformer model with additional phenomena.

Gyrator transducers. Electromechanical transducers, such as rotary/linear motors or voice coils, use gyrators. A current-carrying conductor moving in a magnetic field experiences an induced voltage (Faraday's law, $e = BlV$) and produces a force (Lorentz force, $Bli = F$). The cross-coupling of effort to flow and flow to effort makes this a gyrator ($e/V = GY = F/i$).

For a DC motor/generator with a commutator, the induced voltage is proportional to the angular speed, and the torque is proportional to the current:

$$e = T\omega$$
$$Ti = \tau$$

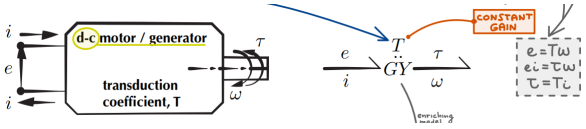


Figure 3.28: Gyrator transducer model for an electromechanical device.

A detailed model is developed by adding the winding resistance and inductance, and the rotor's inertia and friction to the basic GY element.

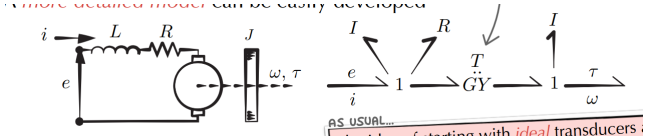


Figure 3.29: Enriched gyrator model with additional phenomena.

Modulated Transducers. If the gain is time-dependent, MTF and MGY can model more complex power transducers.

$$\begin{cases} \lambda = nBl_2l_1 \sin\theta \\ e = (nBl_2l_1 \cos\theta)\omega \\ (nBl_2l_1 \cos\theta)i = \tau \end{cases}$$

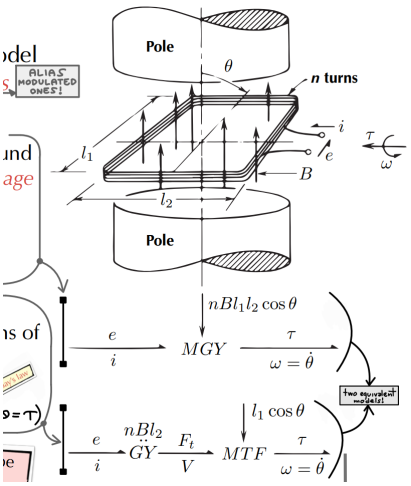


Figure 3.30: Modulated transformer and gyrator transducers.

Part II

Causality, Equations and Advanced Structures

Chapter 4

Causality and System Equations

In the world of state-space equations, the very same system can be described with different (but equivalent) sets of differential equations. The whole order of the system equations will always be the same (equal to the order of the model). For instance, a system can be represented by a single nth-order differential equation in terms of one unknown variable, or n first-order coupled differential equations in terms of n unknown variables (system theory approach), or various combinations of unknowns and equations of appropriate orders obtained with the Lagrangian approach.

One reason for constructing bond graphs is that it is straightforward to derive system equations and to develop numerical simulations. We can also go from a bond graph directly to a simulation without first obtaining the equations of motion manually. The real virtue of bond graph modelling becomes apparent when one considers that all system models use the same symbols. One formulation and solution procedure is needed!

Assigning causality to a bond graph is the step that transforms a structural description into a computational one. It decides which variables are inputs and which are outputs at every port, propagates these choices through the graph, and ultimately determines the mathematical character of the resulting model. The systematic procedure for doing this, the Sequential Causal Assignment Procedure (SCAP), is the workhorse algorithm for deriving state-space equations from bond graphs.

4.1 The augmented bond graph

Before deriving equations, the bond graph must be **augmented**:

1. The bond graph has been designed, and an appropriate power convention has been assigned.
2. All bonds are numbered consecutively. This creates the *computational bond graph*.
3. Each bond is assigned a causal orientation (causal stroke) for the e and f variables.

The causal strokes record, for every bond, whether effort flows left-to-right or right-to-left — i.e. which element is computing each variable. The power half-arrow and the causal stroke are completely independent pieces of information. The half-arrow records the assumed positive power direction; the causal stroke records the computational input–output assignment. Do not confuse them. Power sign convention and causality assignment are totally unrelated.

4.2 Causality constraints by element type

In a causal sense, there are two types of bond graph elements: sources (Se and Sf) and junctions (0, 1, TF, and GY), that must meet certain causal conditions or their basic definitions are no longer valid. If a system cannot meet these causal conditions, the basic physical model on which the bond graph is based must be restudied; an impossible situation that is not capable of sensible mathematical resolution has been created.

Every element imposes constraints on the causalities of its bonds:

- **Se**: must impose effort — the causal stroke is always *away* from the Se.
- **Sf**: must impose flow — the causal stroke is always *toward* the Sf.
- **C (integral causality)**: flow is input, effort is output — stroke away from C. A capacitor in integral causality is used as a charge storage and for this it fills having its own voltage. q is the state variable.

- **C (derivative causality)**: effort is input, flow is output — stroke toward C. This is the *non-preferred* form. A capacitor in derivative causality is used as a current generator to keep up lacks of voltage by discharging $I = C\dot{V}$. The state variable is q, but it is algebraically related to the rest of the system and does not contribute an independent state variable.
- **I (integral causality)**: effort is input, flow is output — stroke toward I. An inductor in integral causality is used as a momentum storage and for this it fills having its own current (speed). p is the state variable.
- **I (derivative causality)**: flow is input, effort is output — stroke away from I. Non-preferred. An inductor in derivative causality is used as a voltage generator to keep up lacks of current by discharging $V = L\dot{I}$. The state variable is p, but it is algebraically related to the rest of the system and does not contribute an independent state variable.
- **R**: generally indifferent (either causality), unless the constitutive law is not invertible (e.g. Coulomb friction, which is not a bijection in e–f).
- **TF**: one port effort-in, the other effort-out (one stroke in, one out).
- **GY**: both ports effort-in, or both effort-out (both strokes on the same side).
- **0-junction**: exactly one bond has its effort directed *into* the junction (one stroke pointing away from the junction); all others have effort directed out (stroke toward junction).
- **1-junction**: exactly one bond has its flow directed *into* the junction (one stroke toward junction); all others have flow directed out (stroke away from junction).

These constraints mean that once any one element in a connected sub-graph gets its causality assigned, that assignment propagates through junctions and 2-port elements like a chain reaction. The constraint elements represent the physical structure in the system, and assigning causality to them means that they are used correctly in a particular "input-output fashion".

4.3 The Sequential Causal Assignment Procedure (SCAP)

Step 1. **Assign source causalities.** For each Se and Sf, assign the only admissible causal form. Propagate the implications through all constrained elements (0, 1, TF, GY) which may be connected to it.

Step 2. Repeat step 1 until all sources have been assigned the required causality.

Step 3. **Assign storage elements (preferred causality).** For each C and I-element not yet causally assigned, assign *integral causality* (preferred). Propagate through constraint elements.

Step 4. Repeat step 3 until all storage elements have been assigned a causality.

Step 5. **Assign R-elements (if still needed).** If bonds remain unassigned after steps 1–4, choose any unassigned R-element and assign an arbitrary causality to it, and extend the causal implications through the graph. Each such arbitrary assignment signals an **algebraic loop**.

Step 6. Repeat step 5 until all R-elements have been used.

Step 7. **Assign remaining bonds arbitrarily.** Any remaining unassigned bond (joined to two constraint elements) can be assigned arbitrarily. Extend the causal implications. Each produces another algebraic loop.

Step 8. Repeat step 7 until all remaining bonds have been assigned.

The procedure is straightforward and orderly, and it can be implemented in a simulation environment. Note that in assigning causality, the results do not depend on the order of bonds chosen (except in special circumstances).

Key Insight: What the SCAP tells you

Several situations can arise when applying the SCAP, revealing the type of mathematical model:

- **All storage elements in integral causality, and the graph is complete after step 4**: the simplest and most common case. The model yields explicit ODEs directly.

- **Causality is completed using steps 5-8**: one or more *algebraic loops* exist. The model involves coupled algebraic–differential equations; some effort or flow variable is algebraically related to itself and must be solved simultaneously.
- **Some storage elements forced into derivative causality (during step 3)**: the model contains *implicit* differential equations (DAEs). The number of derivative-causality storage elements equals the *reduction in system order* relative to the total element count.

4.4 Reading off state-space equations: Basic formulation & Reduction

From a fully augmented bond graph model, the equations for the system can be developed in a very orderly fashion, either by hand or thanks to a simulation package. Very general and powerful procedures are available for producing sets of system equations, consisting of these main steps: select input and energy state variables, formulate the initial set of system equations, and reduce the initial equations to state-space form.

The selection of inputs is straightforward, since they are associated to each source element. These variables will appear in the final state-space equations if they have any effect on system behaviour.

The state variables are:

- The **momentum** p on every I-element in *integral* causality.
- The **displacement** q on every C-element in *integral* causality.

It is useful to indicate the **co-energy variables**: flows on the I-elements ($f = \Phi_I^{-1}(p)$) and efforts on the C-elements ($e = \Phi_C^{-1}(q)$). Co-energy variables appear in the initial formulation, and then they are eliminated during the reduction process since they are function of the state.

The order of the system equals the number of energy storage elements in integral causality. The derivation procedure is systematic:

(1) For each state variable, write its rate of change (time derivative) as an effort or a flow: $\dot{p} = e$ or $\dot{q} = f$.

(2) Causality is used to track the efforts and flows through the bond graph. The causal stroke tells us which element is *providing* this effort or flow. Trace through the bond graph from the state variable's bond, substituting at each junction and element, until only sources and **co-energy variables** remain. It makes no difference which state variable we start with.

(3) Substitute the co-energy variables last to obtain equations purely in state variables and inputs. This leads to explicit first-order equations.

4.5 Algebraic loops

Talking about state-space equations formulation, in situations in which we do NOT have algebraic-loops, and we have ALL storage energy elements in integral causality, the formulation is very straight-forward, obtaining explicit differential equations. Instead, when we have algebraic loops (we have encountered steps 5-6 of the SCAP) and/or we have some energy storage elements in derivative causality, the formulation is not anymore so immediate!

An algebraic loop arises when a chain of elements forms a closed ring with no energy-storage element breaking it. In that loop, one variable (effort or flow) becomes algebraically related to itself (as well as to other inputs and co-energy state variables). Tracing around the loop brings you back to your starting point: $e_1 = F_1(e_2, f_2)$, $e_2 = F_2(e_3, f_3) \ldots$ until we reach again the initial variable.

The procedure for handling algebraic loops is to first make an arbitrary causal assignment on one of the causally unassigned bonds and then propagate the causal information as far as possible. If the causality completes after one arbitrary assignment, then there is one algebraic loop. The more loops involved, the more difficult the final formulation becomes.

Watch out: Cause of algebraic loops

Algebraic loops always arise from a **modelling simplification**. For example, two parallel dampers with no masses to separate them create an algebraic loop in force. In reality, there would be some small compliance or inertia in between; the idealization of perfectly rigid connections removes it and creates the loop.

When SCAP is stuck on step 4, it is necessary to arbitrarily assign causality to one bond. It is recommended that when a resistance element is involved in the algebraic loop, the resistance element be chosen for the arbitrary assignment (step 5). It is customary to choose the variable that was arbitrarily selected to be the output from the R-element and the input to the rest of the system.

In order to solve the algebraic loop, we have to collect all the equations involved in the loop and solve the system to find an expression of e/f that depends *only* on inputs and/or co-energy state variables. This explicit algebraic equation can then be substituted directly into the ODEs.

For linear systems, this procedure will work for multiple loops, although solving by hand becomes formidable rather quickly. For nonlinear systems, it may not be possible to invert some of the algebraic relationships (e.g. Coulomb friction). When these inversions appear to be impossible, we HAVE TO eradicate the problem at the root by passing to less simplistic modelling assumptions!

4.6 Derivative causality

Another possible consequence of certain modelling assumptions is that, after sequential causal assignment, one or more energy storage elements are forced into derivative causality. Storage elements in derivative causality are not dynamically independent of the remaining storage elements, and they do not contribute a state variable to the final equations. **The state equations become implicit**. Implicit state equations can be algebraically transformed to explicit ones, but this is not necessarily a simple task. **The derivative elements do still store energy!**

When a C-element is forced to accept derivative causality (effort-in), it means that its effort (and hence its displacement $q = Ce$) is *algebraically determined* by the rest of the system — it does not contribute an independent state variable to the final equations. This occurs when two C-elements are connected in a configuration that rigidly constrains their displacements (e.g. two springs rigidly connected without an intervening mass). The system order is reduced by one per derivative-causality C.

Similarly, for I-elements in derivative causality (flow-in): their momenta are algebraically determined. The independent state variables remain the energy variables in the energy storage elements with integral causality. (Note: The derivative elements do still store energy!).

The true problem arises when formulating the equations. When we encounter a C/I element in derivative causality, we are guaranteed to end up with its power variable that is NOT the co-energy one ($\dot{q}$ for C-elements and $\dot{p}$ for I-elements). We end up with a double problem: we end up with the derivative of an energy variable, AND this energy variable is NOT a state one!

These situations always signal an **implicit (DAE) model**. Implicit state equations can be algebraically transformed to explicit ones, but this is not necessarily a simple task.

Handling Derivative Causality Algebraically: Given the dependent energy variable of a C/I element in derivative causality, it is always guaranteed that it is algebraically related (constrained) to variables that are independent state ones.

1. Find out (by inspecting the BG) the algebraic relation that gives the energy dependent variable as a function of the state ones.

2. Differentiate that relation to obtain an expression for $\dot{q}$ (for C) or $\dot{p}$ (for I) as a function of differentiated state variables.

3. Substitute this back into your equations. The result is a valid state equation, but it will be in implicit form.

4. To get it in explicit form, further algebraic manipulations are required to isolate the derivatives of the state variables on the left side.

Handling Derivative Causality at the Root: Derivative causality is ALWAYS caused by too simplistic modelling assumptions! It physically indicates an over-idealization (e.g. perfectly rigid connections between compliant elements, or massless inertial elements).

It is always possible to revisit the modelling assumptions that resulted in the formulation problem and change these assumptions to yield a model that is virtually the same as the original one, but without the formulation problems. Introducing a new energy storage element (a tiny mass or compliance) restores integral causality and ordinary ODE structure.

There are some computational penalties that accompany very small added inertias or very stiff added compliances in a model (it introduces a fast timescale, making the system "stiff" for numerical solvers).

However, for some systems, these penalties are a very small price to pay for eliminating formulation problems at the root.

4.7 System order and physical meaning

The order of a bond graph model equals the number of independent energy storage elements. This is a fundamental invariant: it does not matter which set of state variables is chosen (positions vs. momenta, etc.) — the order is always the same. Equivalently, the order equals the minimum number of initial conditions needed to uniquely determine future evolution.

The energy of the system at any instant is the sum of the energies stored in all elements with integral causality:

$$E_{\text{total}}(t) = \sum_{C\text{-elements}} E_C(q) + \sum_{I\text{-elements}} E_I(p). \tag{4.1}$$

This is the **Hamiltonian** of the system (in the linear case, $E = \frac{1}{2}\mathbf{x}^T Q \mathbf{x}$ for an appropriate positive-definite matrix Q).

Chapter 5

Multiport Fields and Junction Structures

Real physical components often couple multiple energy flows simultaneously. A flexible beam deformed by two forces stores energy that depends on *both* displacements simultaneously, not independently. A 3D rigid body has coupled translational and rotational inertia. These multi-port couplings cannot be captured by separate 1-port elements — they require **multiport fields**.

Using fields and junction structures, it is possible to study systems with complex multiport components using bond graphs because they combine structural detail, clarity, and ease of visualization.

Key Insight: Important Assumption for Fields

All multiport elements that we are going to study in this chapter are assumed to be (and are going to be studied as) **ALL INWARDS** in terms of power bond directions. This means power is conventionally positive when flowing into the field from all ports.

5.1 C-fields: multiport potential energy storage

A **C-field** is the multiport generalization of the 1-port C-element. It is itself a distributed compliance element, representing a distributed hole of energy conservation. It has n ports, and its potential energy is a function of all n displacements simultaneously:

$$E(\mathbf{q}) = \int_0^t \sum_{i=1}^n e_i f_i d\tau = \int_0^t \sum_{i=1}^n e_i \dot{q}_i d\tau = \int_0^t \sum_{i=1}^n e_i(\mathbf{q}) dq_i = \int_{\mathbf{q}_0}^{\mathbf{q}} \mathbf{e}(\bar{\mathbf{q}})^T \, d\bar{\mathbf{q}}, \qquad e_i = \frac{\partial E}{\partial q_i}(\mathbf{q}). \tag{5.1}$$

This means that a change in any component is going to cause a change in the stored energy. Because $E(\mathbf{q})$ represents stored potential energy, the C-field inherently conserves energy.

Key Insight

Advantage of Fields The advantage of Fields is that they allow us to capture the physics of multiport energy storage directly, without needing to break it down into multiple 1-port elements and junctions. Moreover, a C-field can be thought of as a **Inertia Compliance Damping** Matrix that captures the coupling between ports in a compact way without the need of explicitly writing the internal structure of the filed.

The constitutive law is a set of n functions $e_i(\mathbf{q})$ giving the efforts at each port as a function of all displacements. By inspection, $\mathbf{e}(\mathbf{q}) = \nabla E(\mathbf{q})$. For the C-field to be physically meaningful (**conservative**), the energy function $E(\mathbf{q})$ must exist, which requires the **Maxwell reciprocal relations** (a structural constraint):

$$\frac{\partial e_i}{\partial q_j} = \frac{\partial^2 E}{\partial q_i \partial q_j} = \frac{\partial e_j}{\partial q_i} \quad \forall\, i, j. \tag{5.2}$$

This is the direct multivariable generalization of the single-port condition $e = \partial E/\partial q$. It is clear that not every arbitrary set of constitutive laws expressing efforts in terms of displacements could come from a valid energy-storing C-field—only those sets that obey this symmetry property!

Compliance and stiffness forms. If the constitutive law can be written $\mathbf{q} = \mathbf{M}_c(\mathbf{e})$ (displacements as functions of efforts), this is the **compliance form**. If it can be written $\mathbf{e} = \mathbf{M}_k(\mathbf{q})$ (or $\mathbf{M}_c^{-1}(\mathbf{q})$), this is the **stiffness form**.

In the linear case, the equations become:

$$\mathbf{e} = \mathbf{K}\,\mathbf{q} \quad \text{(stiffness)}, \qquad \mathbf{q} = \mathbf{C}\,\mathbf{e} \quad \text{(compliance)}, \tag{5.3}$$

where $\mathbf{K} = \mathbf{C}^{-1}$ is the stiffness matrix and $\mathbf{C} = \mathbf{M}_c$ is the compliance matrix. The stored energy is $E = \frac{1}{2}\mathbf{q}^T\mathbf{K}\mathbf{q}$. Maxwell reciprocity requires $\mathbf{K} = \mathbf{K}^T$ and $\mathbf{C} = \mathbf{C}^T$ (symmetric matrices), a familiar result from structural mechanics.

Example 5.1 (Elastic beam loaded at two points)**.** An Euler-Bernoulli beam of length L, stiffness EI, loaded by forces F_1 at $L/3$ and F_2 at $2L/3$ has the compliance matrix:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{L^3}{243EI} \begin{pmatrix} 4 & \frac{7}{2} \\ \frac{7}{2} & 4 \end{pmatrix} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}. \tag{5.4}$$

The symmetric compliance matrix confirms Maxwell reciprocity.

Nonlinear C-fields. Nonlinear C-fields are common in certain application areas. A mechanical element that behaves as a linear 1-port in restricted cases can often be modelled as a nonlinear 2-port C-field when two-dimensional motion is permitted.

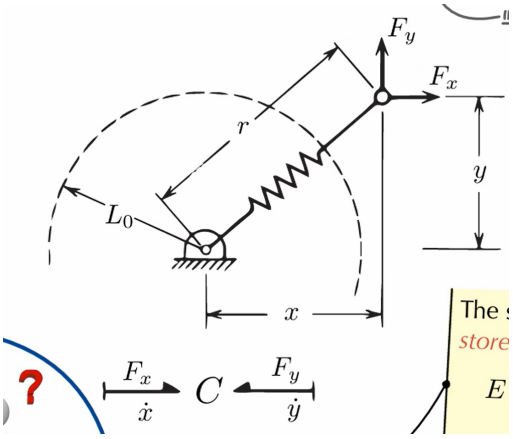


Figure 5.1: A simple linear spring pivoting in a 2D plane is a nonlinear C-field.

For example, a simple linear spring (rest length L_0, stiffness k) pivoting in a 2D plane (x, y) has the energy:

$$E(x, y) = \frac{1}{2}k\left(\sqrt{x^2 + y^2} - L_0\right)^2 \tag{5.5}$$

The gradient of this potential energy gives the highly nonlinear constitutive equations for the x and y efforts:

$$F_x = kx - k\frac{L_0 x}{\sqrt{x^2 + y^2}}, \qquad F_y = ky - k\frac{L_0 y}{\sqrt{x^2 + y^2}} \tag{5.6}$$

5.1.1 Explicit vs. implicit C-fields

An **implicit field** arises when a group of 1-port C-elements, connected by junction structures (0-junctions, 1-junctions, TF), collectively behaves as a C-field. For example, two springs k_1 and k_2 connected in series via a 1-junction (sharing a common force, but adding displacements) or connected in parallel via a 0-junction form an implicit C-field. The implicit form is naturally obtained from the assembly of the bond graph; converting it to explicit form requires solving for the field's constitutive equations algebraically.

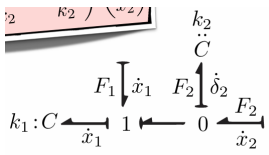


Figure 5.2: Implicit C-field

An **explicit field** is one whose constitutive law is given directly as $\mathbf{e}(\mathbf{q})$ or $\mathbf{q}(\mathbf{e})$ — the multiport is defined from the outset as a single entity (like the beam example above).

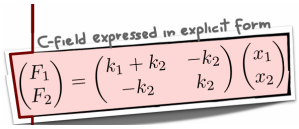


Figure 5.3: Explicit C-field

Key Insight: When to convert implicit to explicit

Converting an implicit field to an explicit form (Viva Explicit C-Fields!) is extremely useful when: (a) Some elements of the field are forced into derivative causality. By pre-solving the algebraic loops and constraints inside a field matrix, you save the computer from crashing during simulation and kill the derivative causality mathematically. (b) Equation formulation is simplified by working directly with the field's stiffness or compliance. The transformation into an explicit form integrates trivial state equations once and for all, avoiding the necessity of handling derivative causality completely.

5.1.2 Causality in C-fields

As in the case of 1-port C-elements, multiport C-fields also admit mixed integral-derivative causalities. A C-field with n ports can accept any mixture of integral and derivative causalities across its ports, as long as the resulting constitutive equations can be inverted appropriately.

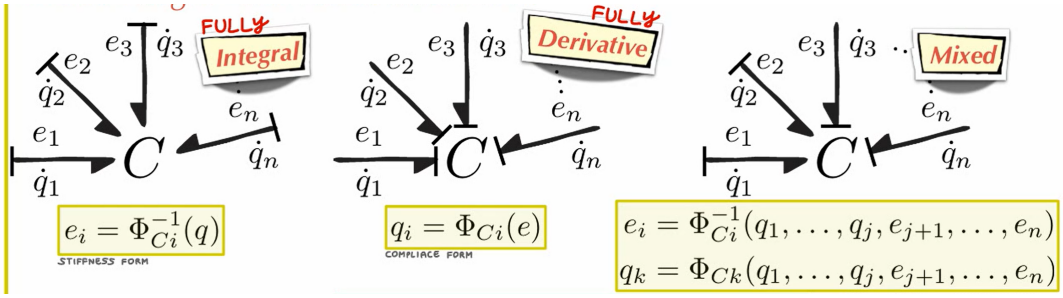


Figure 5.4: Causality in C-fields

Key Insight: Example of causality assignment in a C-field

1. Draw bond graph
2. Assign causality
3. Express the C-field

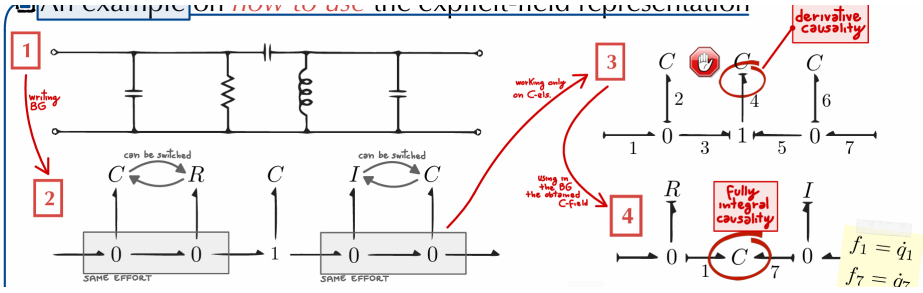


Figure 5.5: Example of causality assignment in a C-field

The manipulation from implicit to explicit field can be of help if one or more elements must be assigned derivative causality.

- The transformation in an explicit form simplifies equation formulation

- This process increases the insight into the **causes** of derivative causality
- The explicit-field representation can be used to avoid the necessity of handling derivative causality.

It's important to note that when assigning causality, even if an element is assigned in derivative causality, it does not necessarily mean that the resulting constitutive equations will be in derivative form. The derivate causallity of the element is holded inside the C-field, but not externally, at the terminal side. From the terminal point of view, this is just a C-field in fully integral causality.
The causality of the inner elements it is not visible from the outside. This is a key point to understand when working with fields.

In **fully integral causality**, all flows are inputs and all efforts are outputs: $\mathbf{e} = \mathbf{K}\,\mathbf{q}$ (stiffness form). In **fully derivative causality**, all efforts are inputs and all displacements are outputs: $\mathbf{q} = \mathbf{C}\,\mathbf{e}$ (compliance form). **Mixed causality** involves both, requiring a mixed-form constitutive law obtained by partially inverting the stiffness/compliance matrix.

For an implicit field, admissible causal forms are easily determined by propagating the standard bond graph causality rules through its constituent elements (e.g., studying conflicts at 0 or 1 junctions). For an explicit field specified in a particular causal form, the question is an algebraic one (whether the constitutive equations can be inverted), and for nonlinear fields, this can be very hard to decide. Notice that, in principle, any strictly valid C-field can always go into fully integral causality!

5.2 I-fields: multiport kinetic energy storage

For I-fields we can develop a complete parallel dual formulation with respect to C-fields. Inertial fields are the exact duals of capacitive fields. The **I-field** is the multiport generalisation of the 1-port I-element. Its kinetic energy is a function of all momenta simultaneously:

$$E(\mathbf{p}) = \int_0^t \sum_{i=1}^n e_i f_i d\tau = \int_0^t \sum_{i=1}^n f_i \dot{p}_i d\tau = \int_0^t \sum_{i=1}^n f_i(\mathbf{p}) dp_i = \int_{\mathbf{p}_0}^{\mathbf{p}} \mathbf{f}(\bar{\mathbf{p}})^T \, d\bar{\mathbf{p}}, \qquad f_i = \frac{\partial E}{\partial p_i}(\mathbf{p}). \tag{5.7}$$

In the linear case, the energy is $E(\mathbf{p}) = \frac{1}{2}\mathbf{p}^T\mathbf{M}^{-1}\mathbf{p}$. The constitutive law maps momenta to flows via the inverse mass/inertia matrix $\mathbf{M}^{-1}$ (i.e. $\mathbf{f} = \mathbf{M}^{-1}\mathbf{p}$).

The Maxwell reciprocal relations for I-fields require $\partial f_i/\partial p_j = \partial f_j/\partial p_i$, which translates to the symmetry of $\mathbf{M}^{-1}$ (and hence of $\mathbf{M}$). Like C-fields, I-fields occur in both implicit and explicit forms, and manipulating them can yield highly useful simplified representations.

The most important mechanical physical I-field is the **3D rigid body**: it has three translational and three rotational momentum ports, with the (constant) mass-inertia tensor providing the constitutive law. In 2D, the I-field reduces to two translational inertias and one rotational inertia, as discussed in Chapter 3. By transforming an implicit rigid-body I-field into an explicit one, we can define the relationship between the center-of-mass velocity and the velocities of the attachment points analytically.

Interacting Coils (Mutual Inductance). Another classic use of I-fields is in electrical circuits containing mutually interacting coils. These are conveniently represented using explicit I-fields in derivative causality. To write the constitutive laws, one needs a sign convention for currents/voltages and a convention for the relative orientation of the coils (often denoted by dots, squares, or triangles on schematic diagrams).

If the currents i_1 and i_2 are defined so that when positive they both enter the dotted ends, the mutual-inductance effects M are positive:

$$\begin{pmatrix}\lambda_1\\ \lambda_2\end{pmatrix} = \begin{pmatrix}L_1 & M_{12}\\ M_{12} & L_2\end{pmatrix}\begin{pmatrix}i_1\\ i_2\end{pmatrix} \tag{5.8}$$

If one current enters a dot and the other leaves a dot, the mutual terms become $-M_{12}$. Independently of the coil orientation convention, the inductance matrix and its inverse will always be symmetric as long as the I-field is conservative.

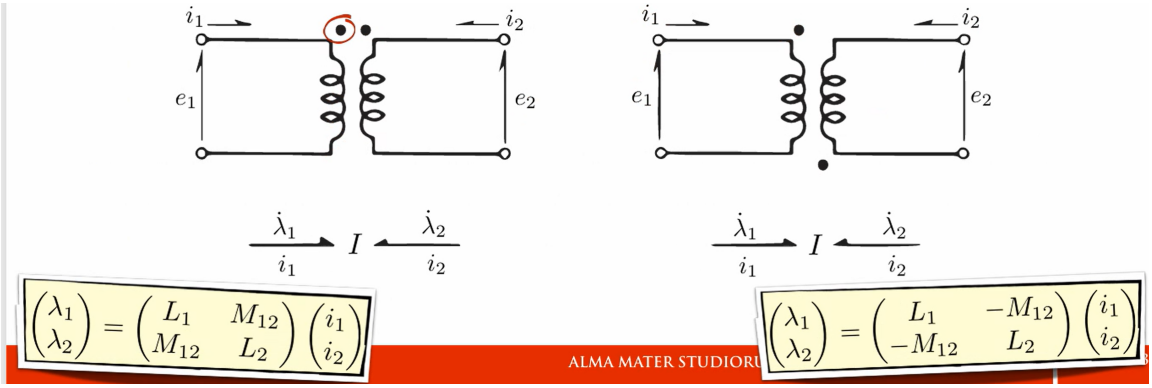


Figure 5.6: Mutually interacting coils represented as an I-field

5.3 Mixed energy-storing fields (IC-fields)

There are cases when an energy-storing device acts as a C-field at some ports and an I-field at others. Within a single energy domain, there is usually no need for such an element. However, energy transducers are often ideally energy conservative but store and release energy across different domains over time (e.g., storing electrical energy and releasing it as mechanical work). Because their electrical and mechanical energies are dynamically coupled, they cannot be modeled by simple 1-port C or I elements. You must use a mixed **IC-field** to mathematically capture the complex physics natively.

For an IC-field with momentum variables p and displacement variables q, the total energy features dynamic coupling (an irrotational case) and contains mixed terms. The constitutive equations are obtained by taking the appropriate partial derivatives of the total stored energy $E(p, q)$.

$$E(\mathbf{p}) = \int_0^t \sum_{i=1}^{n} e_i f_i d\tau = \int_0^t \sum_{i=1}^{j} f_i \dot{p}_i d\tau + \int_0^t \sum_{k=j+1}^{n} e_k \dot{q}_k d\tau \tag{5.9}$$

Maxwell's reciprocal relations for IC-fields require that the mixed second derivatives of the energy function are equal:

$$\frac{\partial^2 E}{\partial p_i \partial q_k} = \frac{\partial^2 E}{\partial q_k \partial p_i} \quad \forall\, i, k. \tag{5.10}$$

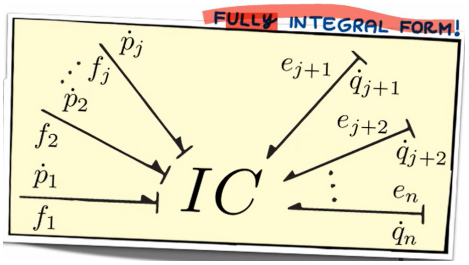


Figure 5.7: IC-field with two I-ports and two C-ports

Example 5.2 (Movable-Plate Capacitor)**.** A capacitor where one plate can move mechanically acts as a C-port electrically (charge q) and a C-port mechanically (position x). The stored energy is $E(q, x) = \frac{1}{2}\frac{q^2}{C(x)}$. The constitutive equations are:

$$e = \frac{\partial E}{\partial q} = \frac{q}{C(x)} \qquad \text{and} \qquad F = \frac{\partial E}{\partial x} = -\frac{1}{2}\frac{q^2}{C^2(x)}\frac{\partial C}{\partial x} \tag{5.11}$$

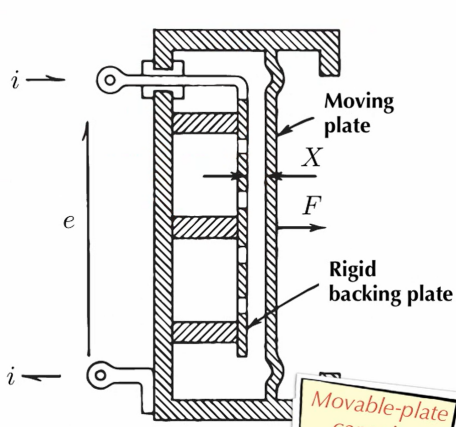


Figure 5.8: Movable-plate capacitor as an IC-field

Example 5.3 (Electrical Solenoid)**.** The electrical solenoid serves as a prototype of an IC-field transducer. From the electrical port, the coil exhibits self-inductance (λ). From the mechanical port, the force on the slug depends on its position (x). The energy is $E(\lambda, x) = \frac{1}{2}\frac{\lambda^2}{L(x)}$. The constitutive equations are:

$$i = \frac{\partial E}{\partial \lambda} = \frac{\lambda}{L(x)} \qquad \text{and} \qquad F = \frac{\partial E}{\partial x} = -\frac{1}{2}\frac{\lambda^2}{L^2(x)}\frac{\partial L}{\partial x} \tag{5.12}$$

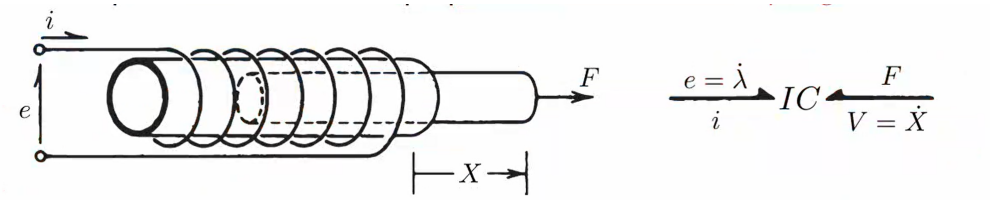


Figure 5.9: Electrical solenoid as an IC-field

5.4 R-fields: multiport dissipation

An **R-field** is an n-port with a static (or algebraic) constitutive law that relates efforts and flows. In general, R-fields are **NOT** strictly related or associated to power losses: their TRUE intrinsic characteristic is to algebraically relate efforts and flows!

This broad definition includes power-conserving elements, such as 0- and 1-junctions, and elements containing sources. However, in practice, most explicit R-fields do dissipate power (e.g. thermal resistors, coupled fluid friction matrices). Explicit R-fields arise frequently in modelling nonlinear devices.

The causality of an R-field is determined by the sources and energy-storing elements attached to it.

- **Resistance causality:** All flows are inputs, efforts are outputs ($e_i = \Phi_{R_i}(f_1 \dots f_n)$).
- **Conductance causality:** All efforts are inputs, flows are outputs ($f_i = \Phi^{-1}_{R_i}(e_1 \dots e_n)$).
- **Mixed causality:** A mixture of effort and flow inputs.

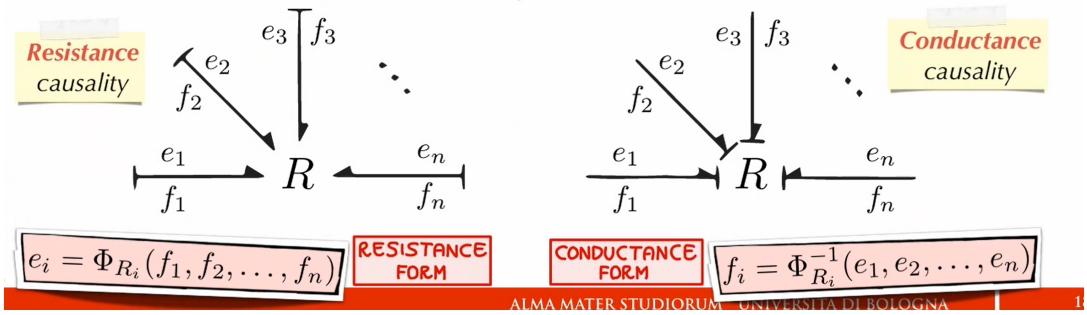


Figure 5.10: Causality in R-fields

Onsager vs. Casimir Forms and Power. Since an R-field has no stored energy function, there is no direct equivalent to Maxwell's Reciprocal Relations. However, structural properties do exist. Any square Constitutive Law Matrix (CLM) M can always be mathematically decomposed into a symmetric part $M_S = \frac{1}{2}(M + M^T)$ and a skew-symmetric (antisymmetric) part $M_A = \frac{1}{2}(M - M^T)$.

The asymmetric contribution M_A is always related to *perfectly conserved power* ($P_w = b^T M_A b = 0$). The symmetric part M_S is associated with power dissipation.

Implicit R-fields composed strictly of linear 1-port resistances, 0-junctions, 1-junctions, and transformers will obey **Onsager Reciprocity**: when expressed in pure resistance or conductance form, their CLM is perfectly symmetric (called the *Onsager form*). When expressed in mixed causality, the matrix is guaranteed to present asymmetric terms.

Importantly, NOT all R-fields obey Onsager Reciprocity. If Gyrators (GY) are allowed inside the implicit R-field, the reciprocity is broken, yielding asymmetric terms even in pure resistance/conductance forms. This is known as the **Casimir form**.

There could be examples of R-fields that are not Onsager and Casimir at the same time, but a mix of the two.

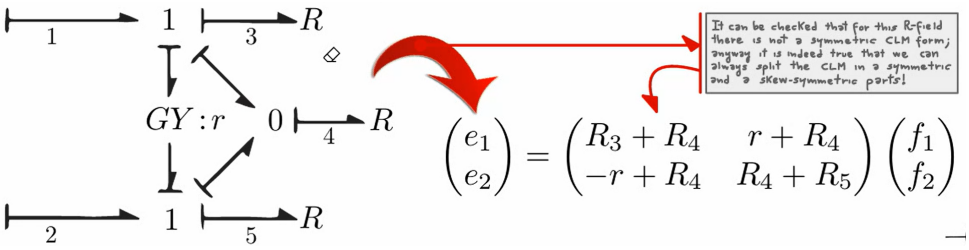


Figure 5.11: R-field without Onsager and Casimir reciprocity

5.5 Modulated 2-port elements

Before discussing pure junction structures, it is essential to introduce the modulated 2-ports. The **modulated transformer (MTF)** and **modulated gyrator (MGY)** are nonlinear, power-conserving generalisations of TF and GY.

The parameters (modulus m or r) of these elements are allowed to be a function of some other parameter in the system (often a displacement or angle, possibly time-varying). These parameters are modulated by means of a signal (active bond) rather than a power bond. Thus, it is characteristic of these elements that the parameters change value without a directly associated power flow!

Examples include the DC motor (Faraday & Lorentz laws modelled as an MGY depending on field current) and the slider-crank or rack-and-pinion mechanisms (modelled as MTF).

5.6 Junction structures

Junction structures are the energy switchyards that enforce the constraints among parts of dynamic systems. They result from the interconnection of 0-junctions, 1-junctions, TF, and GY. They contain no dissipative or storage elements.

Junction structures are essentially special types of R-fields that *never* dissipate power (meaning their CLM is completely skew-symmetric). They are always **anti-symmetric** because they cannot dissipate power. Because they cannot dissipate power, they typically **cannot** accept all possible causal assignments at their ports (e.g., a pure junction structure without gyrators cannot accept resistance or conductance causality on all ports simultaneously).

Junction structures have a key property: they are themselves power-conserving, so $\sum_k e_k f_k = 0$ across all their external ports. Equivalently, the junction structure defines a set of linear constraints (from the junctions) and scaling/exchange rules (from TF and GY) on the port variables.

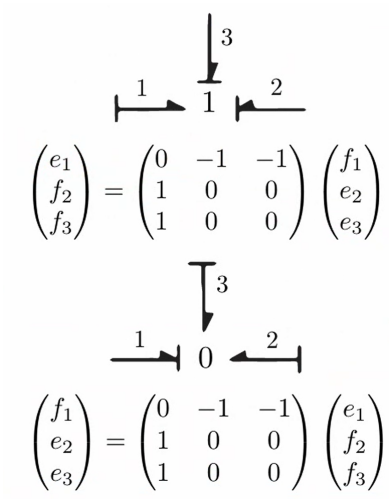


Figure 5.12: Junction structure as a power-conserving R-field

5.6.1 Multiport Transformers (Kinematics)

Another extremely interesting viewpoint about junction structures involves the transformations of variables they provide. A **multiport transformer** is a specific assemblage of 0s, 1s, and TFs that shows more than two ports to the external world.

Mathematically, a multiport transformer is characterised by a matrix $\mathbf{G}$ and its transpose $\mathbf{G}^T$, just as a 2-port transformer is characterised by a single modulus m. By setting a "through" power convention (power in for inputs, out for outputs), the constitutive law can always be written as:

$$\mathbf{e}_{in} = \mathbf{G} \cdot \mathbf{e}_{out} \qquad \text{and} \qquad \mathbf{f}_{out} = \mathbf{G}^T \cdot \mathbf{f}_{in} \tag{5.13}$$

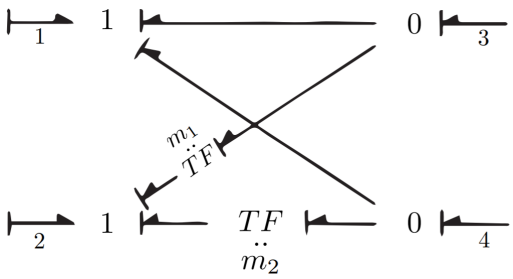


Figure 5.13: Multiport transformer as a power-conserving R-field

This is exactly the formulation used for 3D Kinematics and coordinate transformations! A typical example is implementing a Jacobian $J(q)$ that links a joint-coordinate space to a Cartesian-coordinate space in robotics. The forward kinematics ($\dot{x} = J(q)\dot{q}$) dictates the flow transformations, and by power conservation, the transpose Jacobian dictates the force-torque effort transformations ($T = J^T(q)F$). Grouping these into a multiport MTF field elegantly handles 3D kinematics directly within the bond graph framework.

Part III

Mechanical, Distributed and Thermo-fluid Systems

Chapter 6

Multi-body Systems

Extending bond graph modelling from 2D plane motion to full **3D rigid-body mechanics** brings geometric nonlinearity through the rotation group. These geometric nonlinearities are a primary cause of equation formulation problems. The key modelling choice — whether to express quantities in the inertial frame or a body-fixed frame — has direct consequences for the structure of the equations and the bond graph.

6.1 Dynamics of a 3D rigid body

Consider a rigid body of mass m with moments of inertia J_x, J_y, J_z along the principal axes. The body-fixed coordinate frame $\{G; x, y, z\}$ is attached to the center of mass G with axes aligned with the principal axes of inertia.

Why the body frame? In the inertial frame, the inertia tensor $\mathbf{J}$ is a 3×3 matrix that varies with the body's orientation — making the constitutive equations of the I-field time-varying and complex. In the body frame, $\mathbf{J} = \mathrm{diag}(J_x, J_y, J_z)$ is **constant**, which is the primary advantage of this formulation. The mass m and inertia $\mathbf{J}$ are invariant, making them highly practical coordinates for computation.

The price is that Newton's law in the body frame contains **nonlinear gyroscopic terms** from the frame's own rotation. Newton's law applied in the body frame requires accounting for the rate of change of momentum relative to the moving frame.

Newton–Euler equations. Let $\mathbf{v} = (v_x, v_y, v_z)^T$ and $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)^T$ be the translational velocity of G and the angular velocity, *both expressed in the body frame*. Let $\mathbf{F}$ and $\boldsymbol{\tau}$ be force and torque at G in the body frame. Then:

$$\mathbf{F} = \dot{\mathbf{p}} + \boldsymbol{\omega} \times \mathbf{p}, \qquad \mathbf{p} = m\mathbf{v}, \tag{6.1}$$

$$\boldsymbol{\tau} = \dot{\mathbf{h}} + \boldsymbol{\omega} \times \mathbf{h}, \qquad \mathbf{h} = \mathbf{J}\boldsymbol{\omega}. \tag{6.2}$$

Written component-wise, these are the **Euler equations**:

$$\begin{aligned} F_x &= m\dot{v}_x + m\omega_y v_z - m\omega_z v_y \\ F_y &= m\dot{v}_y + m\omega_z v_x - m\omega_x v_z \\ F_z &= m\dot{v}_z + m\omega_x v_y - m\omega_y v_x \end{aligned}$$

$$\begin{aligned} \tau_x &= J_x\dot{\omega}_x + \omega_y J_z\omega_z - \omega_z J_y\omega_y \\ \tau_y &= J_y\dot{\omega}_y + \omega_z J_x\omega_x - \omega_x J_z\omega_z \\ \tau_z &= J_z\dot{\omega}_z + \omega_x J_y\omega_y - \omega_y J_x\omega_x. \end{aligned} \tag{6.3}$$

Key Insight

Note that we are dealing the body fixed frame here. This is just the derivative wrt to time of a vector, that in this case are the momentum (p) and the angular momentum (h). There is a **hidden frame fixed to the momentum vectors.**
The angular momentum formula is more intuitive since it's more used as we work with fixed body frame with fixed inertia and axis of inertia. This is not usually done with forces, since we see them in the inertial frame.
Note also that it's always possible to measure ω with a gyro, since all the points in a rigid body have the same angular velocity. However, the linear velocity of the CoG is not directly measurable by a sensor fixed to the body, since it depends on the point itself. That's why we don't use the first

equations in practice.

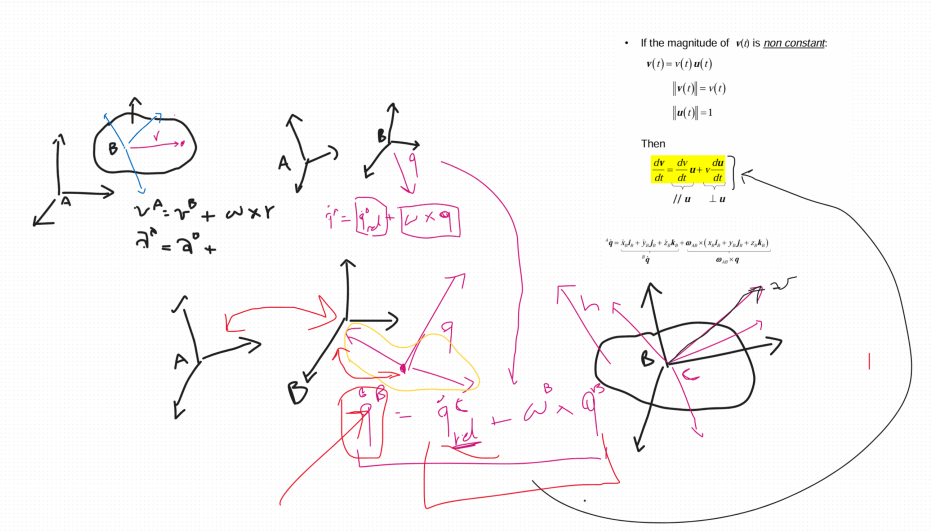


Figure 6.1: Considerations on the body frame and the hidden frame of the momentum vectors.

These nonlinear differential equations have no general analytical solution. Furthermore, they provide velocities with respect to a frame that is aiming in a different direction at every instant. The cross-product terms $\boldsymbol{\omega} \times \mathbf{p}$ and $\boldsymbol{\omega} \times \mathbf{h}$ are gyroscopic: they do not do work (they are perpendicular to velocity), but they redistribute momentum between components. In the bond graph, they appear as **modulated gyrators** (MGY elements) whose moduli depend on the angular velocity — making the model intrinsically nonlinear.

Key Insight: The I-field of a 3D rigid body vs. Independent Elements

The 3D rigid body is conceptually represented by a **6-port I-field** in the body frame: three translational ports and three rotational ports. The constitutive law (flow = $\mathbf{M}^{-1}\mathbf{p}$, with block-diagonal $\mathbf{M} = \mathrm{diag}(m\mathbf{I}_3, \mathbf{J})$) is linear and constant.
However, *attention must be paid to the actual bond graph implementation*. In practice, this block is modelled as **six independent 1-port I-elements**, which are in integral causality. (since the mass matrix is diagonal in the principal body frame), interconnected by an Eulerian junction structure containing the MGYs. The nonlinear gyroscopic interaction is handled by MGY elements in the junction structure, not inside the I-elements themselves. This separation of linear storage from nonlinear routing is elegant and computationally efficient. As long as external forces and torques are causal inputs, integral causality exists for all six I-elements.

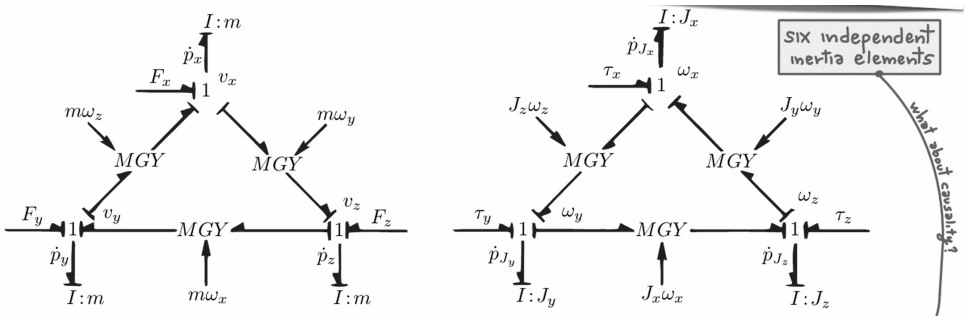


Figure 6.2: The I-field of a 3D rigid body is often implemented as six independent I-elements (three for translation, three for rotation) interconnected by a junction structure that encodes the gyroscopic interactions via MGY elements.

Note that in the bond enters the ($F_{x,y}$, F_z) forces that's important to mention that they are refered to the body frame (which is moving w.r.t. the inertial frame). Forces assign causality effort, and the I-field is in integral causality, so the velocities are outputs. The MGY elements take these velocities and feed them

Figure 6.3: THis is not a single field, this is a double field because they are two separated systems.

back into the I-field to account for the gyroscopic effects.

6.2 Rotation matrices and coordinate transformations

Since all velocities and forces are expressed in the body frame, but external loads and gravity often come from the inertial frame (where e.g., gravity acts strictly downwards), coordinate transformations are necessary. We need an appropriate "junction element" to bridge the body frame and the inertial frame.

The rotation matrix $\mathbf{R}(t)$ (a 3×3 orthogonal matrix with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$, $\det \mathbf{R} = 1$) transforms vectors from the body frame to the inertial frame ($\mathbf{w}_{inertial} = \mathbf{R}\mathbf{w}_{body}$). In bond graphs, $\mathbf{R}$ acts as a Modulated Transformer (MTF) that conserves power.

Yaw, Pitch, and Roll (Cardan Angles). Any spatial transformation consists of a series of coordinate rotations. For spinning bodies, Euler angles are typical, but for multi-body systems like robotic arms or aircraft, Yaw (ψ), Pitch (θ), and Roll (ϕ) (Cardan Angles) are frequently used. The combined rotation matrix is $\mathbf{R}_{\psi\theta\phi} = \mathbf{R}_\psi \mathbf{R}_\theta \mathbf{R}_\phi$.

By mapping the body-frame angular velocity $\boldsymbol{\omega}$ to the angle derivatives, we obtain the kinematic relationships:

$$\dot{\psi} = \frac{\sin\phi}{\cos\theta}\omega_y + \frac{\cos\phi}{\cos\theta}\omega_z \tag{6.4}$$

$$\dot{\theta} = \cos\phi\,\omega_y - \sin\phi\,\omega_z \tag{6.5}$$

$$\dot{\phi} = \omega_x + \sin\phi\frac{\sin\theta}{\cos\theta}\omega_y + \cos\phi\frac{\sin\theta}{\cos\theta}\omega_z \tag{6.6}$$

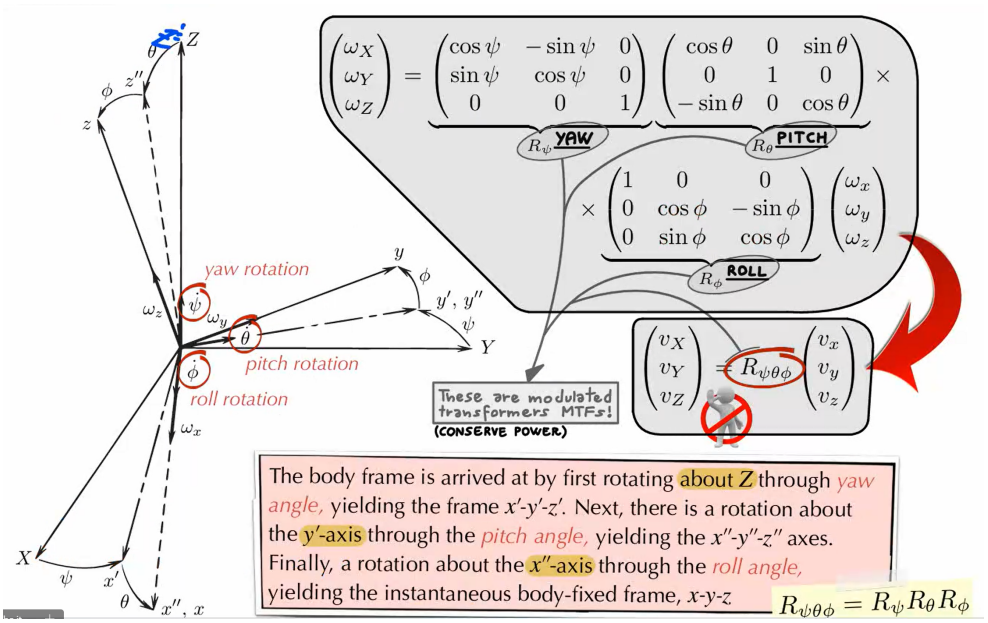


Figure 6.4: Roll pitch and yaw rotation matrices

Care must be taken regarding kinematic singularities (e.g., when $\cos\theta = 0$). Alternatively, one can

directly integrate the differential equation of the rotation matrix:

$$\dot{\mathbf{R}} = \mathbf{R}\,\mathbf{S}(\boldsymbol{\omega}) \tag{6.7}$$

where $\mathbf{S}(\boldsymbol{\omega})$ is the skew-symmetric matrix:

$$\mathbf{S}(\boldsymbol{\omega}) = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}. \tag{6.8}$$

The quantity $\dot{\mathbf{R}}\mathbf{R}^T = \mathbf{R}\,\mathbf{S}(\boldsymbol{\omega})\,\mathbf{R}^T$ is the angular velocity in the inertial frame expressed as a skew-symmetric matrix. In the bond graph, MTF elements with moduli from $\mathbf{R}$ handle these transformations when coupling to the environment or to other bodies.

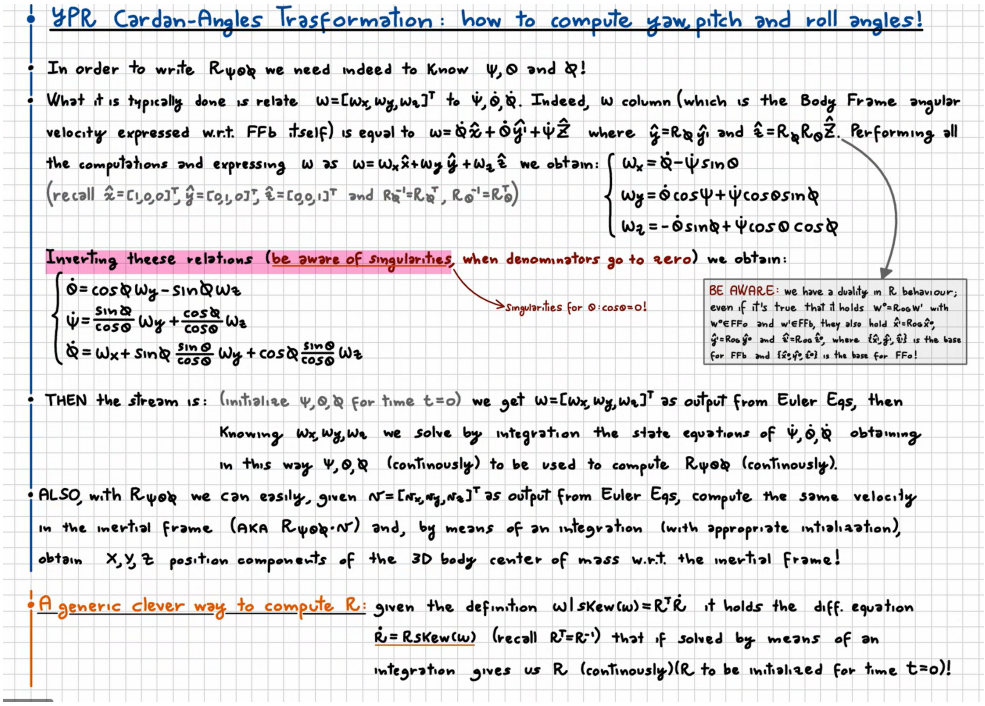


Figure 6.5: Andrea Lombardi insight

Multidimensional dynamics We can always tranform the body-fixed variables into inertial frame variables by using a modulated transformer that handles the rotation.

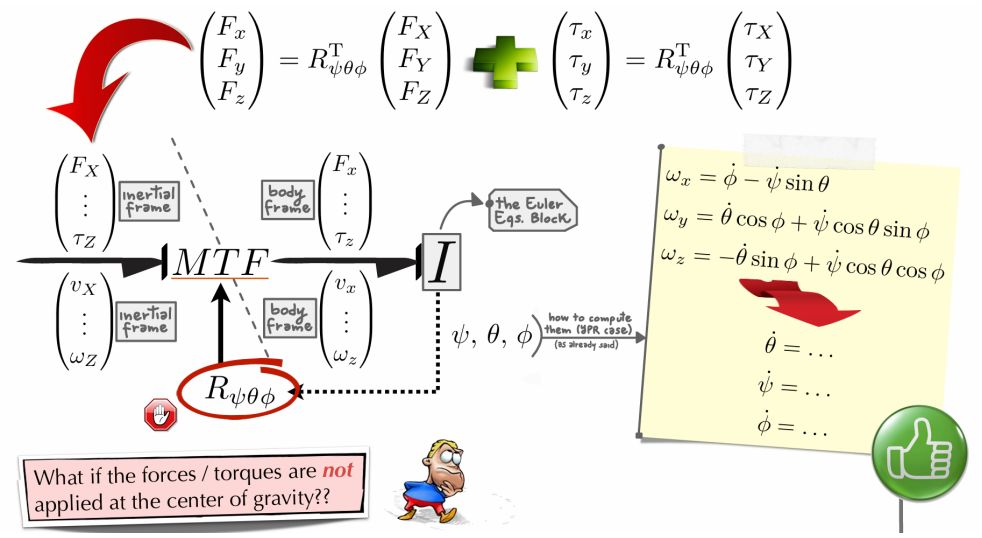


Figure 6.6: The rotation matrix $\mathbf{R}$ acts as a modulated transformer (MTF) that converts body-frame velocities and forces to inertial-frame quantities, allowing us to connect the rigid body I-field to external forces and gravity defined in the inertial frame.

Forces not applied at the Centre of Gravity. Often, connecting forces are applied at a point P offset from the CoG G by a fixed vector $\mathbf{r}_{GP}$ in the body frame. The velocity of point P is given by:

$$\mathbf{v}_P = \mathbf{v}_G + \boldsymbol{\omega} \times \mathbf{r}_{GP} \tag{6.9}$$

In matrix form, this kinematic relationship defines a **Body Frame Configuration Tensor** $\mathbf{T}(P)$:

$$\begin{pmatrix} \mathbf{v}_P \\ \boldsymbol{\omega} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -\mathbf{r}_{GP}\times \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{v}_G \\ \boldsymbol{\omega} \end{pmatrix} \tag{6.10}$$

By power conservation, the transposed tensor maps the forces applied at P back to equivalent forces and torques at the CoG:

$$\begin{pmatrix} \mathbf{F}_G \\ \boldsymbol{\tau}_G \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{r}_{GP}\times & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{F}_P \\ \boldsymbol{\tau}_P \end{pmatrix} \tag{6.11}$$

This allows us to freely attach any number of forces or velocities applied at generic points using an MTF block.

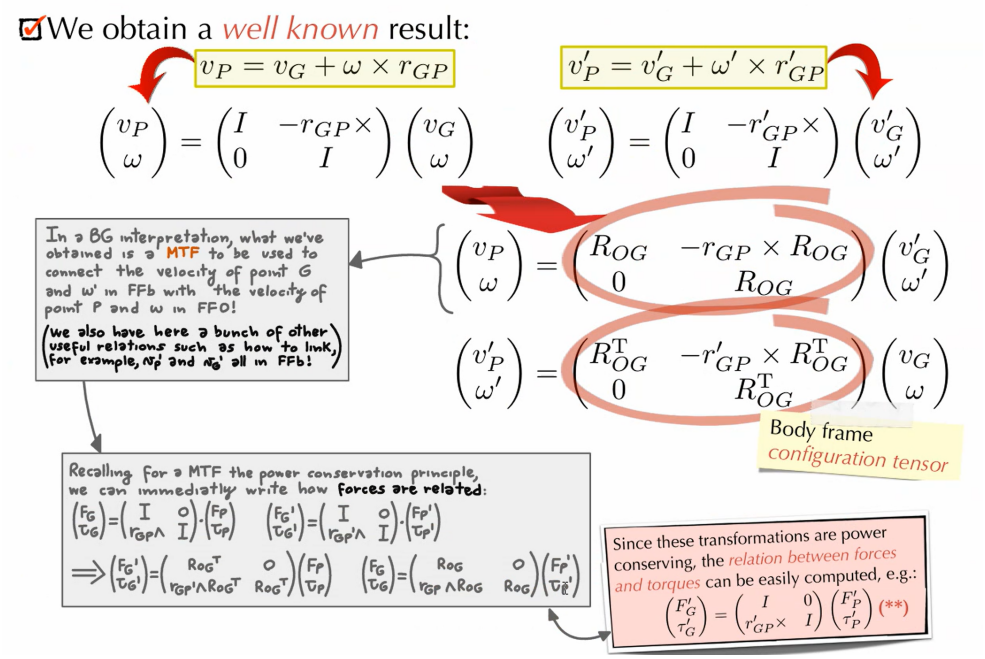


Figure 6.7: Forces applied at a point P offset from the CoG G can be transformed to equivalent forces and torques at G using a Body Frame Configuration Tensor $\mathbf{T}(P)$, which accounts for the lever arm effect of the offset.

We can write the whole rigid link bond graph in this manner:

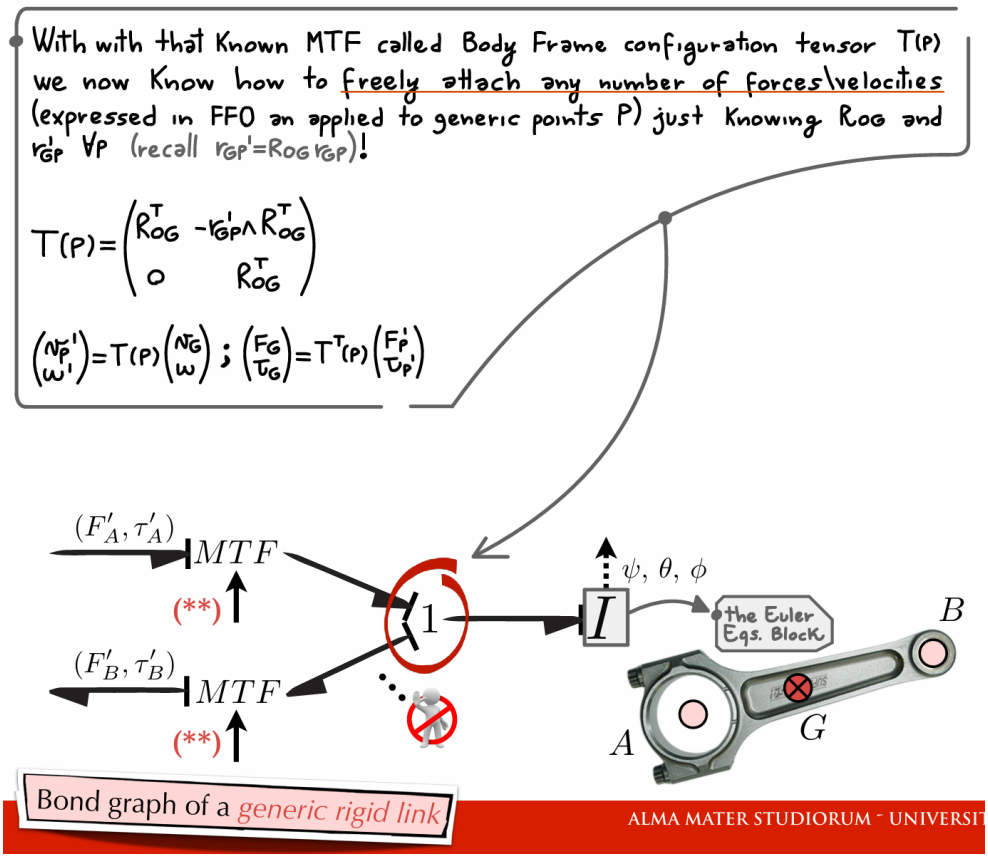


Figure 6.8: The bond graph of a single rigid body in the inertial frame.

They are used to move from a point A in the rigid body to a point B in the same rigid body.

6.3 Joints between rigid links

A **joint** connects two rigid links by constraining their relative motion to a subspace. In the inertial frame:

Rotational (Revolute) joint. The relative linear velocity between the two attachment points is zero; the relative angular velocity $\omega_r = \omega_B - \omega_A$ is free (the joint can rotate). A torque τ_r acts along the joint axis. A revolute joint fully constrains linear velocities at the joint centre P, which is fixed with respect to both body frames. The bond graph imposes the velocity constraint via an MTF ($\mathbf{v}_P^A = \mathbf{R}_B^A \mathbf{v}_P^B$), and identifies the relative angular velocity:

$$\mathbf{v}_A = \mathbf{v}_B \quad \text{(at the joint location)}, \qquad \boldsymbol{\omega}_{AB}^A = \boldsymbol{\omega}_B^A - \boldsymbol{\omega}_A^A \text{ (free)}. \tag{6.12}$$

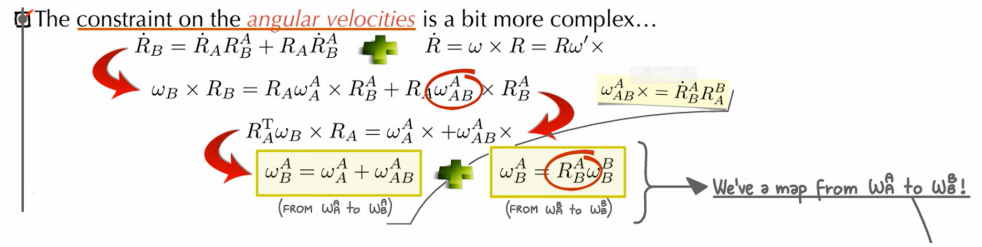


Figure 6.9: Derivation of angular velocity formula

Assuming the z-axis is oriented along the rotation axis, $\boldsymbol{\omega}_{AB}^A = (0, 0, \dot{\varphi})^T$.
Wrt inertial frame, the rotation matrix is:

$$\mathbf{R}_B = \mathbf{R}_A + \mathbf{R}_B^A \tag{6.13}$$

The revolute joint can be modelled as followed, following the Denavit–Hartenberg convention for the joint axis and the relative angle

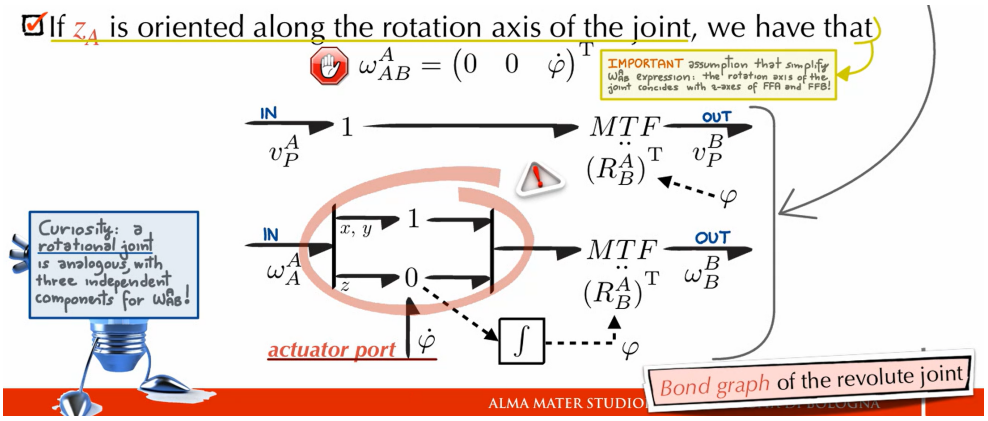


Figure 6.10: Bond graph of a revolute joint between two rigid bodies, showing the velocity constraints and the free rotational degree of freedom.

Translational (Prismatic) joint. A prismatic joint fully constrains angular velocities among the bodies ($\boldsymbol{\omega}_A = \boldsymbol{\omega}_B$), meaning they always have the same relative orientation. The relative linear velocity is constrained to one direction $\hat{\mathbf{d}}(\theta)$ (which itself varies as the mechanism rotates). A force F_r acts along this direction. Assuming the translation axis coincides with the z-axis of both frames, $\mathbf{v}_{AB}^A = (0, 0, \dot{s})^T$. The constraint and the direction transformation involve MTF elements with time-varying moduli that depend on the current configuration angle θ.

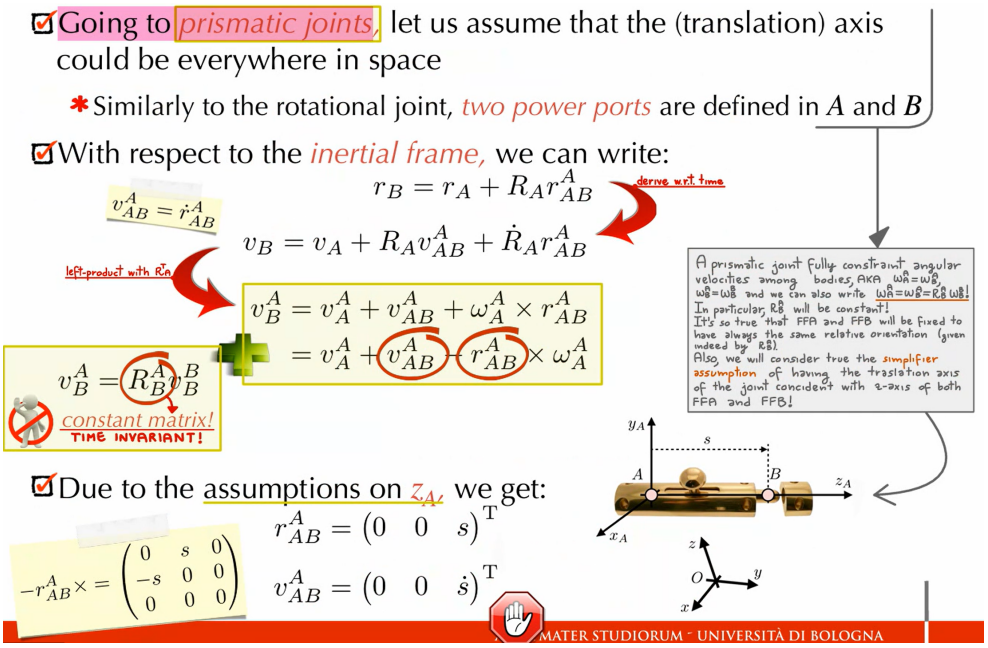


Figure 6.11: Derivation of equations for a prismatic joint, plus assumption of Denavit–Hartenberg convention for the joint axis.

The key practical point is: in the inertial-frame formulation, joint directions $\hat{\mathbf{d}}(\theta)$ are time-varying, making the MTF moduli time-varying. This is why the **body-frame formulation** is typically preferred for multi-body dynamics — joint directions remain constant in the body frame.

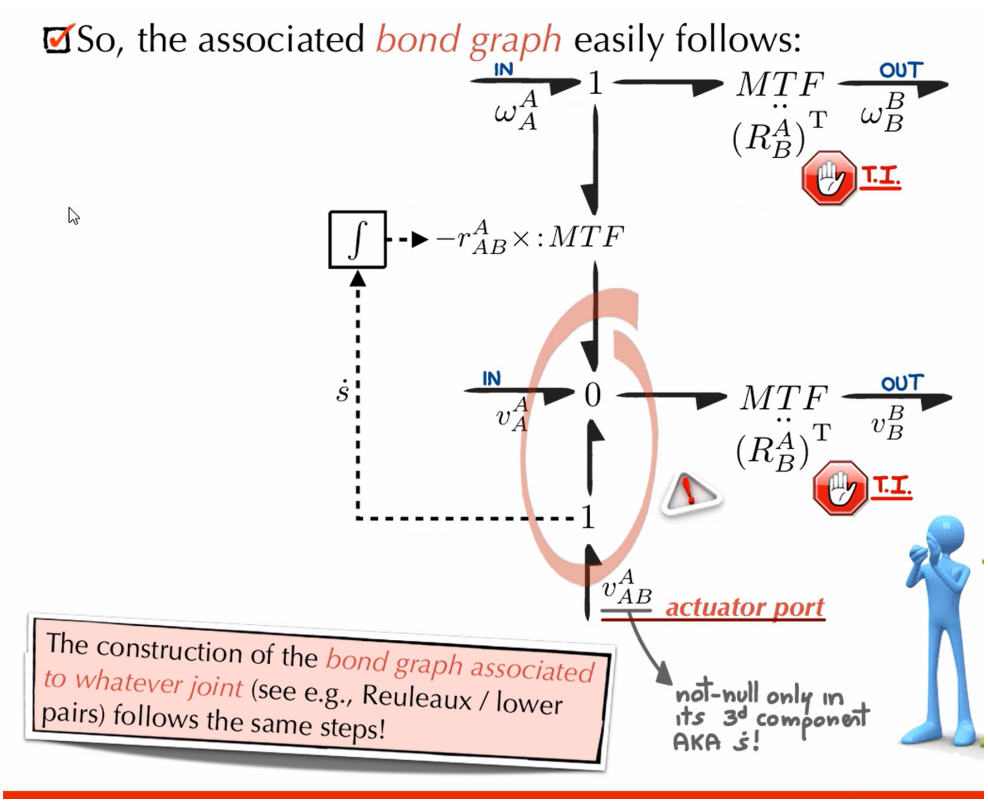


Figure 6.12: Bond graph of a prismatic joint between two rigid bodies, showing the angular velocity constraints and the linear velocity constraint along the prismatic axis.

6.4 Kinematic nonlinearities in the dynamics

Main drawback is that the final equations are nonlinear and with (integrable) constraint. Up to now we just used the kinematic constraints to write the bond graph, we haven't yet introduced the dynamics of the system. For simpler systems using the proposed method is too complex.

We must find a convenient way to describe the system without the rigid body dynamics. We'll use a **standard model**.

For any problem in mechanics involving particles, rigid bodies and elastic elements, the mathematical model consists of a set of first order ODEs in terms of variables with some physical insight.

Equations are **coupled** and **explicit**.

$$f_C = \Phi_C^{-1}(q_C) \tag{6.14}$$
$$v_I = \Phi_I^{-1}(p_I) \tag{6.15}$$
$$\dot{q}_C = T_{CI}(q_C) v_I \tag{6.16}$$
$$\dot{p}_I = -T_{CI}^T(q_C) f_C \tag{6.17}$$

The most difficult aspect is the nonlinear coupling of C- and I- fields introduced by the geometry of large-scale motions, i.e. $T_{CI}(q_C)$ by that is associated with multiport displacement-modulated transformers

Forces and velocities (f_C, v_I) can be eliminated which gives:

$$\dot{q}_C = T_{CI}(q_C)\Phi_I^{-1}(p_I), \qquad \dot{p}_I = -T_{CI}^T(q_C)\Phi_C(q_C)$$

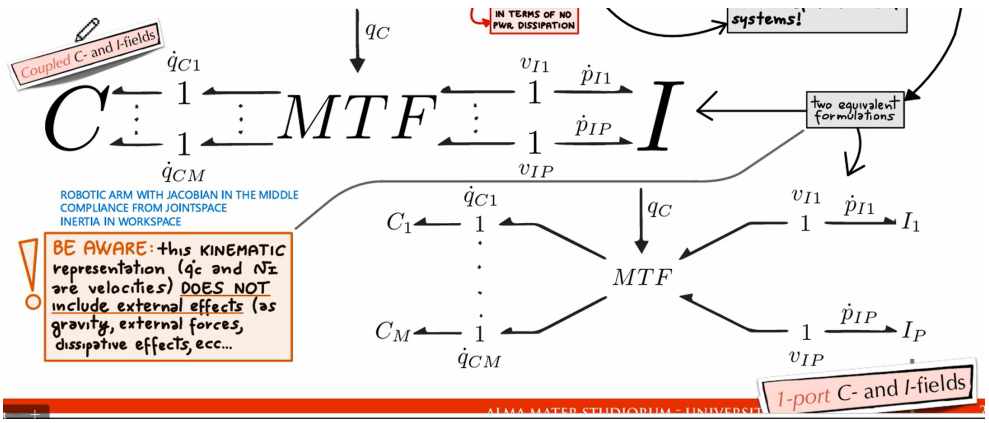


Figure 6.13: Basic nonlinear conservative system

Here is an example with the spring-pendulum. Notice that it's possible to add external forces.

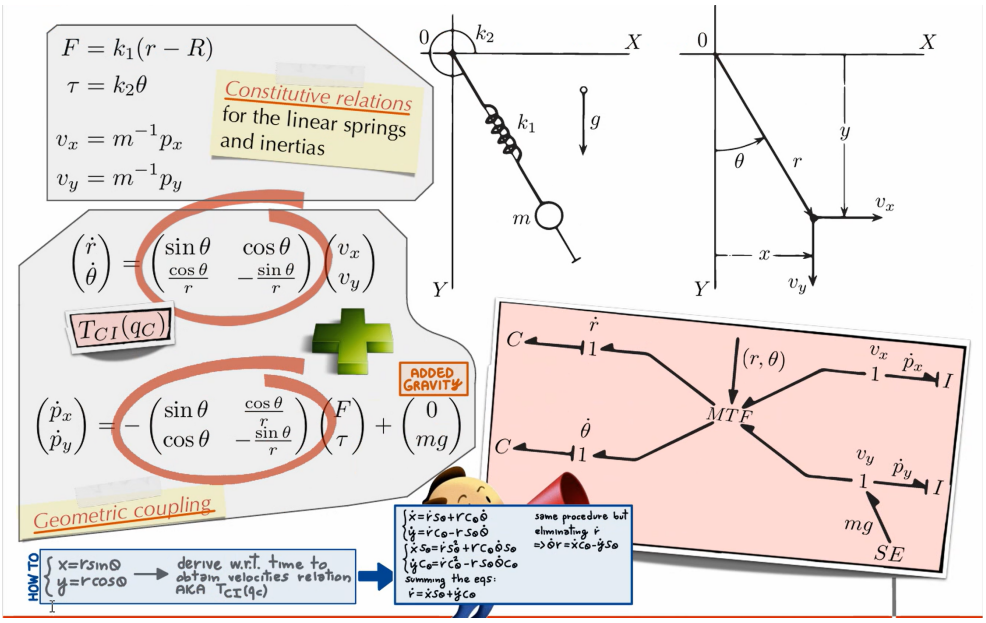


Figure 6.14: Spring pendulum example

This is a conservative system, so there are no damping terms.

6.5 The K.A.M. (Kinematic Analysis Method)

Before addressing closed loops and derivative causality, it is useful to formalize how to build a mechanism's bond graph starting purely from kinematics.

1. **Identify key variables**: Determine the vector of generalized coordinates $\mathbf{q}_k$, the C-field displacements $\mathbf{q}_C$, and the I-field velocities $\mathbf{v}_I$. Note that this method works with velocities (flows).

   (a) If the vector containing the time derivatives of the generalised coordinates q_k represents a necessary and sufficient set of velocities to fix all the motions at any instant, then the system is holonomic

2. **C-field constraints**: Relate $\mathbf{q}_C$ to $\mathbf{q}_k$ and differentiate to get $\dot{\mathbf{q}}_C = \mathbf{T}_{Ck}(\mathbf{q}_k)\dot{\mathbf{q}}_k$. Write this into the bond graph using MTFs and 0-junctions.

3. **I-field constraints**: Obtain the velocity transformation $\mathbf{v}_I = \mathbf{T}_{Ik}(\mathbf{q}_k)\dot{\mathbf{q}}_k$ and append to the graph via MTFs.

4. **Attach elements**: Append C- and I-fields to their respective junctions. Add dissipation, gravity (as SE), and simplify the MTFs where possible.

This kinematic description uses a junction structure that links flows (assumed to depend on C-energy variables) to couple C-fields and I-fields. However, this procedure often leads to bond graphs with algebraic loops or derivative causality when bodies are rigidly constrained.

The procedure is often successful at deriving a bond graph what will yield state equations easily but it can lead to algebraic loops or with derivative causality. A serious problem occurs when two rigid bodies are constrained in a non-flexible way.

6.6 Introduction to DAEs vs ODEs

In standard Ordinary Differential Equations (ODEs), the system states are defined by explicit derivatives (e.g., accelerations), which a numerical solver integrates to find velocities and positions. A Differential-Algebraic Equation (DAE) introduces algebraic constraints alongside differential equations. The solver must satisfy both the integration steps and the strict algebraic rules simultaneously.

6.7 Major Problems with DAEs

6.7.1 The Index Problem (Hidden Derivatives)

Standard solvers integrate derivatives, meaning they cannot directly handle pure algebraic equations (which lack time derivatives). To solve the system, the algebraic constraint must be differentiated with respect to time until the required derivatives appear. The number of differentiations required is called the **Differential Index**. Most rigid multibody systems are Index-3 DAEs, which are mathematically heavy and unstable to solve directly.

6.7.2 Constraint Drift

When an algebraic constraint (e.g., position) is differentiated into a velocity or acceleration constraint, the solver ensures only the differentiated constraint is met. Due to tiny numerical truncation errors during integration, the uncorrected position variables will slowly diverge from the original algebraic constraint, causing the physical system to literally drift apart.

6.8 Classic Example: The Rigid Pendulum

Consider a pendulum with mass m attached to a rigid, massless rod of length L. Modeled in Cartesian coordinates (x, y), this forms a DAE.

1. Differential Equations (Newton's Laws): Using λ for the unknown tension magnitude:

$$m\ddot{x} = -x\lambda \; m\ddot{y} \qquad\qquad = -y\lambda - mg \tag{6.18}$$

2. Algebraic Equation (The Constraint): The rigid rod forces the mass to remain on a circle of radius L:

$$x^2 + y^2 - L^2 = 0 \tag{6.19}$$

Solving and Drift: Since λ does not appear in the constraint, we must differentiate the constraint twice (Index-3) to find the accelerations:

$$2x\ddot{x} + 2\dot{x}^2 + 2y\ddot{y} + 2\dot{y}^2 = 0 \tag{6.20}$$

Substituting Newton's laws into this allows us to solve for λ. However, because the solver is now strictly enforcing the *acceleration* constraint, microscopic integration errors accumulate over time. The solver forgets the original $x^2 + y^2 = L^2$ rule, leading to constraint drift where the rigid rod slowly stretches and breaks apart in the simulation.

Constraints might be enforced by a joint between the bodies. The degrees of freedom are reduced by the constraints and this always lead to derivative causality.

6.9 Multi-body problem and constraints

When two rigid bodies are rigidly connected (e.g. a robotic link assembly), the degrees of freedom are reduced by the constraints. A rigid body requires six I-elements, but if rigidly connected, their velocities cannot all be independent. The naive bond graph has *velocity constraints* that force some storage elements into derivative causality, signalling an inconsistency in the model. This is the classic **Multi-Body Problem**.

6.9.1 The "Gaps Method" and Lagrange Multipliers (DAE formulation)

If we treat the constraints mathematically, we define "gap variables" $\boldsymbol{\delta}$ (e.g., horizontal and vertical distances between two theoretically pinned joints) which must vanish ($\boldsymbol{\delta} = \mathbf{0}$). The forces required to enforce these gaps are **Lagrange Multipliers** ($\boldsymbol{\lambda}$). The system is written in a *descriptor form* (a Differential-Algebraic Equation, DAE system), where $E\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ and matrix E is singular. The Jacobian of the constraints $\boldsymbol{\Delta} = \partial\boldsymbol{\delta}/\partial\mathbf{q}_k$ relates the velocities ($\dot{\boldsymbol{\delta}} = \boldsymbol{\Delta}\dot{\mathbf{q}}_k$) and the constraint forces ($\mathbf{F}_e = \boldsymbol{\Delta}^T\boldsymbol{\lambda}$). In the bond graph, these appear as external forces opposing the natural motion to keep $\dot{\boldsymbol{\delta}} = 0$.

6.9.2 Handling Derivative Causality Physically and Algebraically

To resolve the derivative causality natively in the simulation, two distinct tricks are widely used:

Method 1: The "Parasitic Spring" (Breaking the Constraint). This is a highly practical engineering approach. Physically, two perfectly rigid bodies connected by a rigid joint represent an over-constrained idealisation. In reality, nothing is perfectly rigid; everything flexes. The resolution is to introduce a small **compliance** at the connection. Rather than a rigid bond forcing equal velocities via a strict 1-junction, the joint is modelled as a stiff spring (C-element with large k) and a small damper (R-element) between them. This:

1. Breaks the strict kinematic constraint, restoring all C- and I-elements to integral causality because the masses can now vibrate at slightly different speeds.
2. Introduces a fast time scale (stiff ODE) associated with the high-frequency compliance vibration. Stiff ODE solvers are required.
3. Is physically correct — real connections always have some finite stiffness.

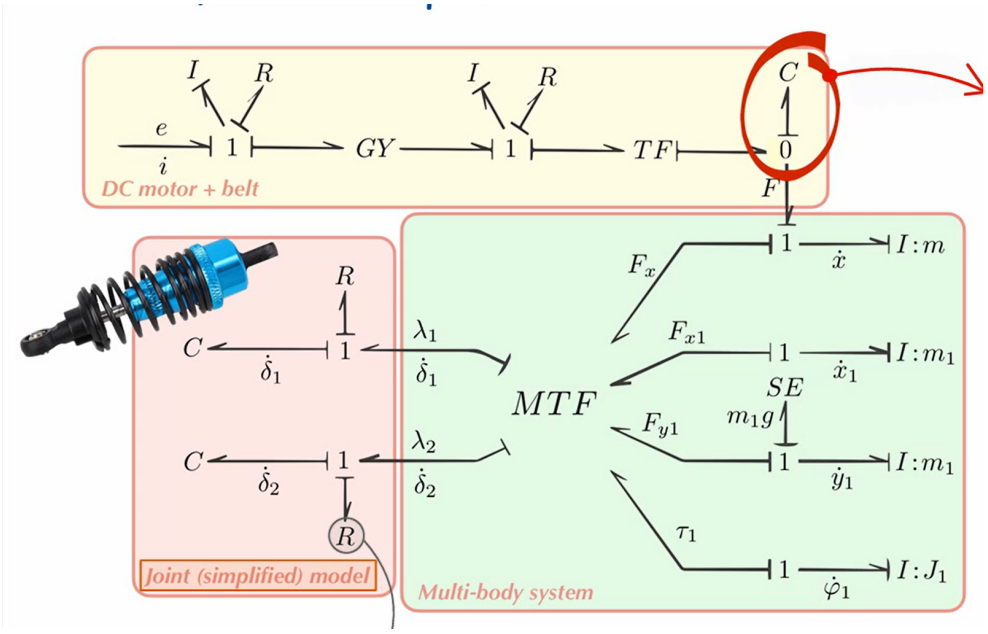


Figure 6.15: Example of RC simplified model for joint

6.10 Lagrangian mechanics and IC-fields (D.A.M.)

The combination of I-fields (kinetic energy) and C-fields (potential energy) in a bond graph with a junction structure encoding kinematic constraints is exactly equivalent to a Lagrangian description.

If we have a holonomic, lagrange-regular mechanical system, the kinetic energy is a quadratic-affine form: $T = \frac{1}{2}\dot{\mathbf{q}}_k^T \mathbf{M}(\mathbf{q}_k)\dot{\mathbf{q}}_k + \dot{\mathbf{q}}_k^T \mathbf{a}(\mathbf{q}_k)$. If $\mathbf{q}_k$ are generalised coordinates, then the Lagrange equations:

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = E_i \tag{6.21}$$

(where E_i includes potential energy derivatives and external forces) are automatically encoded by the bond graph. The IC-field representation makes this connection explicit and is particularly useful for systems with many degrees of freedom where writing Lagrange equations by hand is error-prone.

It's possible to write the Lagrangian Equations into Hamiltonian equations by the following formulas:

$$p_i = \frac{\partial T}{\dot{q}_i} \tag{6.22}$$

$$\dot{p}_i = \frac{\partial T}{q_i} + E_i \tag{6.23}$$

$$\tag{6.24}$$

where E_i are the generalized forces

The first relation has always a special form:

$$p_k = M(q_k, t)\dot{q}_k + a(q_k, t)$$

Therefore we can write:

$$\dot{q}_k = M^{-1}(q_k, t)[p_k - a(q_k, t)]$$

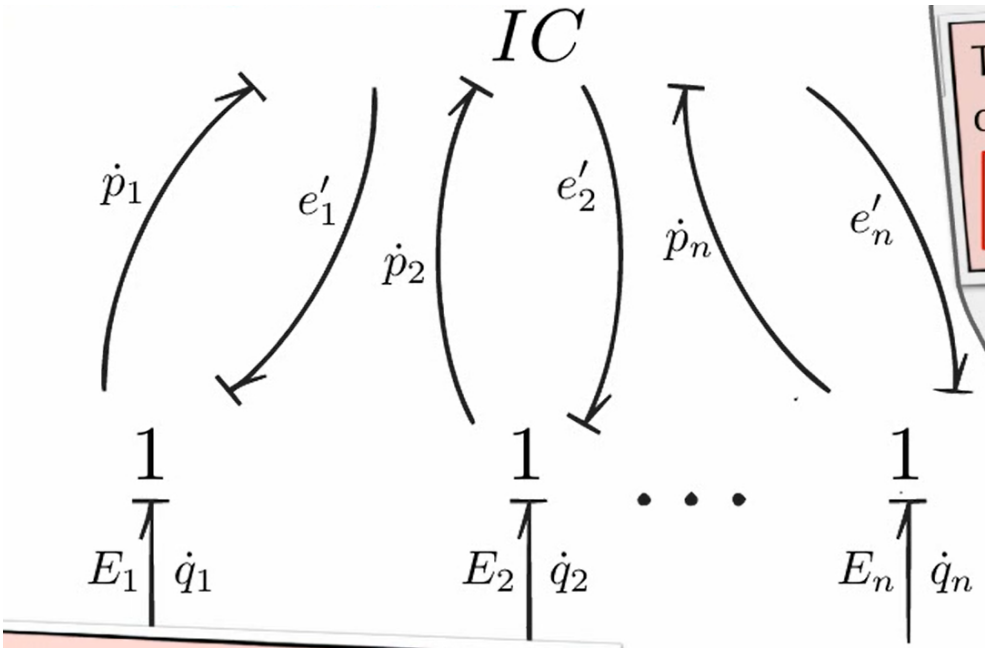


Figure 6.16: IC field representing the Hamiltonian form of Lagrangian equations

The Hamiltonian Form and the D.A.M. The **Dynamic Analysis Method (D.A.M.)** replaces the kinematic junction structure with a true dynamic IC-field based on Hamilton's equations. The Hamiltonian is defined as $\mathcal{H} = \mathbf{p}_k^T \dot{\mathbf{q}}_k - \mathcal{L}$. By defining generalised momenta $\mathbf{p}_k = \partial T / \partial \dot{\mathbf{q}}_k$, we can invert the mass matrix to find velocities:

In the bond graph, the IC-field acts with an I-part computing $\dot{\mathbf{q}}_k$ (flows) from $\mathbf{p}_k$ (momenta), and a C-part computing the internal efforts $\mathbf{e}'_k$ from $\mathbf{q}_k$ (displacements).

The *main advantage* of this Hamiltonian IC-field approach is that it mathematically guarantees integral causality everywhere, avoiding algebraic loops entirely. The *major drawback* is that the configuration-dependent mass matrix $\mathbf{M}(\mathbf{q}_k, t)$ must be non-singular and continuously inverted during the simulation.

Chapter 7

Models of Distributed Parameter Systems

All models considered so far are **lumped-parameter models**: the state is a finite-dimensional vector, components are point-like, and the number of degrees of freedom is finite. Real structures, however, have **continuously distributed** inertia, compliance, and damping — a vibrating rod, a flexible beam, a hydraulic line, a heat-conducting bar. These are described by **partial differential equations** (PDEs), and the number of degrees of freedom is in principle infinite. This chapter develops two strategies for obtaining tractable finite-dimensional bond-graph models: *simple lumping* and *separation of variables (normal modes)*.

7.1 Distributed-parameter systems

A **distributed-parameter system** is represented by a PDE: it cannot be accurately approximated by a lumped-parameter model without significant loss of information. The overarching philosophy is the same that governs all engineering modelling: use the *simplest model* that captures the behaviour of interest, but resist oversimplification. In the memorable words often attributed to Einstein, *"Everything should be made as simple as possible, but not simpler."*

So far it has been assumed that systems are built from components exhibiting inertia-like, compliance-like, or resistance-like effects in isolated, point-like fashion: a mass is perfectly rigid (no finite stiffness), and a spring is perfectly compliant (no mass). Of course, all real materials simultaneously exhibit inertial and compliance effects; it is our modelling *assumptions* that justify separating these effects into specific lumps. When those assumptions break down — for example when excitation frequencies become comparable to the first natural frequency of the distributed body — a distributed-parameter description is unavoidable.

7.2 Simple lumping techniques

7.2.1 Derivation of the wave equation for a bar

The simplest and most physically transparent route to a PDE model is to start with **spatially distributed finite lumps** and then take the limit as the lump size becomes infinitesimal. We illustrate this approach with the canonical example: the **longitudinal vibrations of a uniform elastic bar**.

Consider a small element of the bar located between x and $x + \Delta x$. Applying **Newton's second law** to this element gives:

$$A\big[\sigma(x+\Delta x) - \sigma(x)\big] = \rho A\, \Delta x\, \frac{\partial^2 \xi}{\partial t^2}(x), \tag{7.1}$$

where $\sigma(x)$ is the axial stress and the right-hand side is the inertia force of the element. The **definition of**

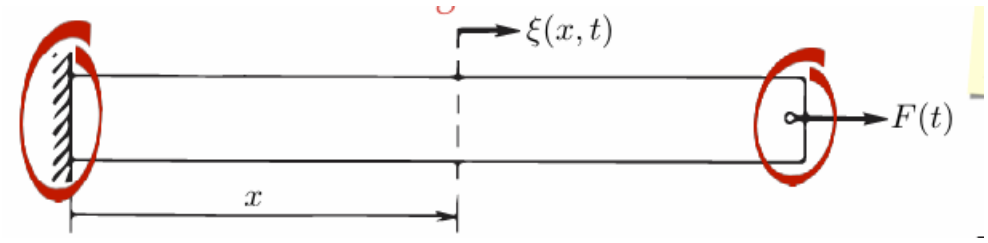


Figure 7.1: Elastic bar of length L, cross-sectional area A, and density ρ, clamped at $x = 0$ and subjected to an axial force $F(t)$

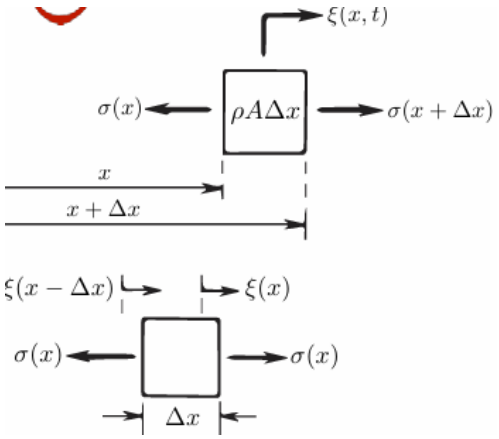


Figure 7.2: Image of a single Lump

linear strain and Hooke's law connect the stress to the displacement gradient:

$$\sigma(x) = E\,\frac{\xi(x) - \xi(x-\Delta x)}{\Delta x}, \tag{7.2}$$

where E is Young's modulus. Substituting (7.2) into (7.1) and **taking the limit** $\Delta x \to 0$ one obtains the **wave equation**:

$$\boxed{E\,\frac{\partial^2 \xi}{\partial x^2} = \rho\,\frac{\partial^2 \xi}{\partial t^2}}. \tag{7.3}$$

Equivalently, defining the stress as $\sigma = E\,\partial\xi/\partial x$, the two-field form reads:

$$A\,\frac{\partial \sigma}{\partial x} = \rho A\,\frac{\partial^2 \xi}{\partial t^2}, \qquad \sigma = E\,\frac{\partial \xi}{\partial x}. \tag{7.4}$$

7.2.2 The lumped approximation and its bond graph

Instead of taking the limit, one can **retain finite lumps** and construct a discrete approximation. Dividing the bar into n elements of length $\Delta x = L/n$, the state variables for the i-th lump are:

$$p_i = \rho A\,\Delta x\,\dot{\xi}_i \quad \text{(momentum)}, \qquad q_i = \xi_i - \xi_{i-1} \quad \text{(relative elongation)}. \tag{7.5}$$

The discrete Newton and Hooke laws are:

$$\frac{\mathrm{d}p_i}{\mathrm{d}t} = \frac{EA}{\Delta x}\big(q_{i+1} - q_i\big), \qquad \frac{\mathrm{d}q_i}{\mathrm{d}t} = \frac{1}{\rho A\,\Delta x}\big(p_i - p_{i-1}\big), \tag{7.6}$$

with the auxiliary relations:

$$F_i = \frac{EA}{\Delta x}\,q_i, \qquad A\sigma = F, \qquad p_i = \rho A\,\Delta x\,\dot{\xi}_i. \tag{7.7}$$

In bond-graph language each I-element represents the momentum of lump i and each C-element represents its elastic compliance. The 1-junctions impose Newton's law (common velocity = $\dot{\xi}_i$), while the 0-junctions impose Hooke's law (common force = F_i).

7.2.3 Boundary elements and admissible configurations

The first element ($i = 1$) and the n-th element require special treatment since they are **boundary elements**: they are not surrounded by lumps on both sides. The modified boundary equations are:

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = \frac{p_1}{\rho A\,\Delta x}, \qquad \frac{\mathrm{d}p_n}{\mathrm{d}t} = F(t) - \frac{EA}{\Delta x}\,q_n. \tag{7.8}$$

To use this approach one first selects the **number of lumps** n; this choice then sets the I and C parameters. The number of lumps should be chosen based on accuracy criteria, but it is generally not possible to determine *a priori* the rate of convergence to the continuous model, especially when the discrete model is embedded in a larger system.

The causal structure of the overall bond graph dictates which of the following configurations is appropriate at each boundary junction:

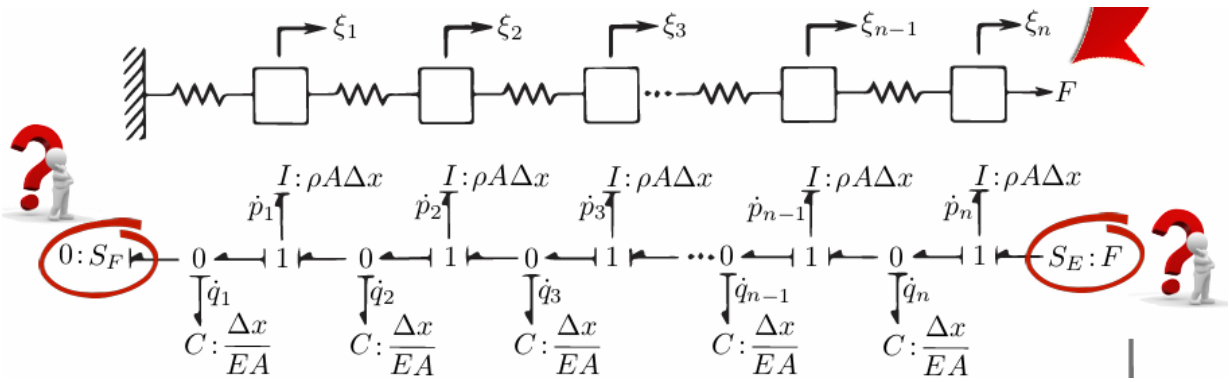


Figure 7.3: bond graph for the n-lump bar

7.2.4 Summary and limitations of simple lumping

Taking the limit $n \to \infty$ in the discrete model recovers the wave equation (7.3). External forcing or active boundary conditions can be incorporated in several ways. For a point force $F(t)$ at $x = L$, the PDE reads:

$$\frac{1}{A} F(t)\, \delta(x - L) + E \frac{\partial^2 \xi}{\partial x^2} = \rho \frac{\partial^2 \xi}{\partial t^2}, \tag{7.9}$$

where $\delta(\cdot)$ is the Dirac delta distribution representing a **distributed force per unit length** concentrated at the boundary.

Two representations are therefore available for the bar:

- **Finite-lump representation.** Physically transparent and easily embedded in an overall system bond graph. Non-linear effects pose no particular problems. However, the state space grows very quickly with n while the accuracy does not: each additional lump adds a new, very inaccurate high-frequency mode while improving the existing low-frequency predictions only mildly. The resulting large disparity in time constants can create severe stiffness problems for numerical integration.
- **(Linear) PDE representation.** Very accurate; admits analytical solutions. However, it is difficult to incorporate into an overall system model, because non-linear effects introduce serious complications and coupling the PDE to the remainder of the system via boundary conditions is non-trivial.

Key Insight: The way forward: normal modes

The PDE representation can be manipulated using *normal mode techniques* to yield an **accurate and computationally efficient finite-order model**. This is the subject of the rest of this chapter. The key idea is that the continuous bar can oscillate in an infinite set of orthogonal *mode shapes*, and any motion — forced or unforced — can be expressed as a linear combination of those shapes. Truncating the series at N modes gives an exact model up to the N-th natural frequency.

Watch out: Use simple lumping with caution

The simple lumping approach *must be used cautiously* and with awareness on the part of the modeller. Even for seemingly low-frequency excitations a large number of lumps may be required, and convergence is hard to predict a priori. Moreover, the high-frequency modes introduced by each additional lump are highly inaccurate and may corrupt both the numerical integration and the interaction with the rest of the system.

7.3 Lumped models via separation of variables

7.3.1 Motivation and overview

Many physical elements possess a **continuous distribution of inertia and compliance** properties. When linearity can be assumed, these distributed elements can be represented by linear PDEs in space and time. The actual dynamics of these distributed systems are composed of **propagating waves** that reflect from boundaries and interfere to produce the observed response at any point and time. This wave-superposition property can be formalised and exploited for modelling purposes via the **separation of variables** technique.

The outcome of this procedure is a **lumped-parameter bond-graph representation** of the distributed element which can be seamlessly coupled to bond-graph models of the rest of the system. Crucially:

- The method works for *general linear PDEs*, not only for wave-like systems.
- It provides a spectrally exact representation up to the retained number of modes.
- The resulting bond graph has all storage elements in integral causality, enabling efficient numerical simulation.

7.3.2 Separation of variables: the unforced bar

Consider again the longitudinal vibration of a bar governed by equation (7.9). The method of separation of variables assumes first that the external force is zero, $F(t) = 0$, and that the displacement can be written as a **product of a purely spatial and a purely temporal function**:

$$\xi(x,t) = Y(x)\, f(t). \tag{7.10}$$

This is the central (and daring) ansatz: it asserts that space and time dependences factorise. Substituting (7.10) into the unforced wave equation gives:

$$\rho\, Y(x)\, \frac{\mathrm{d}^2 f}{\mathrm{d}t^2} - E\, f(t)\, \frac{\mathrm{d}^2 Y}{\mathrm{d}x^2} = 0. \tag{7.11}$$

Dividing through by $Y(x)f(t)$:

$$\frac{1}{f}\,\frac{\mathrm{d}^2 f}{\mathrm{d}t^2} = \frac{E}{\rho}\,\frac{1}{Y}\,\frac{\mathrm{d}^2 Y}{\mathrm{d}x^2} = -\omega^2, \qquad \omega \in \mathbb{C}. \tag{7.12}$$

The left side depends only on t and the right side depends only on x; hence both must equal the **same constant**, $-\omega^2$. This yields **two decoupled ODEs**:

$$\frac{\mathrm{d}^2 f}{\mathrm{d}t^2} + \omega^2\, f = 0, \tag{7.13}$$

$$\frac{\mathrm{d}^2 Y}{\mathrm{d}x^2} + \frac{\rho}{E}\,\omega^2\, Y = 0, \tag{7.14}$$

which must be solved once proper boundary and initial conditions are provided. The temporal ODE (7.13) is a simple harmonic oscillator: once ω is known, $f(t) = C_1 \cos\omega t + C_2 \sin\omega t$. The spatial ODE (7.14) determines the *allowable* values of ω.

7.3.3 Eigenvalue problem and mode shapes

The general solution of the spatial ODE (7.14) is:

$$Y(x) = A\cos kx + B\sin kx, \qquad k^2 = \frac{\rho}{E}\,\omega^2, \tag{7.15}$$

where the constants A and B and the wave number k depend on the boundary conditions. For the bar **clamped at** $x = 0$ **and free at** $x = L$:

- At $x = 0$ the displacement vanishes: $\xi(0,t) = Y(0)\, f(t) = 0 \;\Rightarrow\; Y(0) = 0$.
- At $x = L$ the stress $A\sigma(L,t) = EA\,\partial\xi/\partial x\,(L,t) = F(t)$ balances the applied force. In the *unforced* case, $F(t) = 0$, this reduces to: $\partial\xi/\partial x\,(L,t) = \mathrm{d}Y/\mathrm{d}x\,(L)\, f(t) = 0 \;\Rightarrow\; Y'(L) = 0$.

Applying $Y(0) = 0$ to (7.15) gives $A = 0$, so $Y(x) = B\sin kx$. Applying $Y'(L) = 0$:

$$k\, B\cos kL = 0. \tag{7.16}$$

The non-trivial solution ($B \neq 0$) requires:

$$\cos kL = 0 \quad\Longrightarrow\quad k_n L = (2n-1)\,\frac{\pi}{2}, \qquad n = 1, 2, 3, \ldots \tag{7.17}$$

This is the **frequency equation**, entirely analogous to the characteristic equation for the eigenvalues of a lumped system. The corresponding **natural frequencies** are:

$$\omega_n^2 = \frac{E}{\rho}\,k_n^2 = \frac{E}{\rho}\,\frac{(k_n L)^2}{L^2} \quad\Longleftrightarrow\quad \omega_n = \sqrt{\frac{E}{\rho}}\,\frac{2n-1}{L}\,\frac{\pi}{2}, \qquad n = 1, 2, 3, \ldots \tag{7.18}$$

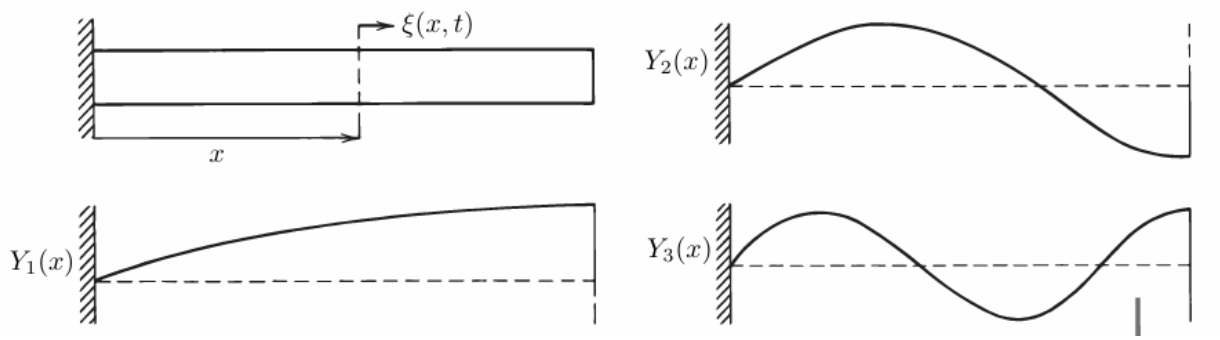


Figure 7.4: First three mode shapes $Y_1(x)$, $Y_2(x)$, $Y_3(x)$

and the corresponding **eigenfunctions** (or **mode shapes**), with normalisation $B_n = 1$, are:

$$Y_n(x) = \sin\!\left(k_n L\,\frac{x}{L}\right) = \sin\!\left((2n-1)\,\frac{\pi}{2}\,\frac{x}{L}\right), \qquad n = 1, 2, 3, \ldots \tag{7.19}$$

If the bar is displaced in the shape of mode n and released, it oscillates purely in that **modal configuration** at the corresponding modal frequency ω_n. Any motion of the bar, from *any* initial conditions, is a **linear combination** of the mode shapes oscillating at their respective natural frequencies.

7.3.4 Orthogonality of mode shapes

The mode shapes are **orthogonal** with respect to the mass distribution:

$$\int_0^L Y_n(x)\,Y_m(x)\,\mathrm{d}x = 0, \qquad n \neq m. \tag{7.20}$$

For the specific shapes (7.19) this evaluates to:

$$\int_0^L \sin\!\left((2n-1)\,\frac{\pi}{2}\,\frac{x}{L}\right)\sin\!\left((2m-1)\,\frac{\pi}{2}\,\frac{x}{L}\right)\mathrm{d}x = \begin{cases} 0, & n \neq m, \\ L/2, & n = m. \end{cases} \tag{7.21}$$

This orthogonality property is the cornerstone of modal analysis: it will allow us to *decouple* the modal equations when the forced case is treated.

7.3.5 Forced response and modal decomposition

We are ultimately interested in the **forced response** of the bar. We now exploit the fact that the response of the bar, whether forced or unforced, can *always* be expressed as a linear combination of the mode shapes. Starting from the forced PDE (7.9), we write:

$$\xi(x,t) = \sum_{n=1}^{\infty} Y_n(x)\,\eta_n(t), \tag{7.22}$$

where $\eta_n(t)$ are the **modal coordinates** (time-dependent amplitudes to be determined). Substituting (7.22) into the forced PDE:

$$\sum_n \rho A\,Y_n\,\ddot{\eta}_n - \sum_n AE\,\frac{\mathrm{d}^2 Y_n}{\mathrm{d}x^2}\,\eta_n = F(t)\,\delta(x-L). \tag{7.23}$$

Using the fact that the mode shapes satisfy $\mathrm{d}^2Y_n/\mathrm{d}x^2 = -(\rho/E)\,\omega_n^2\,Y_n$, multiplying both sides by $Y_m(x)$ and integrating over $[0, L]$:

$$\sum_n \left(\int_0^L \rho A\,Y_n\,Y_m\,\mathrm{d}x\right)\ddot{\eta}_n + \sum_n \left(\int_0^L \rho A\,Y_n\,Y_m\,\mathrm{d}x\right)\omega_n^2\,\eta_n = \int_0^L F(t)\,\delta(x-L)\,Y_m\,\mathrm{d}x. \tag{7.24}$$

The orthogonality relation (7.21) eliminates all cross-terms ($n \neq m$), leaving one equation for each mode index m:

$$\left(\int_0^L \rho A\,Y_m^2\,\mathrm{d}x\right)\ddot{\eta}_m + \left(\int_0^L \rho A\,Y_m^2\,\mathrm{d}x\right)\omega_m^2\,\eta_m = \int_0^L F(t)\,\delta(x-L)\,Y_m\,\mathrm{d}x. \tag{7.25}$$

Modal mass, stiffness and forcing. The **modal mass** is:

$$m_m = \int_0^L \rho A\, Y_m^2 \,\mathrm{d}x = \frac{\rho A L}{2}. \tag{7.26}$$

For a point force $F(t)$ applied at $x = L$, the Dirac delta collapses the integral on the right-hand side:

$$\int_0^L F(t)\,\delta(x-L)\,Y_m(x)\,\mathrm{d}x = F(t)\,Y_m(L), \tag{7.27}$$

yielding the **modal forcing function** $F(t)\,Y_m(L)$ — the force modulated by the value of the mode shape at the forcing point.

The **decoupled modal equation** for mode m is therefore:

$$\boxed{m_m\,\ddot{\eta}_m + k_m\,\eta_m = F(t)\,Y_m(L)}, \qquad k_m = m_m\,\omega_m^2 \quad \text{(modal stiffness)}. \tag{7.28}$$

This is the equation of a **simple harmonic oscillator** with mass m_m and stiffness k_m, driven by a force proportional to $Y_m(L)$. Due to mode orthogonality, all equations (7.28) are *completely decoupled*: each $\eta_m(t)$ can be solved independently and the actual physical response is then reconstructed as:

$$\xi(x,t) = \sum_{m=1}^{\infty} Y_m(x)\,\eta_m(t). \tag{7.29}$$

In practice, only a **finite number of modes** can be retained in the simulation, and the truncation level is chosen based on the highest frequency of interest in the input.

7.3.6 Bond-graph representation of the modal model

To embed the modal model in a bond graph we introduce the **modal momentum** and **modal displacement**:

$$p_m = m_m\,\dot{\eta}_m, \qquad q_m = \eta_m. \tag{7.30}$$

Equation (7.28) then reads, in first-order form:

$$\dot{p}_m = -k_m\,q_m + F(t)\,Y_m(L), \qquad \dot{q}_m = \frac{p_m}{m_m}. \tag{7.31}$$

These are exactly the state equations of an **I-element** ($I = m_m$) and a **C-element** ($C = 1/k_m$) joined at a 1-junction, with the external force connected through a **transformer** of modulus $Y_m(L)$. The velocity at the forcing point $x = L$ is recovered from the bond-graph structure as:

$$\frac{\partial \xi}{\partial t}(L,t) = \sum_{m=1}^{\infty} Y_m(L)\,\dot{\eta}_m(t) = \sum_{m=1}^{\infty} Y_m(L)\,\frac{p_m}{m_m}, \tag{7.32}$$

which corresponds to a 0-junction collecting the TF outputs.

The construction above answers the question that naturally arises: *why build a bond graph when the PDE has already been "solved" analytically?* The answer is that the force $F(t)$ is in general **not** a specified input but rather *results from the interaction with some external sub-system*. The bond-graph modal model provides a **port** at $x = L$ with well-defined effort and flow variables, so that any external subsystem (a mass–spring–damper, an actuator, another flexible body) can be connected without further analysis.

7.3.7 Reading outputs and applying forces at interior locations

A fundamental strength of the modal bond-graph approach is that **any location** on the rod can be monitored as an output, and forces or moments can be applied at arbitrary interior points, without any structural modification to the modal model.

To read the velocity at an arbitrary interior location x_0, recall:

$$\frac{\partial \xi}{\partial t}(x_0,t) = \sum_{m=1}^{\infty} Y_m(x_0)\,\frac{p_m}{m_m}. \tag{7.33}$$

In bond-graph terms this is a new 0-junction connected to the m TF elements with moduli $Y_m(x_0)$, reading velocities from each modal I-element.

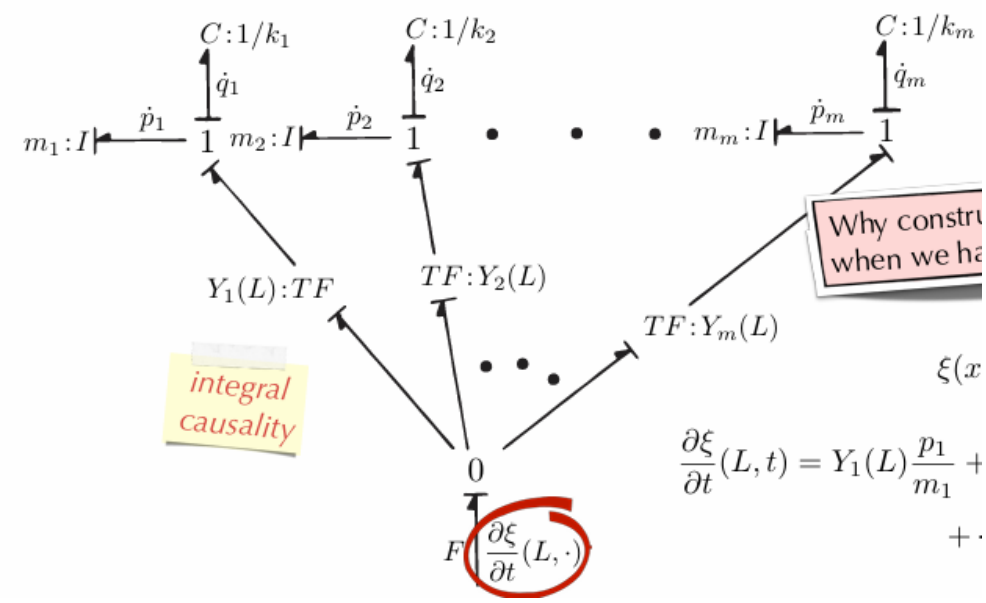


Figure 7.5: Bond-graph for the N-mode bar

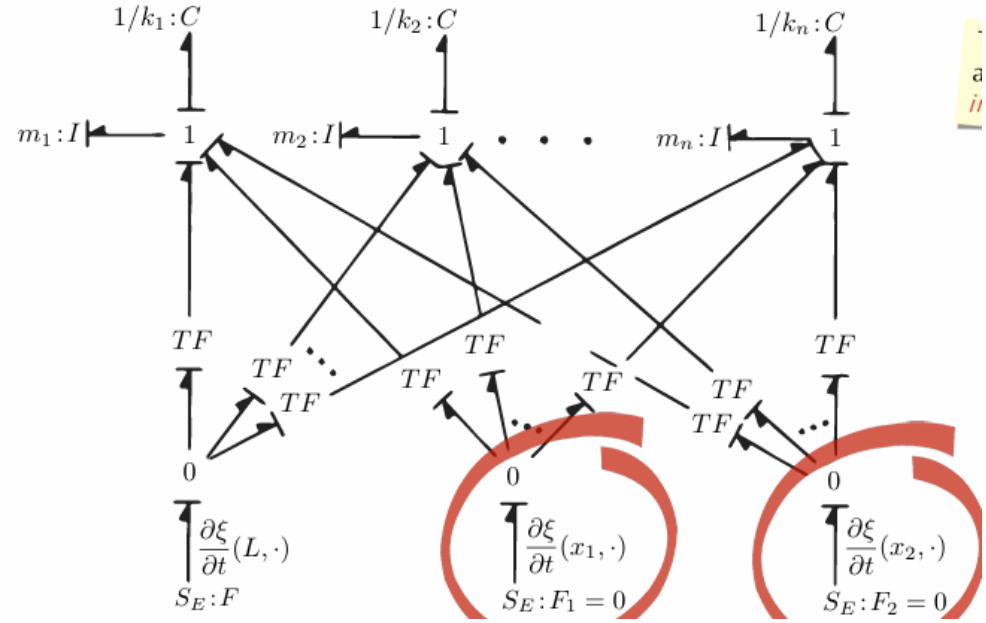


Figure 7.6: Rod with three interface points

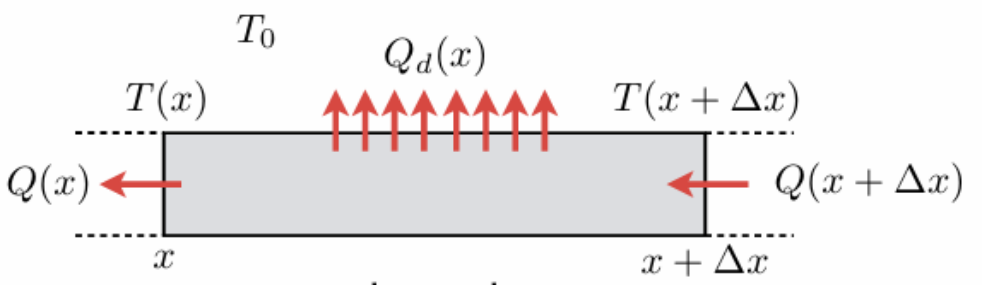


Figure 7.7: Schematic of the heated moving bar

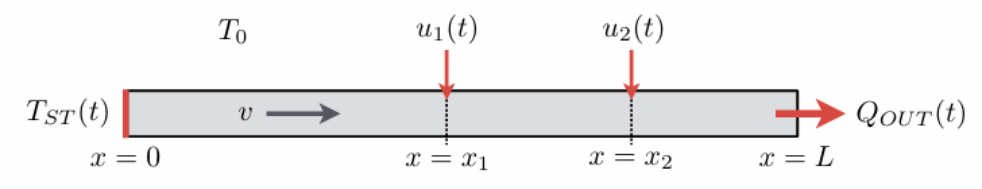


Figure 7.8: Thermal element between x and $x + \Delta x$

To **apply a force at an interior point** x_0, one introduces a new 0-junction at x_0 with its own S_E source (or interface), and adds to each mode m an additional TF of modulus $Y_m(x_0)$. Alternatively, if only output is desired (no power injected), the source is set to $S_E : 0$: this defines the location without disturbing the dynamics.

This "cross-coupling" of all modes to each interface point is visible in the bond graph as a set of TF elements that fan out from each modal 1-junction to all 0-junctions. The structure resembles a web: each node is connected to every other through the modal transformers. While the visual complexity grows with the number of modes and ports, the underlying mathematics remains fully decoupled at the modal level — the coupling only reappears at the physical output level through the linear combination of mode shapes.

Key Insight: Exploiting the zero-force trick

Defining each output point as a zero-force input also allows one to *act* on the beam at that location later, simply by replacing the zero source with a finite one. This provides a clean and modular way to change the actuation architecture without rebuilding the modal model.

7.4 Application: heat transport in a bar

7.4.1 Problem statement

As a second major application of the modal approach we consider the simulation of the **temperature distribution** $T(t, x)$ in a bar of length L moving at a constant transport speed ν_0 (material convection), with:

- External distributed heat sources $u_i(t)$ applied at discrete locations x_i inside the bar.
- Radiation/convection from the bar surface to the ambient temperature T_0 with heat-transfer coefficient h.
- **Time-varying boundary conditions**: a prescribed temperature $T_{ST}(t)$ at $x = 0$ (inlet) and an output heat flux $Q_{OUT}(t)$ at $x = L$ (outlet).

7.4.2 Derivation of the governing PDE

The PDE is obtained via lumping over a small element $[x, x + \Delta x]$, exactly as in the mechanical case, using the **pseudo-bond-graph framework** introduced in the module on thermal systems. The element is shown in the following figure.

For the **heat storage part**:

$$\rho c \, \Delta x \, \frac{\partial T}{\partial t}(t, x) = Q(t, x + \Delta x) - Q(t, x) - Q_d(t, x), \tag{7.34}$$

where ρ is the bar density and c its specific heat. The **heat flows** are:

$$\Delta x\, Q(t,x) = k\big[T(t,x) - T(t,x-\Delta x)\big], \tag{7.35}$$

$$Q_d(t,x) = h\,\Delta x\big[T(t,x) - T_0\big], \tag{7.36}$$

where k is the thermal conductivity and h is the surface heat-transfer coefficient.

Starting from the simplified case $Q_d = 0$ and letting $\Delta x \to 0$, one recovers the classical **heat equation**:

$$\rho c\,\frac{\partial T}{\partial t}(t,x) = \frac{\partial Q}{\partial x}(t,x), \qquad Q(t,x) = k\,\frac{\partial T}{\partial x}(t,x), \tag{7.37}$$

which combined gives:

$$\rho c\,\frac{\partial T}{\partial t}(t,x) = k\,\frac{\partial^2 T}{\partial x^2}(t,x) - h\big[T(t,x) - T_0\big]. \tag{7.38}$$

The full problem — including heat transport at velocity ν_0, distributed point heat sources $u_i(t)\,\delta(x-x_i)$, and radiation — is governed by:

$$\rho c\,\frac{\partial \tilde T}{\partial t}(t,x) = \alpha\,\frac{\partial^2 \tilde T}{\partial x^2}(t,x) - \nu_0\,\frac{\partial \tilde T}{\partial x}(t,x) - \beta\,\tilde T(t,x) + \sum_{i=1}^{m} \delta(x-x_i)\,\tilde u_i(t), \tag{7.39}$$

with boundary conditions:

$$\tilde T(t,0) = \tilde T_{ST}(t) \quad \text{(inlet temperature)}, \tag{7.40}$$

$$\alpha\left[\frac{\partial \tilde T}{\partial x}(t,L) + \gamma_2\,\tilde T(t,L)\right] = \tilde Q_{OUT}(t) \quad \text{(outlet flux)}, \tag{7.41}$$

where $\alpha = k/(\rho c)$ is the thermal diffusivity, $\beta = h/(\rho c)$ accounts for surface radiation, and γ_2 is a parameter defined below.

7.4.3 Coordinate transformation to remove transport and radiation

A key step is to **"remove"** the first-order transport term and the radiation term from the PDE, so that the remaining equation is in the standard diffusion form to which separation of variables applies. This is achieved by the **exponential change of coordinates**:

$$\tilde T(t,x) = \mathrm{e}^{\gamma_1 t + \gamma_2 x}\, w(t,x), \tag{7.42}$$

with the two parameters:

$$\gamma_2 = \frac{\nu_0}{2\alpha}, \qquad \gamma_1 = -\beta - \nu_0\gamma_2 + \alpha\gamma_2^2 = -\beta - \frac{\nu_0^2}{4\alpha}. \tag{7.43}$$

Substituting (7.42) into (7.39), the transport and radiation terms cancel and the new function $w(t,x)$ satisfies the **standard diffusion equation** with source terms:

$$\frac{\partial w}{\partial t}(t,x) = \alpha\,\frac{\partial^2 w}{\partial x^2}(t,x) + \sum_{i=1}^{m} \delta(x-x_i)\, v_i(t,x), \tag{7.44}$$

with transformed inputs:

$$v_i(t,x) = \mathrm{e}^{-(\gamma_1 t + \gamma_2 x)}\,\tilde u_i(t), \qquad i = 1,\dots,m, \tag{7.45}$$

and transformed boundary conditions:

$$w(t,0) = W_{ST}(t) = \mathrm{e}^{-\gamma_1 t}\,\tilde T_{ST}(t), \tag{7.46}$$

$$\alpha\left[\frac{\partial w}{\partial x}(t,L) + \gamma_2\, w(t,L)\right] = \alpha\, H_{OUT}(t) = \mathrm{e}^{-(\gamma_1 t + L\gamma_2)}\,\tilde Q_{OUT}(t). \tag{7.47}$$

The transformation (7.42) is *not* a change of independent coordinates: w and $\tilde T$ are evaluated at the same (t,x). It is rather a **dependent-variable scaling** that absorbs the exponential spatial and temporal decay introduced by transport and radiation.

7.4.4 Separation of variables for the heat equation

We now apply separation of variables to (7.44) in two steps.

Step 1: handling time-varying boundary conditions. Time-varying boundary conditions prevent a direct application of the method. Following a standard approach, we split w as:

$$z(t,x) = \bar{z}(t,x) + z_0(t) + x\, z_L(t), \tag{7.48}$$

where the lifting functions $z_0(t)$ and $x\, z_L(t)$ absorb the boundary data, so that $\bar{z}$ satisfies *homogeneous* boundary conditions:

$$\bar{z}(t,0) = 0, \qquad \frac{\partial \bar{z}}{\partial x}(t,L) = 0. \tag{7.49}$$

Note that this splitting is **not** a change of coordinates in the strict sense: it is an additive decomposition that simplifies the boundary value problem.

Step 2: modal expansion of the homogeneous problem. The homogeneous problem for $\bar{z}$:

$$\frac{\partial \bar{z}}{\partial t}(t,x) = \alpha\, \frac{\partial^2 \bar{z}}{\partial x^2}(t,x), \qquad \bar{z}(t,0) = 0, \quad \frac{\partial \bar{z}}{\partial x}(t,L) = 0, \tag{7.50}$$

has the same boundary conditions as the unforced bar (zero temperature at $x = 0$, zero flux at $x = L$). The **mode shapes** are therefore:

$$Z_n(x) = \sin\!\left[(2n+1)\,\frac{\pi}{2}\,\frac{x}{L}\right], \qquad n = 0, 1, 2, \ldots, \tag{7.51}$$

and the solution takes the form:

$$\bar{z}(t,x) = \sum_{n=0}^{\infty} Z_n(x)\, f_n(t). \tag{7.52}$$

7.4.5 Final modal equations for the heat problem

After substituting the expansion (7.52) into the full PDE (including the boundary-lifting terms, the transformed sources v_i, and the boundary output condition), and exploiting the orthogonality of the mode shapes, one arrives at the **coupled modal ODE system**:

$$w(t,x) = \sum_{n=0}^{\infty} Z_n(x)\, \eta_n'(t), \tag{7.53}$$

$$\begin{aligned} w_L(t) &= -\gamma_2\, w(t,L) + H_{OUT}(t) = -\frac{\nu_0}{2\alpha} \sum_{n=0}^{\infty} Z_n(L)\, \eta_n'(t) + H_{OUT}(t) \\ &= -\frac{\nu_0}{2\alpha} \sum_{\bar{n}=0}^{\infty} (-1)^{\bar{n}}\, \eta_{\bar{n}}'(t) + H_{OUT}(t), \end{aligned} \tag{7.54}$$

and the evolution equation for each modal amplitude:

$$\begin{aligned} \dot{\eta}_n'(t) = &-\frac{(2n+1)^2\pi^2\alpha}{4L^2}\, \eta_n'(t) - \frac{\nu_0}{L} \sum_{\bar{n}=0}^{\infty} (-1)^{(n+\bar{n})}\, \eta_{\bar{n}}'(t) + \frac{2}{L} \sum_{i=1}^{m} Z_n(x_i)\, v_i(t,x_i) \\ &+ \frac{(2n+1)\pi\alpha}{L^2}\, W_{ST}(t) + (-1)^n\, \frac{2\alpha}{L}\, H_{OUT}(t). \end{aligned} \tag{7.55}$$

Equation (7.55) is the central result of the heat-problem analysis. Let us examine its structure term by term:

- The first term, $-(2n+1)^2\pi^2\alpha/(4L^2)\ \eta_n'$, is the standard **modal decay** of the n-th diffusion mode: it comes from the spatial second derivative and gives the mode a characteristic relaxation time $\tau_n = 4L^2/[(2n+1)^2\pi^2\alpha]$. Higher modes decay faster.
- The second term, $-(\nu_0/L)\sum_{\bar{n}}(-1)^{(n+\bar{n})}\eta_{\bar{n}}'$, arises from the outlet flux boundary condition which, after the coordinate change, introduces a **coupling between all modes**. This is the price paid for non-zero transport: the modes are no longer fully decoupled.

- The third term, $\frac{2}{L}\sum_i Z_n(x_i)\, v_i(t, x_i)$, is the **modal forcing** from the distributed point-wise heat sources. The mode-shape value $Z_n(x_i)$ plays the same role as $Y_n(L)$ in the mechanical bar.
- The fourth term involves $W_{ST}(t)$: the **inlet temperature boundary condition** drives each mode with a gain proportional to $(2n + 1)$.
- The fifth term involves $H_{OUT}(t)$: the **outlet flux** drives mode n with an alternating sign $(-1)^n$.

Key Insight: Coupling introduced by transport

In the purely diffusive case ($\nu_0 = 0$, i.e. $\gamma_2 = 0$), the boundary condition at $x = L$ becomes simply $\partial w/\partial x = 0$ (zero flux), and the second term in (7.55) disappears. The modal equations become fully decoupled — each mode evolves independently, as in the mechanical bar. Transport introduces inter-modal coupling: physically, the advection of heat modifies the effective boundary condition in a way that mixes modes. The coupling is weighted by $(-1)^{(n+\bar{n})}$, so adjacent modes interact most strongly with alternating signs.

Watch out: Linearity is required

The entire separation-of-variables and modal-decomposition machinery applies **only to linear PDEs**. The coordinate transformation (7.42) is linear; the modal expansion (7.52) is linear; and the decoupling by orthogonality is linear. If the heat-transfer coefficient h depends on temperature, or if the transport velocity ν_0 is state-dependent, the approach breaks down and one must resort to finite-difference, finite-element, or other nonlinear numerical schemes.

7.5 Summary

Two complementary approaches for modelling distributed-parameter systems have been presented. Their salient properties are summarised below.

Simple (finite) lumping. A lumped-parameter model can be found for a distributed system by describing the behaviour of a small element; however, such a model may not be accurate without retaining a large number of small elements. The lumped-parameter representations have the advantage that they can be *easily combined* into an overall system model and non-linear effects pose no particular problems. However, the model has a **large state space** and can cause computational problems due to *disparities in the time scales* of the distributed portion of the model and the remainder of the system.

Continuum (PDE) models. Continuum models can yield insight into system behaviour when analytical solutions are available. However, it is difficult to incorporate a continuum model into an overall system model, mostly because of the difficulties in solving the PDEs for the continuum part subject to complicated boundary conditions, and the inability to handle non-linear effects analytically.

Modal (separation-of-variables) models. The modal approach bridges the two: it takes the PDE as its starting point (full accuracy of the continuum), applies separation of variables to decouple the spatial and temporal dependences, and produces a **finite-order lumped-parameter bond graph** that retains spectral accuracy up to the number of modes included. The resulting model:

1. Has all storage elements in *integral causality*.
2. Can be seamlessly coupled to bond-graph models of the rest of the system.
3. Allows forces, moments, or heat sources to be applied (and outputs to be read) at arbitrary locations by evaluating the mode shapes at those points.
4. Admits the addition of modal damping (R-elements on each modal junction).
5. Applies to *general linear PDEs* — wave equation, heat equation, beam equation — not only to wave-like systems.

The trade-off is that **linearity is mandatory**: non-linear distributed effects force a return to finite lumping or full numerical PDE solvers.

Chapter 8

Thermo-fluid Systems

Thermodynamics is the **universal science of physical processes**: every model constructed so far — mechanical, electrical, hydraulic, thermal — can be studied through its lens. When C- and I-fields are constrained to conserve energy, or R-fields are arranged to dissipate power, thermodynamic arguments are implicitly at work. Most dynamic models can be characterised as **isothermal models**, and much of engineering thermodynamics is concerned with changes between equilibrium states rather than with the transient path of change. When the path itself is relevant — when we care about irreversibility and entropy production — we enter the domain of **irreversible thermodynamics**, which involves genuinely non-equilibrium systems.

The modelling strategy adopted here rests on the assumption that physical extents and time scales are such that **local pseudo-equilibrium conditions** exist: at each instant every identifiable lump of substance can be treated as if it were in thermodynamic equilibrium, even if the overall system is not. Under this assumption it becomes possible to define meaningful state variables at every instant and to write first-order ODEs for their evolution.

This chapter develops two complementary threads: the *true bond-graph representation* of thermodynamic systems (which identifies temperature T as an effort and entropy rate $\dot{S}$ as a flow, so that $T\dot{S}$ is genuine power), and a critical re-examination of *fluid-dynamic systems*, including the lumped modelling of fluid inertia and compliance, Bernoulli's equation, and the challenges introduced by compressibility and mass exchange through a control volume.

8.1 Basic thermodynamics in true bond-graph form

8.1.1 Assumptions and notation

We begin with a set of simplifying assumptions that frame the entire discussion:

- **Closed system (Lagrangian viewpoint).** No mass passes through the boundary. The total mass m of the substance is fixed.
- **Pseudo-equilibrium.** All parts of the substance are at essentially identical conditions of pressure, temperature, and density, so that a single set of state variables describes the whole lump.
- **No motion, no electromagnetic or surface-tension forces.** Under these conditions a *pure substance* has only **two independent properties**; all others are given by the *equations of state*.

Extensive quantities (volume, internal energy, entropy) for a *unit mass* are written in lowercase: v (specific volume), u (specific internal energy), s (specific entropy). When a mass m of the substance is involved the corresponding total quantities are:

$$V = mv, \qquad U = mu, \qquad S = ms. \tag{8.1}$$

8.1.2 The Gibbs equation and the internal energy function

It is assumed that the **internal energy per unit mass** u is a function of the *specific volume* v and the *specific entropy* s:

$$u = u(v, s). \tag{8.2}$$

The **Gibbs equation** relates infinitesimal changes in u to changes in v and s:

$$\mathrm{d}u = T\,\mathrm{d}s - p\,\mathrm{d}v, \tag{8.3}$$

Figure 8.1: True bond-graph C-field for the pure substance

where T is the *thermodynamic temperature* (conjugate to entropy) and p is the *pressure* (conjugate to volume, with a sign because expansion does work on the surroundings). In rate form:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = T\,\frac{\mathrm{d}s}{\mathrm{d}t} - p\,\frac{\mathrm{d}v}{\mathrm{d}t}. \tag{8.4}$$

The first term ($T\dot{s}$) is associated with a flow of heat or, sometimes, with dissipative work; the second term ($p\dot{v}$) is associated with *reversible* mechanical power.

Comparing (8.3) with the total differential $\mathrm{d}u = (\partial u/\partial s)\,\mathrm{d}s + (\partial u/\partial v)\,\mathrm{d}v$ immediately identifies:

$$T = \frac{\partial u}{\partial s}, \qquad -p = \frac{\partial u}{\partial v}. \tag{8.5}$$

A useful consistency check is the **Maxwell reciprocal relation**, which follows from the symmetry of mixed second partial derivatives:

$$\left.\frac{\partial T}{\partial v}\right|_s = \frac{\partial^2 u}{\partial v\,\partial s} = \left.\frac{\partial(-p)}{\partial s}\right|_v. \tag{8.6}$$

8.1.3 The pure substance as a true bond-graph C-field

The constitutive laws of the pure substance may be represented by a **true bond-graph C-field**, where T and p are the *efforts* and s and v are the *displacements*:

The identification of T as an effort and s as a displacement is the crucial distinction between a **true bond graph** and the **pseudo-bond graph**: in the true formulation $T\dot{s}$ is genuine thermodynamic power, whereas in the pseudo-bond graph the product of conjugate variables is merely a useful analogy (heat flow $\dot{Q}$) and not a true power.

Under **integral causality** the C-field is expressed as:

$$T = T(s, v), \qquad p = p(s, v), \tag{8.7}$$

i.e. the efforts are computed from the stored displacements. For *mixed-* or *all-derivative causality* the constitutive laws must be switched around, expressing the displacements as functions of the efforts. This inversion is accomplished systematically through the **Legendre transformations** of the internal energy u.

8.1.4 Legendre transformations and causal patterns

Different causal patterns at the two ports of the C-field correspond to different Legendre transforms of u:

Enthalpy. Replace s (displacement) by p (effort) as independent variable:

$$h(s,p) = u + pv, \qquad \frac{\partial h}{\partial s} = T(s,p), \qquad \frac{\partial h}{\partial p} = v(s,p). \tag{8.8}$$

Here the T-port retains integral causality (T computed from s) while the p-port has derivative causality (p is now an input, v an output).

Helmholtz free energy. Replace s by T:

$$f(T,v) = u - Ts, \qquad \frac{\partial f}{\partial T} = -s(T,v), \qquad \frac{\partial f}{\partial v} = -p(T,v). \tag{8.9}$$

Gibbs free energy. Replace both s by T and v by p:

$$\phi(T,p) = u + pv - Ts, \qquad \frac{\partial \phi}{\partial T} = -s(T,p), \qquad \frac{\partial \phi}{\partial p} = v(T,p). \tag{8.10}$$

Key Insight: Why use state functions?

Constitutive laws can in principle be written without reference to any state function. The danger is that *arbitrary* constitutive laws would not, in general, allow the existence of an internal energy function: a substance with such laws could, in a cycle, produce net energy and thus act as a *perpetual motion machine of the first kind*. Deriving constitutive laws by differentiating u, h, f, or ϕ is therefore always safer, since conservation of energy is *automatically* built into the relations. The C-field derived from a Legendre transform is guaranteed to be conservative.

8.1.5 The perfect gas: Helmholtz free energy route

A classical example of a constitutive law *not* stated in terms of state-function derivatives is the **perfect gas law** (per unit mass):

$$pv = RT, \tag{8.11}$$

where R is the specific gas constant and u is a function of temperature T only (i.e. it is independent of v). Equation (8.11) alone is *not* a complete characterisation of the C-field: it also admits only differential causality on the $T\dot{S}$ port, meaning it cannot be used in integral causality without supplementary information.

Since the natural causality has T and v as independent variables, it is appropriate to work with the **Helmholtz free energy**. Starting from $-p(T,v) = \partial f/\partial v = -RT/v$:

$$f(T,v) = -RT\ln v + \psi(T), \tag{8.12}$$

where $\psi(T)$ is an unknown function of temperature alone. The specific entropy follows as:

$$-s(T,v) = \frac{\partial f}{\partial T} = -R\ln v + \frac{\mathrm{d}\psi}{\mathrm{d}T}(T), \tag{8.13}$$

and the internal energy is recovered via $u = f + Ts$:

$$u(T,v) = f(T,v) + Ts = \psi(T) - T\frac{\mathrm{d}\psi}{\mathrm{d}T}(T). \tag{8.14}$$

Since u must be independent of v (a consequence of the perfect gas model), this expression is consistent: the volume dependence cancels. To determine $\psi(T)$ completely, additional thermal data are needed.

8.1.6 Complete equations of state: constant specific heats

To evaluate the complete equations of state it is necessary to express u in terms of s and v. This requires more information than just equation (8.11) because the gas is a **2-port C-field** and two independent constitutive relations are needed. The standard assumption is that the **specific heats are constant**:

$$c_p = \frac{\partial h}{\partial T} \quad \text{(specific heat at constant pressure)}, \qquad c_v = \frac{\partial u}{\partial T} \quad \text{(specific heat at constant volume)}. \tag{8.15}$$

Choosing the reference state (T_0, v_0, p_0) with $u = s = h = 0$:

$$u = c_v(T - T_0), \qquad h = c_p(T - T_0). \tag{8.16}$$

Taking the partial derivative with respect to T of the enthalpy relation $h = u + pv = u + RT$ yields $c_p = c_v + R$, the well-known **Mayer relation**.

From the Gibbs equation (8.3) written in terms of T and v:

$$\mathrm{d}s = \frac{\mathrm{d}u}{T} + \frac{p}{T}\,\mathrm{d}v = c_v\,\frac{\mathrm{d}T}{T} + R\,\frac{\mathrm{d}v}{v}, \tag{8.17}$$

which integrates to:

$$s = c_v \ln\frac{T}{T_0} + R\ln\frac{v}{v_0}. \tag{8.18}$$

Solving (8.18) for the temperature gives the first complete equation of state:

$$\boxed{(*) \quad T = T_0\,\mathrm{e}^{s/c_v}\left(\frac{v}{v_0}\right)^{-R/c_v}} \tag{8.19}$$

Further manipulations of the Gibbs equation written in terms of v and p yield:

$$\mathrm{d}s = c_p \frac{\mathrm{d}v}{v} + c_v \frac{\mathrm{d}p}{p} \quad \Longrightarrow \quad s = c_p \ln \frac{v}{v_0} + c_v \ln \frac{p}{p_0}, \tag{8.20}$$

which inverts to give the second complete equation of state:

$$\boxed{(**) \quad p = p_0 \, \mathrm{e}^{s/c_v} \left(\frac{v}{v_0} \right)^{-c_p/c_v}} \tag{8.21}$$

Equations (8.19) and (8.21) are the **complete relations for the C-field**. Substituting (8.19) back into $u = c_v(T - T_0)$ yields the internal energy as an explicit function of the state variables (s, v):

$$u(s, v) = c_v T_0 \left[\mathrm{e}^{s/c_v} \left(\frac{v}{v_0} \right)^{-R/c_v} - 1 \right]. \tag{8.22}$$

One can verify directly that $T = \partial u/\partial s$ and $-p = \partial u/\partial v$, confirming consistency with the Gibbs equation.

Watch out: The perfect gas law is not enough

Remember that $pv = RT$ is *not* a complete characterisation of the C-field. It also only allows differential causality on the $T\dot{S}$ port. The complete description requires both equations (8.19) and (8.21), and these in turn require knowledge of c_v in addition to R.

8.2 True bond graphs for heat transfer

8.2.1 The $T\dot{S}$ port and irreversibility

The $T\dot{S}$ port of the pure-substance C-field can be used to model all types of power flow *associated with entropy increase* — in other words, all irreversible effects. Changes in entropy of a pure substance can be accomplished by a variety of dissipative mechanisms, for example:

- Stirring with a paddle wheel.
- Heating with an electrical resistor immersed in a fluid.

In each case, the power dissipated can be identified with $T\dot{S}$ in order to find the change of state of the substance.

An important special use of $T\dot{S}$ ports is in modelling the effects of **heat flow**. If $\dot{Q}$ denotes the rate of heat transfer in power units, then:

$$\dot{Q} \;\leftrightarrow\; T\dot{S}. \tag{8.23}$$

This equivalence is exact in the reversible (quasi-static) limit and remains a good model for slow, near-equilibrium processes.

8.2.2 Two thermal reservoirs and the thermal resistor

Consider two **reservoirs of thermal energy** at temperatures T_1 and T_2 communicating through a thermal resistance only (the sides are perfectly insulated). Heat $\dot{Q}$ flows from the hotter to the cooler reservoir.

Since the same heat $\dot{Q}$ leaves reservoir 1 and enters reservoir 2, the power balance at each reservoir gives:

$$T_1\dot{S}_1 = \dot{Q} = T_2\dot{S}_2. \tag{8.24}$$

The **net entropy production rate** in the resistor is:

$$\dot{S}_2 - \dot{S}_1 = \frac{\dot{Q}}{T_2} - \frac{\dot{Q}}{T_1} = \dot{Q} \, \frac{T_1 - T_2}{T_1 T_2} \geq 0, \tag{8.25}$$

which is always non-negative (zero only when $T_1 = T_2$), confirming the second law. The entropy flow *leaving* the field is *greater* than the entropy flow *entering*, regardless of the direction of heat flow. This means the thermal resistor is neither a transformer nor a gyrator: it is an *entropy-generating* element whose entropy production is independent of the sign of the heat flow.

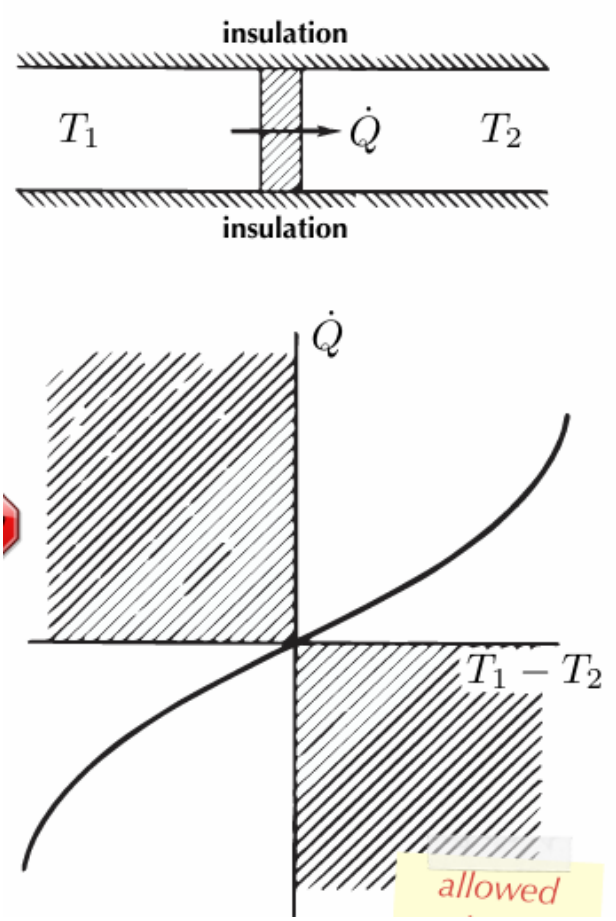


Figure 8.2: Two perfectly insulated thermal reservoirs at temperatures T_1 and T_2 ($T_1 > T_2$) connected by a thermal resistor

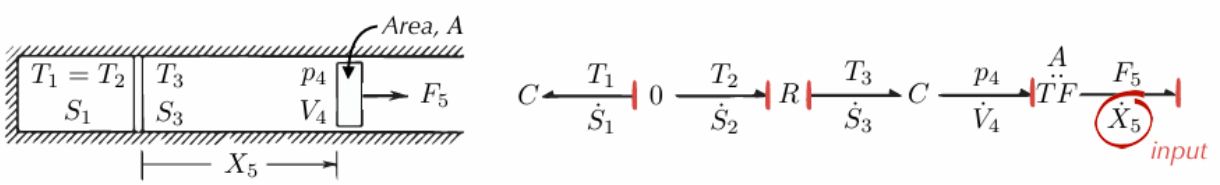


Figure 8.3: Two-chamber system

The thermal resistor is modelled as a **2-port R-field**:

$$\frac{T_1}{\dot{S}_1} \quad R \quad \frac{T_2}{\dot{S}_2} \tag{8.26}$$

This representation differs fundamentally from the pseudo-bond-graph treatment, where a single scalar resistance connected temperatures and heat flows. Here the two entropy-flow ports must be tracked separately because $\dot{S}_1 \neq \dot{S}_2$.

Key Insight: The electrical RC analogy has limits

It is tempting to model a thermal system exactly like an electrical RC network, replacing voltage by temperature and current by heat flow. This analogy is useful but *not* complete: in an electrical resistor the current entering equals the current leaving, while in a thermal resistor the entropy leaving exceeds the entropy entering. The analogy between electrical and thermal RC systems is therefore not as complete as is often assumed.

Constitutive law for the thermal resistor. A natural and physically consistent choice is:

$$\dot{Q} = H(T_1 - T_2), \tag{8.27}$$

where $H > 0$ is the **heat transfer coefficient**. The individual entropy flow rates follow from (8.24):

$$\dot{S}_1 = \frac{H(T_1 - T_2)}{T_1}, \qquad \dot{S}_2 = \frac{H(T_1 - T_2)}{T_2}. \tag{8.28}$$

The **only admissible causal form** has both temperatures (T_1 and T_2) as inputs and both entropy flows ($\dot{S}_1$ and $\dot{S}_2$) as outputs: we cannot impose arbitrary entropy flow rates, because the constitutive relation (8.28) must hold and temperatures can never be negative.

8.2.3 Two compressible chambers: a worked example

As a worked example, consider a system incorporating **two chambers containing compressible fluids**:

- One chamber has *constant volume* (the gas is heated or cooled only through the thermal resistance).
- The other chamber has a *variable volume* whose boundary is a piston of area A that can move.

The two chambers are separated by the thermal resistance R. The constitutive laws for the fluids are derived from the total internal energy $U(S, V)$ of the perfect gas — specifically, temperature T and pressure p are found by differentiating U with respect to S and V respectively.

The model represented by this bond graph is appropriate only for **slow changes**: the pseudo-equilibrium assumption requires that processes occur slowly enough for temperature to be uniform throughout each chamber at each instant.

Although the entire system is **power conservative** (no energy is created or destroyed), the entropy of the two C-elements increases over time: it appears as if energy is lost. What the bond graph reveals is that energy is *not* lost, but rather converted into thermal energy that *cannot be completely converted back* into mechanical energy — a direct manifestation of the second law. The unique limiting case in which X_5 moves so slowly that T_2 and T_3 are virtually identical is the only scenario in which net entropy production almost vanishes.

8.2.4 Including Coulomb friction

Let us extend the two-chamber model to include **Coulomb friction** in the piston. The friction force is:

$$F_6 = A \operatorname{sgn} \dot{X}_5, \tag{8.29}$$

so the dissipated mechanical power is:

$$F_6 \dot{X}_5 = A\dot{X}_5 \operatorname{sgn} \dot{X}_5 = A|\dot{X}_5| \geq 0. \tag{8.30}$$

This power is converted entirely into entropy flowing into the adjacent fluid at temperature T_3:

$$\dot{S}_4 = \frac{1}{T_3} A|\dot{X}_5| \geq 0. \tag{8.31}$$

In bond-graph terms, the mechanical friction is represented as a **2-port R-field** connecting a mechanical port and a thermal port. The R-field can accept either causality at the mechanical port (force or velocity input), but *only the causality shown* at the thermal port — temperature must be an input and entropy flow an output. The frictional energy reaches the fluid instantaneously, which is an idealisation valid when thermal conduction within the fluid is fast compared to the timescale of interest.

8.2.5 Electrothermal resistor

Power dissipation results in thermal effects. To illustrate the true bond-graph treatment of mixed energy domains, consider a resistor that simultaneously carries **electrical current** and is heated by **Joule losses**. The resistor has a uniform temperature T and dissipates electrical power:

$$ei \geq 0, \tag{8.32}$$

where e is the voltage and i the current. If this inequality holds then both e and i have a "some" temperature dependence, because the resistance is a function of T. The power lost electrically is entirely converted into heat:

$$ei = \dot{Q} = T\dot{S}. \tag{8.33}$$

The R-field has two ports: one electrical (e, i) and one thermal (T, $\dot{S}$). There are **two admissible causal forms**:

$$\text{(a)} \quad e = e(i, T), \qquad \dot{S} = \frac{i\, e(i,T)}{T}, \tag{8.34}$$

$$\text{(b)} \quad i = i(e, T), \qquad \dot{S} = \frac{e\, i(e,T)}{T}. \tag{8.35}$$

The temperature of the resistor depends on how much thermal energy has been stored, so a **1-port C-element** is necessary to model the thermal capacity. It is common to assume T is a linear function of the stored energy Q:

$$T = T_0 + \frac{Q}{C}, \tag{8.36}$$

with $\dot{Q} = T\dot{S}$. However, for the true bond-graph C-element the constitutive law must express T as a function of the *entropy* S:

$$S = \int_0^S \mathrm{d}\sigma = \int_0^Q \frac{\mathrm{d}q}{T_0 + q/C} = C \ln\left[\frac{1}{T_0}\left(T_0 + \frac{Q}{C}\right)\right]. \tag{8.37}$$

Inverting gives the true C-field constitutive law:

$$T = T_0\, \mathrm{e}^{S/C}, \tag{8.38}$$

which is the correct entropy-based form. Note the nonlinearity: even in the simple case of constant heat capacity C, the true bond-graph description requires an exponential constitutive law, not a linear one.

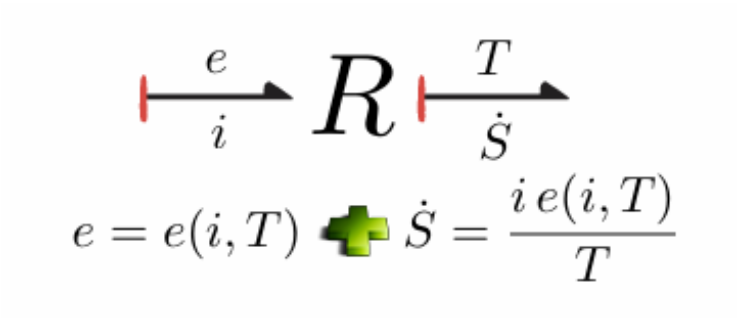


Figure 8.4: Bond graph for the electrothermal resistor

■ *: the electrical R-field (effort e, flow i) is connected via a 0-junction to the thermal C-element (effort T, flow $\dot{S}$), with an ambient temperature*

Watch out: Only closed systems so far

Up to this point, only **closed systems** have been modelled — no mass crossed the system boundary. When mass can flow into and out of a control volume, the situation becomes significantly more complex and requires the introduction of enthalpy convection terms. This is addressed in Section 8.3.7.

8.2.6 The missing element: no thermal inertia

In every energy domain encountered so far (mechanical, electrical, hydraulic) there is an I-element dual to the C-element. Let us ask whether a **thermal I-element** exists. In the thermodynamic domain the effort is temperature T (or temperature difference ΔT) and the flow is the entropy flow $\dot{S}$. A hypothetical I-element would obey:

$$\Delta T = I \frac{\mathrm{d}f}{\mathrm{d}t}, \tag{8.39}$$

where I is some parameter (a "thermal inductance"). Equation (8.39) implies that $f = \dot{S}$ may have a *constant non-zero value* even when $\Delta T = 0$: there would be a sustained entropy (heat) flow in the absence of any temperature difference.

However, the **second law of thermodynamics** states that a *spontaneous* heat or entropy flow is *always* directed from a point of higher temperature to a point of lower temperature. With an entropy flow there *must* be a temperature difference. A thermal I-element would violate this postulate. Therefore, **the I-element does not exist in the thermodynamic domain**. This is not a modelling choice but a fundamental physical constraint.

Key Insight: "Spontaneous" means no cross-effects

The word "spontaneous" in the second-law statement means that no other gradients (chemical potential, electric field, etc.) are present that might cause cross-effects and drive entropy flow in the absence of a temperature difference. Cross-effects (e.g. the Seebeck effect) can couple entropy flow to non-thermal driving forces, but these are modelled by multi-port elements, not by a thermal I-element.

8.3 Fluid-dynamic systems revisited

8.3.1 Scope and limitations of the bond-graph approach

The bond-graph models of fluid systems introduced earlier were of the **closed-circuit hydraulic type**: static pressure times volume flow rate (pQ) represents most of the transmitted power. We now take a deeper look at fluid systems in general and identify the limitations of these models.

Full **fluid mechanics** deals with field problems in two or three dimensions described by PDEs (Navier–Stokes equations), for which finite-difference or finite-element techniques are required and bond graphs offer little advantage. The focus here is therefore:

- The *rough lumping* appropriate when a fluid sub-system is a small part of a larger mechatronic system.
- *Interior flows* and the fluid-dynamic interaction with solid mechanical elements.

8.3.2 Estimating fluid inertia

The I-element represents the inertia of the fluid column. An elementary derivation starts from the 1-D assumption: treat the fluid as moving like a rigid body (quasi-one-dimensional flow) with uniform velocity Q/A across the cross-section. Newton's second law for the fluid column of length L and area A gives:

$$\rho A L \,\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Q_2}{A}\right) = A(p_1 - p_3) = A p_2, \tag{8.40}$$

which can be rearranged to:

$$Q_2 = \frac{1}{\rho L/A}\, p_{P_2}, \qquad I_{\text{fluid}} = \rho L/A \quad \text{(inertia coefficient)}. \tag{8.41}$$

Several caveats attach to this result:

- The uniform velocity assumption is valid for *transient* conditions only approximately; a correcting factor can improve accuracy.
- Unless the fluid is *incompressible*, the density ρ is not constant and the two ends of the fluid column do not necessarily move with the same velocity.
- Momentum fluxes at the pipe ends cancel only when the two ends have *identical cross-sectional areas*; this derivation does not generalise readily to pipes of varying area.

The compliance element C_6. The compliance element models the combined compliance of the fluid and pipe walls. For small pressure changes the linear capacitor relation is adequate:

$$p_6(t) = p_6(0) + \frac{B}{V_0}\int_0^t Q_6\,\mathrm{d}\tau = p_6(0) + \frac{B}{V_0}\,V_6(t), \tag{8.42}$$

where B is the bulk modulus and V_0 the nominal volume. For *large* density changes the thermodynamics of the situation must be studied — replacing the linear law with a nonlinear one requires a full thermodynamic analysis (see Section 8.3.6).

The friction element R_4. The friction element R_4 represents a pressure loss. The relation between p_4 and Q_4 is in general *nonlinear*: friction factor data from experimental fluid mechanics are almost exclusively for *fully developed steady flow*, making the transient nonlinear behaviour difficult to capture accurately within a simple lumped model.

8.3.3 Bernoulli's equation from the bond-graph viewpoint

To understand the connection between standard fluid mechanics and lumped bond-graph representation, we derive **Bernoulli's equation** for quasi-one-dimensional flow. This is an extension of the models of M03 and does not involve thermodynamic effects.

We adopt the **Eulerian method**: let s denote the arc-length coordinate along the pipe centreline, $v(s,t)$ the average fluid velocity, $p(s,t)$ the pressure, $A(s)$ the cross-sectional area, and $z(s)$ the height above a datum.

Newton's law applied to a fluid element gives the Euler equation (the *material derivative* D/Dt accounts for both local and convective acceleration):

$$\rho\,\frac{Dv}{Dt} = \rho\,\frac{\partial v}{\partial t} + \rho v\,\frac{\partial v}{\partial s} = -\frac{\partial p}{\partial s} - \rho g\,\frac{\mathrm{d}z}{\mathrm{d}s}. \tag{8.43}$$

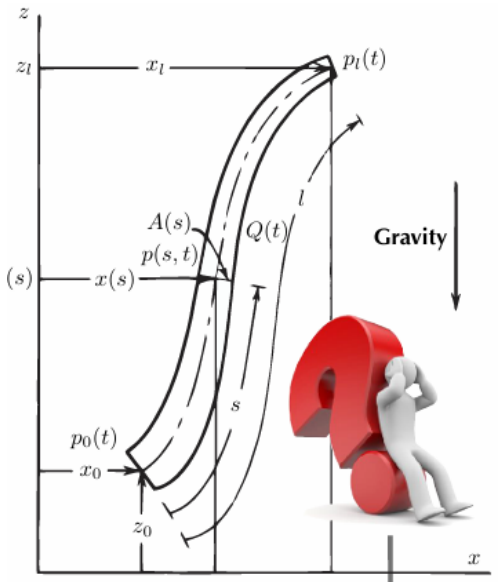


Figure 8.5: Pipe of varying cross-section

For an **incompressible fluid**, continuity requires a constant flow rate along the pipe:

$$v(s,t) = \frac{Q(t)}{A(s)}. \tag{8.44}$$

Substituting (8.44) into (8.43) and integrating along the pipe from $s = 0$ to $s = l$:

$$\rho\left[\frac{\dot{Q}(t)}{A(s)} + \frac{Q^2(t)}{A(s)}\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{A(s)}\right)\right] = -\frac{\partial p}{\partial s} - \rho g\frac{\mathrm{d}z}{\mathrm{d}s}, \tag{8.45}$$

which integrates to the **generalised Bernoulli equation**:

$$\boxed{I\dot{Q} + \frac{\rho Q^2}{2}\left(\frac{1}{A_l^2} - \frac{1}{A_0^2}\right) = p_0(t) - p_l(t) - \rho g(z_l - z_0)}, \qquad I = \rho\int_0^l \frac{\mathrm{d}s}{A(s)}. \tag{8.46}$$

The first term is the inertia (I, with the fluid inertia coefficient matching (8.41) for constant A); the second term involves Q^2 and requires further discussion.

8.3.4 Dynamic pressure and the Bernoulli resistance

The Q^2 term in (8.46) is the **dynamic pressure** term:

$$p_{\mathrm{dyn}} = \frac{\rho Q^2}{2}\left(\frac{1}{A_l^2} - \frac{1}{A_0^2}\right). \tag{8.47}$$

If $A(s)$ is constant, this term vanishes (the kinetic energies at inlet and outlet are equal). When A varies, the term represents the conversion of static pressure to kinetic energy (nozzle) or vice versa (diffuser), and is represented in the bond graph as a **Bernoulli resistance** R_B.

This resistance is *unusual*: no matter whether $A_0 > A_l$ or $A_0 < A_l$, there exist regimes in which R_B *appears to supply* rather than dissipate power. Furthermore, although the pressure p_{dyn} is uniquely determined for a given flow rate Q, given a pressure p_{dyn} there is *either no* corresponding flow rate *or two* of them — making the inverse causality (pressure input, flow output) problematic. This gives rise to a **strong preference for the flow-input causality** when the Bernoulli resistance is included in a bond graph.

8.3.5 Example: tank emptying through a pipe

To illustrate the utility of the bond-graph representation, consider estimating the time for a **tank to empty through a pipe**. The system consists of a large tank (area A_T, volume V) connected to a pipe of length L and exit area $A_2 \ll A_1 \approx A_T$. The pipe discharges to a zero-pressure environment ($p_L = 0$).

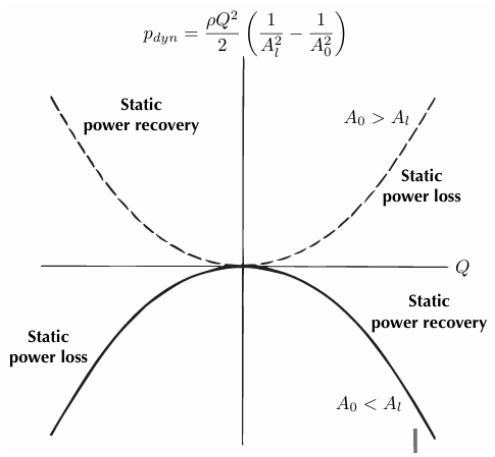


Figure 8.6: The Bernoulli resistance characteristic

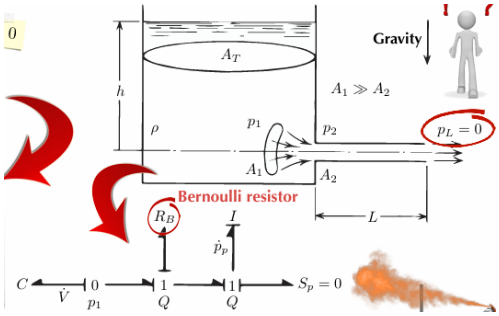


Figure 8.7: Tank emptying

The key assumption is that the fluid fills the pipe at all times ($v = Q/A$), and the derived equations are valid for *one direction of flow only*.

The bond-graph state equations, derived from (8.46) with a Bernoulli resistance and a fluid inertia $I = \rho L/A_2$, are:

$$\dot{V} = -\frac{A_2}{\rho L}\, p_p, \qquad V \geq 0, \tag{8.48}$$

$$\dot{p}_p = -\frac{\rho}{2A_2^2}\left(\frac{A_2}{\rho L} p_p\right)^2 + \frac{\rho g V}{A_T}. \tag{8.49}$$

Eliminating p_p in favour of V:

$$-\frac{\rho L}{A_2}\,\ddot{V} = -\frac{\rho}{2A_2^2}\,\dot{V}^2 + \frac{\rho g}{A_T}\,V, \qquad V \geq 0. \tag{8.50}$$

This is a second-order nonlinear ODE that can be integrated numerically. Note that, although no static pressure power flows into the zero-pressure source at the pipe exit, there *is* a flow; with a diffuser at the exit one could theoretically recover power from this flow.

8.3.6 Compressible fluids and the transition to pseudo-bond graphs

For **small density changes** compressibility is readily modelled by linear capacitors (as in (8.42)). For **large density changes** the thermodynamics of the fluid must be studied explicitly. Maintaining a true bond-graph representation for compressible fluids leads to further complications; this fact motivates the development of a new form of **pseudo-bond graph**.

The internal energy per unit mass depends on density ρ and entropy s through the Gibbs equation:

$$\mathrm{d}u = T\,\mathrm{d}s - p\,\mathrm{d}\left(\frac{1}{\rho}\right), \qquad u = u\left(s, \frac{1}{\rho}\right), \tag{8.51}$$

where $1/\rho$ plays the role of specific volume. This corresponds to a C-field in the thermodynamic sense. Using the C-field representation it is straightforward to model *closed* systems in which fixed amounts of a

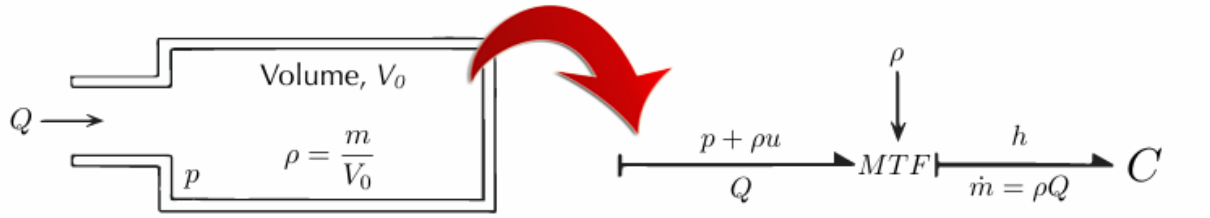


Figure 8.8: Bond graph for a fixed-volume open control volume

fluid are compressed, expanded, and heated. When the fluid **passes into and out of a control volume** the situation is more complex.

8.3.7 Open volume: convection of internal energy

Consider a fluid compressed in a **fixed volume** V_0 by allowing it to pass slowly in and out of a port with volumetric flow rate Q and pressure p:

$$\rho = \frac{m}{V_0}, \qquad \dot{m} = \rho Q. \tag{8.52}$$

The total energy U in the control volume depends on the mass m and the internal energy per unit mass u. Since U changes not only because of the *flow work* pQ but also because of the *convection of internal energy*, the correct flow variable is the **mass flow rate** $\dot{m} = \rho Q$. The energy balance is:

$$\mathrm{d}U = u\,\mathrm{d}m + \frac{p}{\rho}\,\mathrm{d}m = h\,\mathrm{d}m, \qquad h = u + \frac{p}{\rho} \quad \text{(specific enthalpy)}, \tag{8.53}$$

so that the power entering the control volume is:

$$h\dot{m} = u\dot{m} + \frac{p}{\rho}\dot{m} = u\rho Q + \frac{p}{\rho}\,\rho Q = (\rho u + p)Q. \tag{8.54}$$

In bond-graph terms, it is necessary to **supplement the pressure** with an extra term ρu to account for the convected internal energy. This is modelled by a **modulated transformer** (MTF) whose modulus is ρ, converting the port $(p + \rho u,\ Q)$ to the port $(h,\ \dot{m} = \rho Q)$, which then drives the C-field of the control volume:

Key Insight: Dynamic pressure neglected

The derivation above has *not* incorporated the dynamic pressure term $\rho Q^2/(2A^2)$. The model is therefore valid only for low flow rates where the kinetic energy of the incoming stream is negligible compared to its enthalpy.

In the **isentropic case** ($\dot{s} = 0$) pressure and internal energy are determined solely by the density, so the enthalpy at the port is determined entirely by conditions *inside* the vessel. In more complicated cases (non-isentropic flow) the enthalpy at the port is determined by:

- Conditions *inside* the vessel for *outflow*.
- *External* conditions for *inflow*.

This causal switching with changes in the direction of flow is not readily apparent in the true bond graph; it will be explicitly addressed in the pseudo-bond-graph formulations developed in the following chapters.

8.3.8 Variable-volume compression: the piston–cylinder

As a final extension, consider **isentropic compression** occurring simultaneously due to variable volume (a piston) and to inflow/outflow. The cylinder has area A, piston displacement x, and volume $V = Ax$. The total energy U depends on both the mass m and the volume V, so the C-field satisfies:

$$h = \frac{\partial U}{\partial m}, \qquad -p = \frac{\partial U}{\partial V}. \tag{8.55}$$

The bond graph is a more sophisticated version of the compressibility models used in servomechanisms, where the volume of fluid is variable.

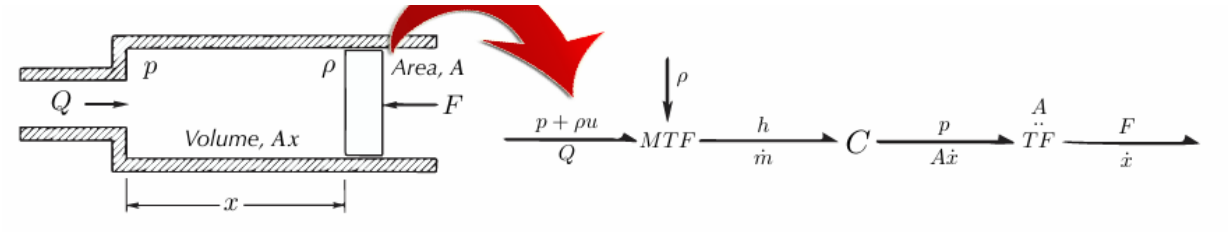


Figure 8.9: Piston–cylinder with inflow/outflow

This bond graph is a powerful, general representation: it handles simultaneously the thermodynamics of compression, the convection of mass and energy through the port, and the mechanical work done by or on the piston. Linearised compressibility parameters (constant bulk modulus) are a special case, but it is **virtually impossible** to proceed from a linearised model to a more accurate nonlinear one without returning to the physics and reconstructing the bond graph from scratch. The systematic derivation from $U(m, V)$ ensures that the nonlinear model is available from the outset and that no energy-conservation errors are introduced by *ad hoc* linearisation.

8.4 Summary

This chapter has developed the bond-graph treatment of thermo-fluid systems at two levels of depth.

True bond graphs for thermodynamics. The pure substance is a 2-port C-field with T (effort) and $\dot{s}$ (flow) at one port and p (effort) and $\dot{v}$ (flow) at the other. The product $T\dot{s}$ is genuine thermodynamic power. The four causal patterns of this C-field correspond to the four classical thermodynamic potentials (u, h, f, ϕ) via Legendre transformations, and all four automatically ensure conservation of energy. Heat transfer is modelled by a 2-port R-field that generates entropy; its only admissible causality has temperatures as inputs. A thermal I-element is physically impossible (it would violate the second law). With these elements a rich variety of thermal and electrothermal systems can be modelled rigorously.

Fluid-dynamic lumped models. Standard lumped models of fluid systems (fluid inertia $I = \rho L/A$, compliance B/V_0, friction R, leakage R) are useful for short, linear elements but carry significant philosophical caveats for long pipes, nonlinear flows, and compressible fluids. The Bernoulli equation reveals an unusual Q^2 term (dynamic pressure / Bernoulli resistance) that can supply as well as dissipate power and creates multi-valuedness under pressure-input causality. For compressible fluids with mass exchange through a control volume, the correct power variable at the port is the enthalpy flux $h\dot{m}$, and the bond graph requires an MTF to account for convected internal energy. The resulting nonlinear C-field formulation is the natural starting point for accurate models of pneumatic actuators, compressors, and variable-volume reservoirs.